Computer Architecture

Lecture 14b: ColumnDisturb & PuDHammer

Ismail Emir Yuksel
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ETH Zürich

Fall 2025

7 November 2025

Brief Self Introduction

- **Ismail Emir Yuksel**
  - Researcher @ SAFARI Research Group since September 2022
    - PhD @ ETH Zurich (Feb 2024 – ongoing)
  - ismail.yuksel@safari.ethz.ch

- **Research interests:**
  - Computer architecture
  - Processing-in-Memory (specifically Processing-using-DRAM)
  - DRAM Safety, Security and Reliability
  - Hardware security, safety, reliability, performance, availability, fairness, energy efficiency
  - …

Today's Agenda

MICRO 2025

ColumnDisturb: Understanding Column-based Read Disturbance in Real DRAM Chips and Implications for Future Systems

İsmail Emir Yüksel[1] Ataberk Olgun[1] F. Nisa Bostancı[1]
Haocong Luo[1] A. Giray Yağlıkçı[1,2] Onur Mutlu[1]
[1]ETH Zürich [2]CISPA

ISCA 2025

PuDHammer: Experimental Analysis of Read Disturbance Effects of Processing-using-DRAM in Real DRAM Chips

İsmail Emir Yüksel Akash Sood Ataberk Olgun Oğuzhan Canpolat Haocong Luo
F. Nisa Bostancı Mohammad Sadrosadati A. Giray Yağlıkçı Onur Mutlu

ETH Zürich

ColumnDisturb

Understanding Column-based Read Disturbance in Real DRAM Chips and Implications for Future Systems

İsmail Emir Yüksel

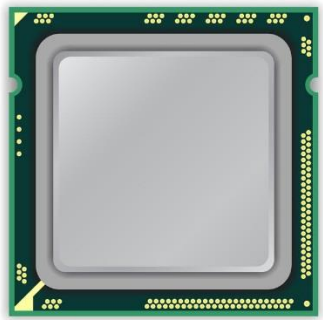
Ataberk Olgun F. Nisa Bostancı Haocong Luo

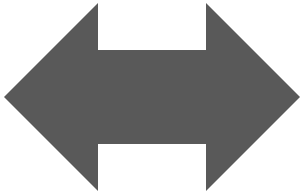
A. Giray Yağlıkçı Onur Mutlu

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A Typical DRAM-Based Computing System

DRAM Chip



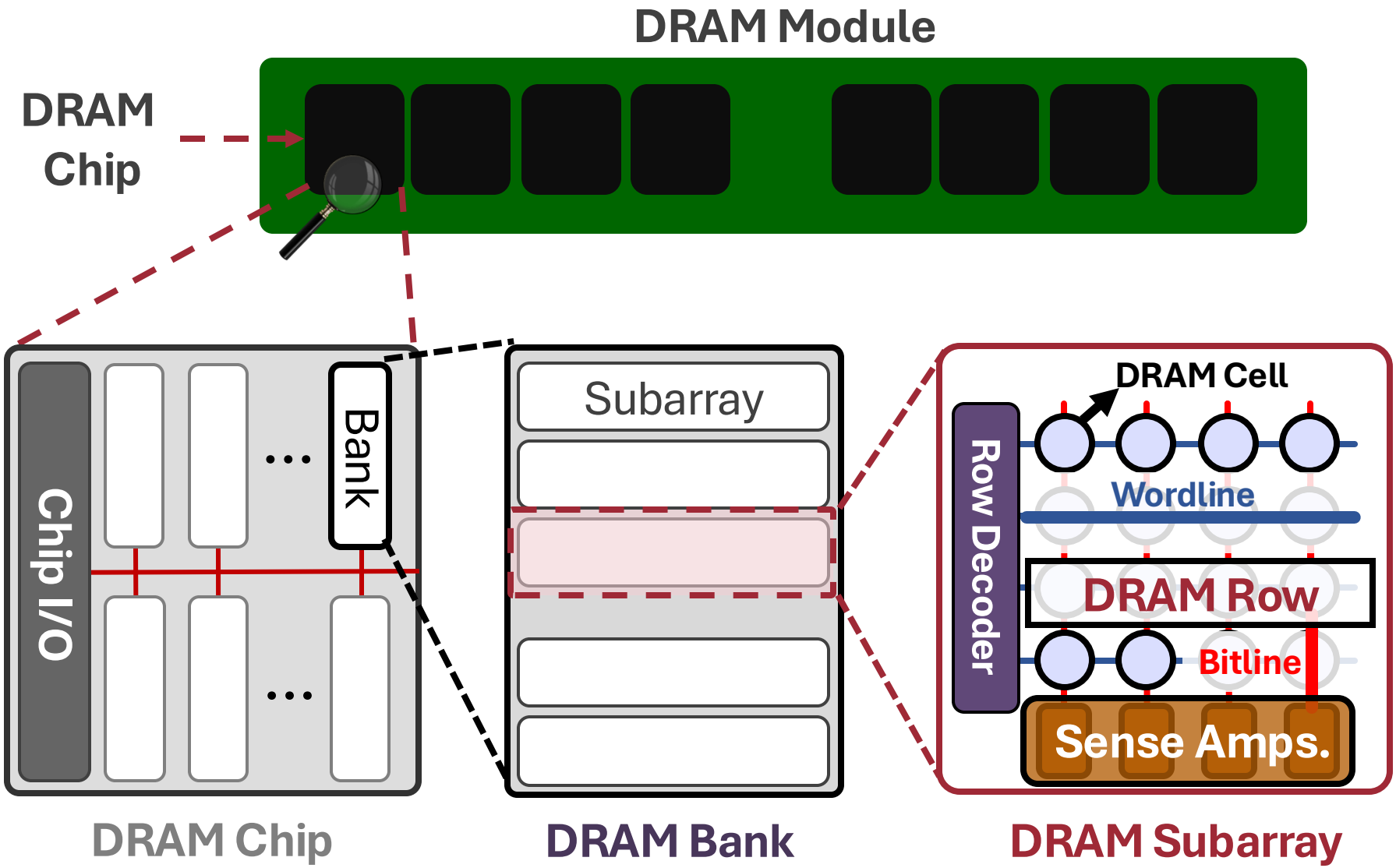




CPU

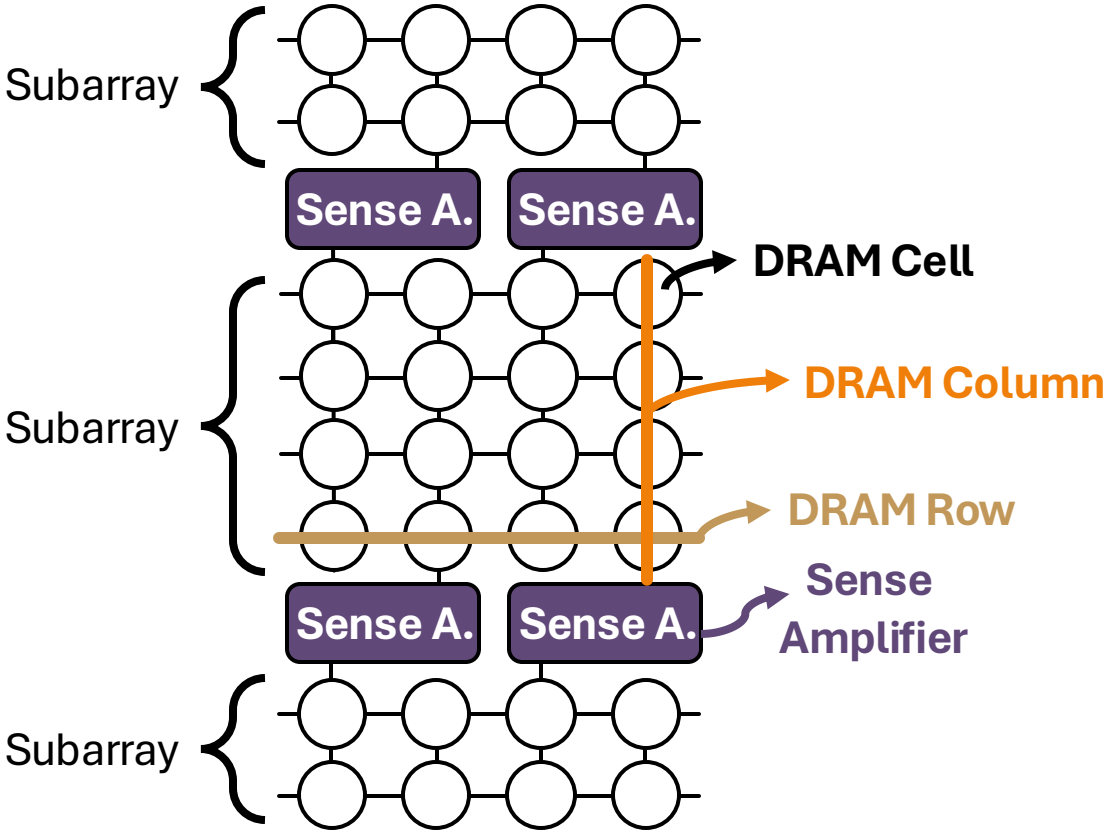
DRAM Module

DRAM Organization



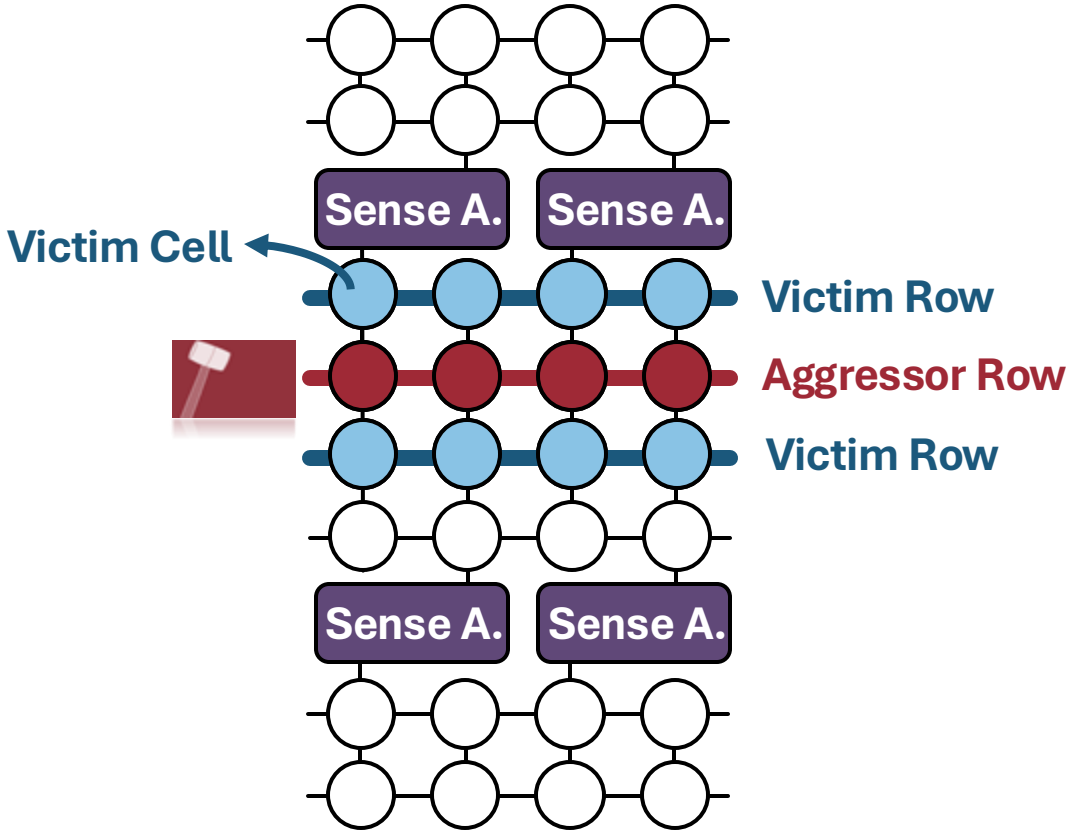
Read Disturbance in DRAM (I)

- Read disturbance in DRAM breaks memory isolation
- Prominent example: RowHammer



Read Disturbance in DRAM (I)

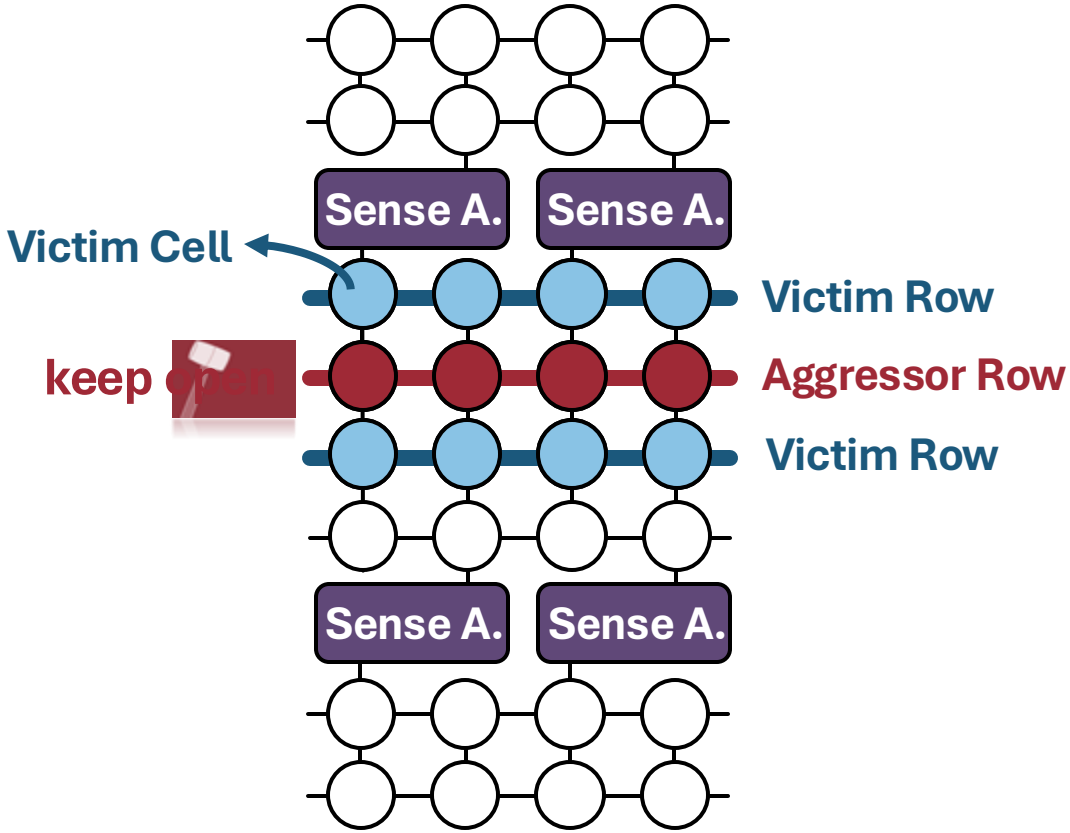
- Read disturbance in DRAM breaks memory isolation
- Prominent example: RowHammer



Repeatedly opening and closing a DRAM row many times causes bitflips in adjacent rows in the same subarray

Read Disturbance in DRAM (II)

- Read disturbance in DRAM breaks memory isolation
- A new read disturbance phenomenon: RowPress



Keeping a DRAM row open for a long time
causes bitflips in adjacent rows in the same subarray

Read Disturbance in DRAM (III)

- RowHammer and RowPress are **row-based read disturbance phenomena** that affect only a few neighboring victim rows

- Most of the mitigations work by
  1) detecting high row activation count or high row open time
  2) refreshing a few neighboring rows (e.g., +/-4)

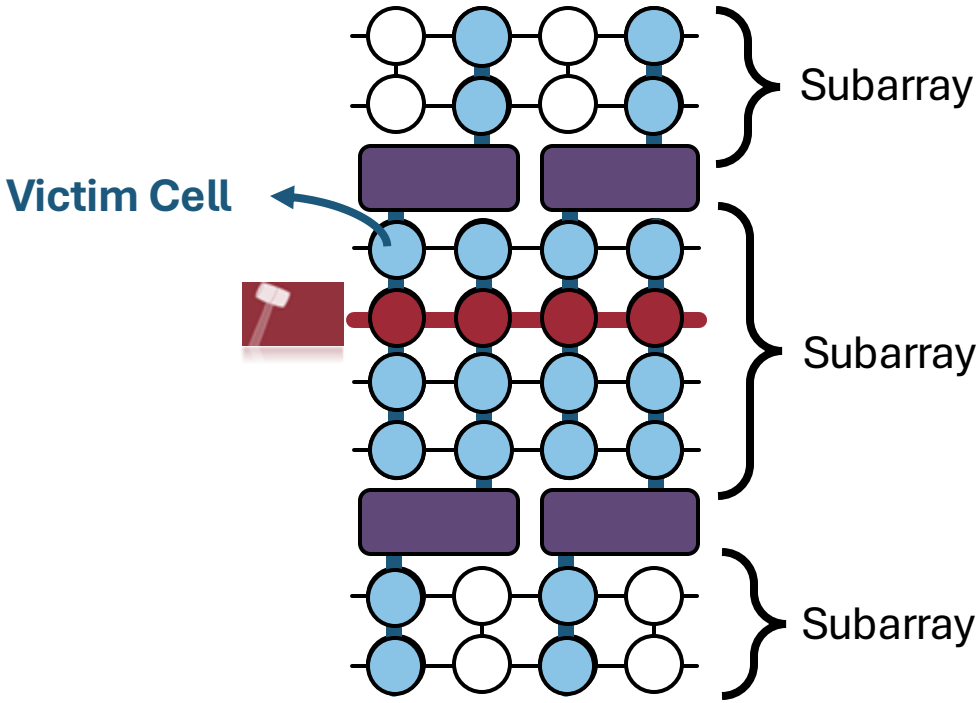
Detecting and refreshing **even only a few** neighboring rows is shown to be **a difficult problem**

Research Question

What if there is a much worse read disturbance phenomenon that affects thousands of DRAM rows through the bitlines?

ColumnDisturb

By repeatedly opening or keeping a DRAM row open, it is possible to disturb many DRAM cells on the same DRAM column



With ColumnDisturb, the activation of a single row concurrently affects DRAM cells across as many as three DRAM subarrays (e.g., up to 3072 DRAM rows)

Outline

What is ColumnDisturb?

Testing Infrastructure

Real DRAM Chip Results

Implications

Conclusion

Outline

What is ColumnDisturb?

Testing Infrastructure

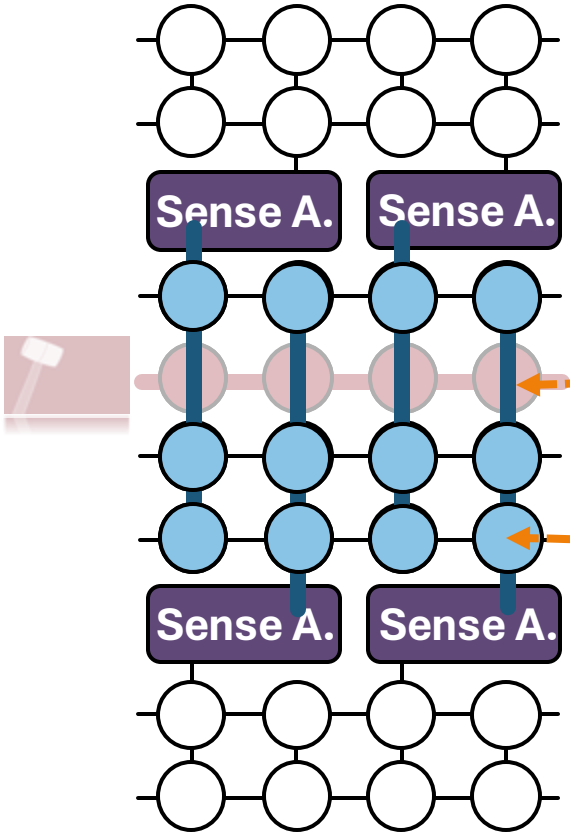
Real DRAM Chip Results

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What is ColumnDisturb?

- ColumnDisturb is a **column-based** read disturbance phenomenon that **affects thousands of rows**

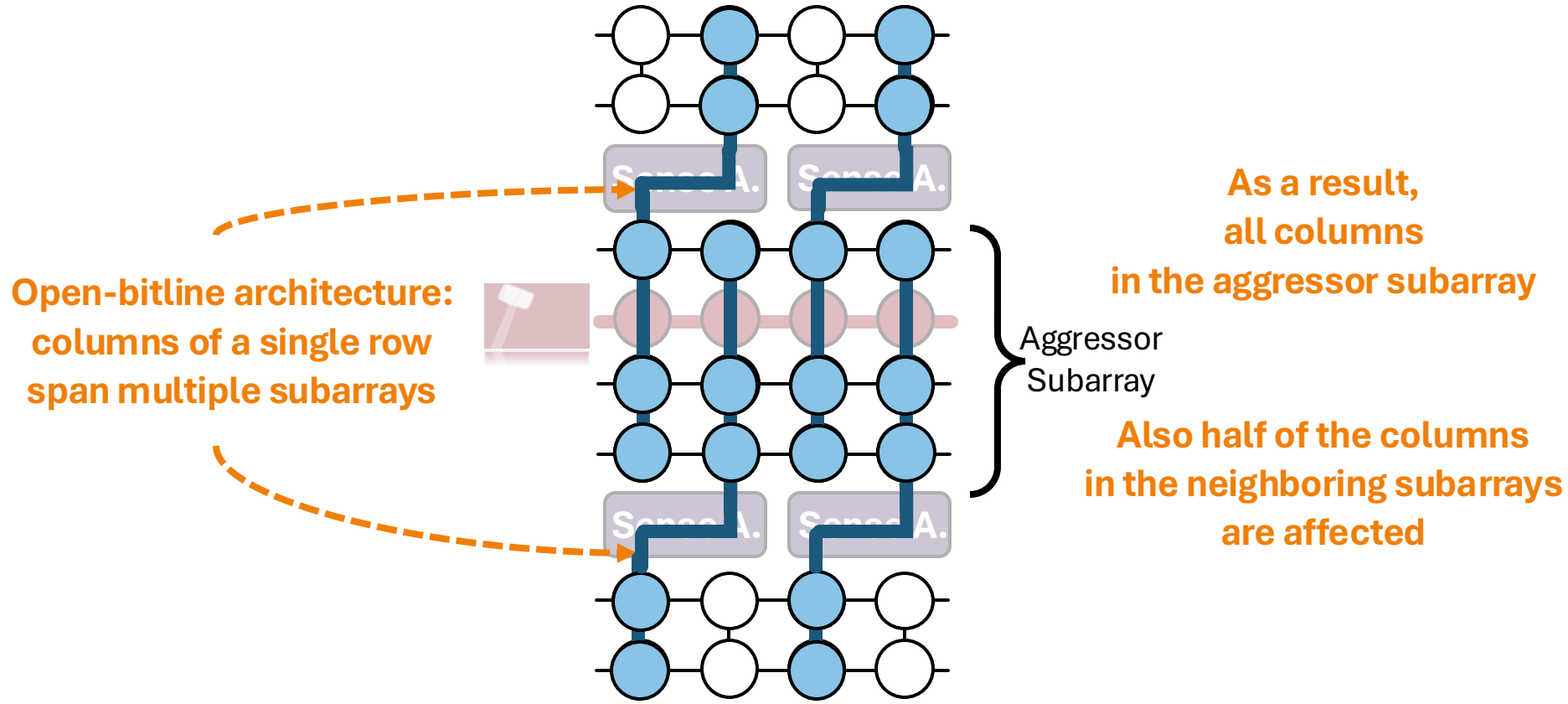


An activation of a row perturbs all the columns connected to that row

Cells in all perturbed columns are disturbed

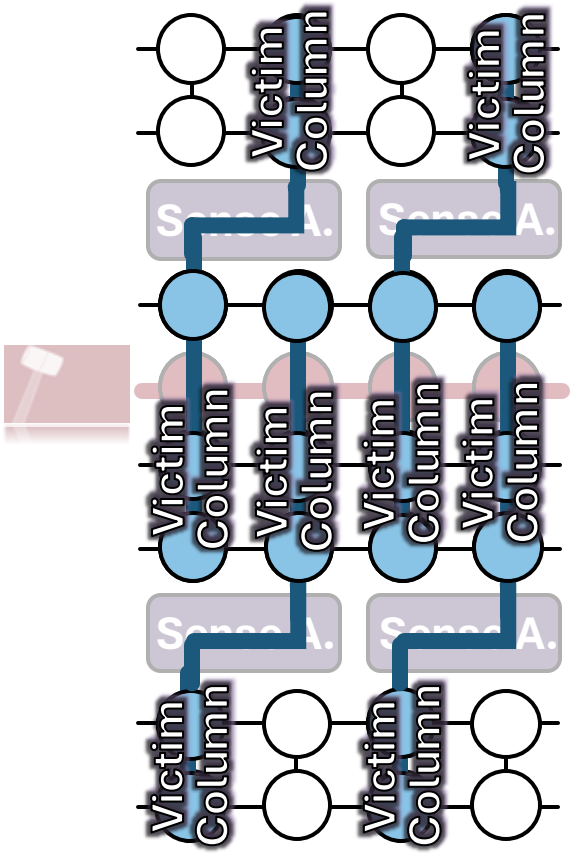
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What is ColumnDisturb?

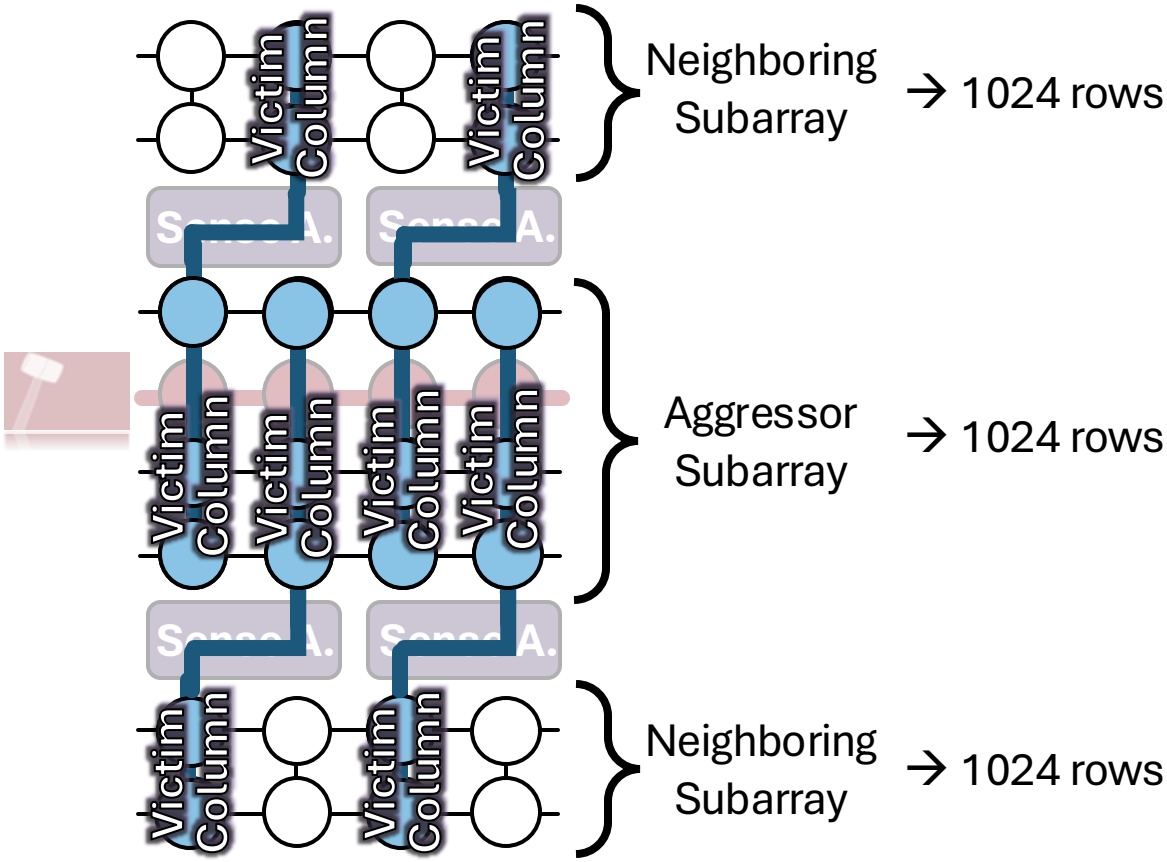
- ColumnDisturb is a **column-based** read disturbance phenomenon that **affects thousands of rows**



ColumnDisturb affects cells **vertically connected to the same bitline.** **DRAM columns** are the victims

What is ColumnDisturb?

- ColumnDisturb is a **column-based** read disturbance phenomenon that **affects thousands of rows**



ColumnDisturb affects cells
across both **the same subarray and multiple neighboring subarrays**

Outline

What is ColumnDisturb?

Testing Infrastructure

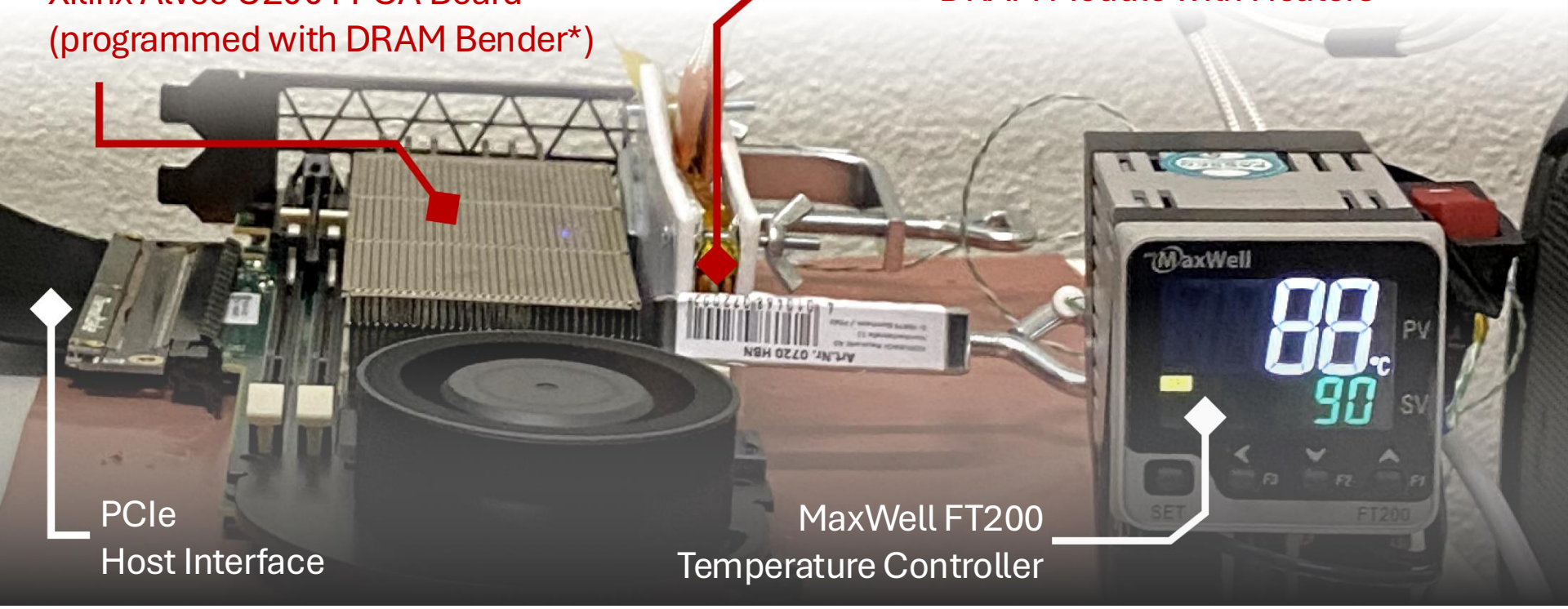
Real DRAM Chip Results

Implications for Future Systems & Mechanisms

Conclusion

DDR4 DRAM Testing Infrastructure

DRAM Bender on a Xilinx Virtex UltraScale+ XCU200

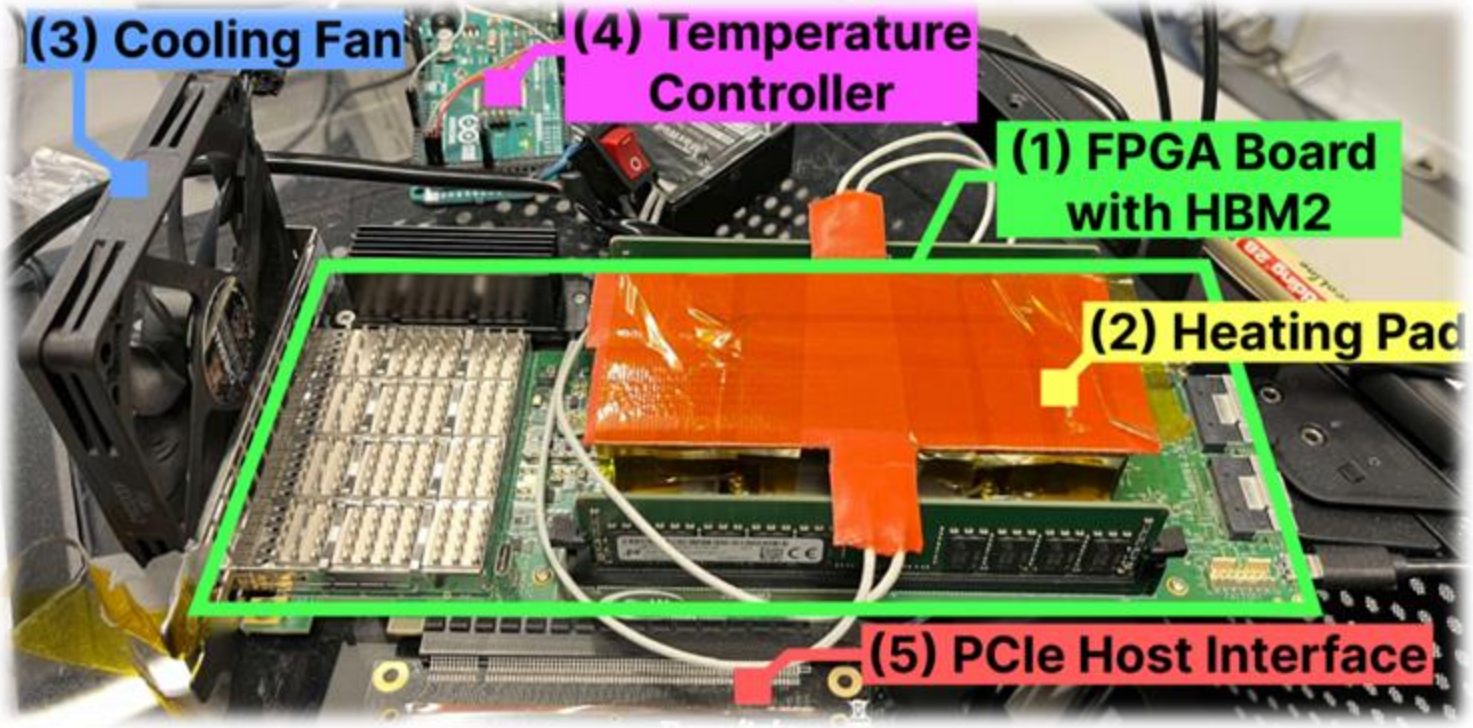


Fine-grained control over DRAM commands,
timing parameters (±1.5ns), and temperature (±0.5°C)

*Olgun et al., "DRAM Bender: An Extensible and Versatile FPGA-Based Infrastructure to Easily Test State-of-the-Art DRAM Chips," TCAD, 2023. [GitHub: https://github.com/CMU-SAFARI/DRAM-Bender]

HBM2 DRAM Testing Infrastructure

DRAM Bender on a Bittware XUPVVH

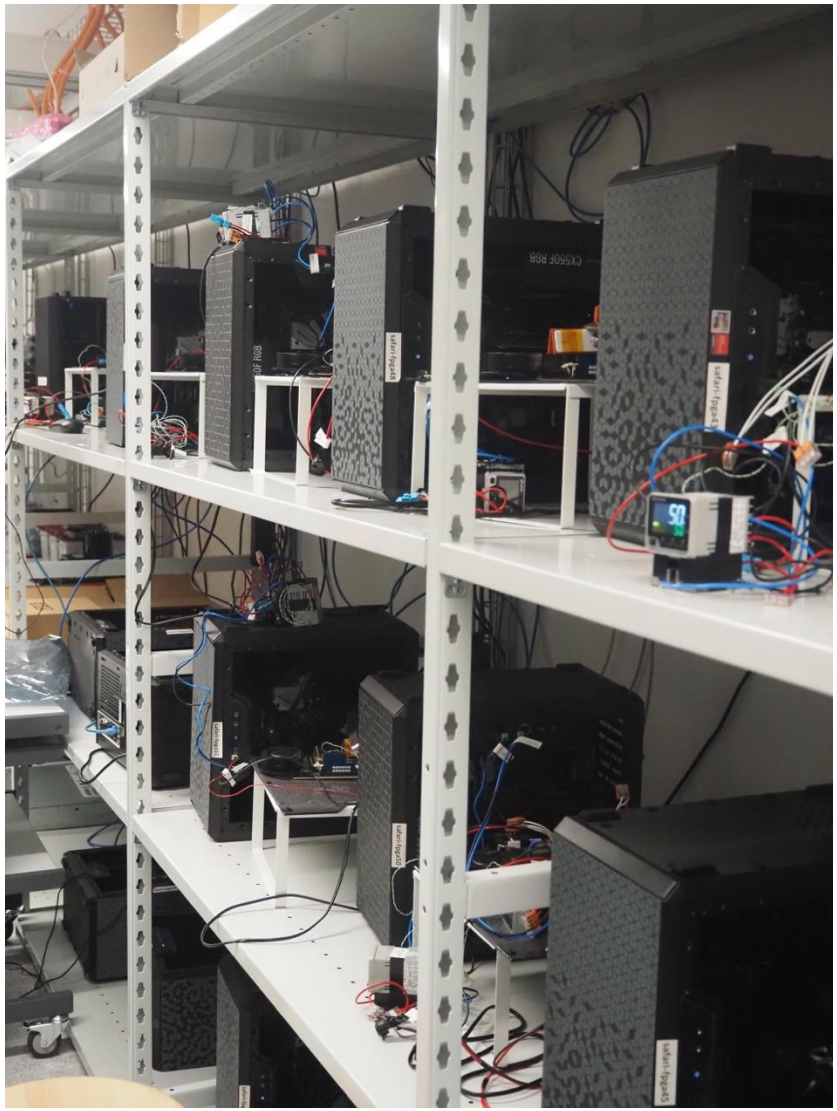


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Laboratory for Understanding Memory

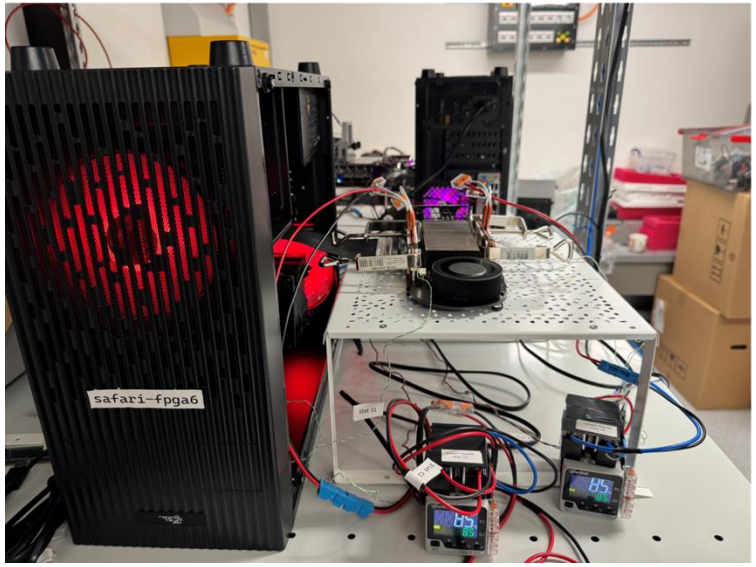




Laboratory for Understanding Memory









Tested DRAM Chips

216 DDR4 and 4 HBM2 Chips from SK Hynix, Micron, Samsung

Chip Mfr.	Module IDs	#Chips	Die Rev.	Density	Org.
SK Hynix	H0,H1,H2	24	A	8Gb	x8
	H3,H4,H5,H6	32	D	8Gb	x8
	H7	8	A	16Gb	x8
	H8,H9	16	C	16Gb	x8
Micron	M0	8	B	4Gb	x8
	M1,M2,M3	24	R	8Gb	x8
	M4, M5	16	B	16Gb	x8
	M6, M7	8	E	16Gb	x16
	M8,M9,M10,M11	32	F	16Gb	x8
Samsung	S0, S1	16	A	16Gb	x8
	S2, S3	16	B	16Gb	x8
	S4, S5	16	C	16Gb	x16
Samsung	HBM2 Chips	4	N/A	N/A	N/A

[a] We report "N/A" if the information is not publicly available.

Testing Methodology: Sources of Interference

Prevent sources of interference during core test loop

- **No DRAM refresh**
  - to avoid refreshing victim rows
- **No read disturbance mitigation mechanisms**
  - to observe circuit-level effects
- **No error correcting codes (ECC)**
  - to observe all bitflips
- **Filter out Retention & RowHammer/RowPress bitflips**
  - to demonstrate that ColumnDisturb bitflips are new and unique

Testing Methodology: Sources of Interference

ColumnDisturb: Understanding Column-based Read Disturbance in Real DRAM Chips and Implications for Future Systems

İsmail Emir Yüksel[1] Ataberk Olgun[1] F. Nisa Bostancı[1]
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We experimentally demonstrate a new widespread read disturbance phenomenon, ColumnDisturb, in real commodity DRAM chips. By repeatedly opening or keeping a DRAM row (aggressor row) open, we show that it is possible to disturb DRAM cells through a DRAM column *(i.e., bitline) and induce bitflips in DRAM cells sharing the same columns as the aggressor row (across multiple DRAM subarrays). With ColumnDisturb, the activation of a single row concurrently disturbs DRAM cells across as many as* three DRAM subarrays *(e.g., up to 3072 DRAM rows in tested DDR4 DRAM chips) as opposed to RowHammer & RowPress, which affect only a few neighboring rows of the aggressor row in a single subarray. We rigorously and comprehensively characterize ColumnDisturb and its characteristics under various operational conditions (i.e., temperature, data pattern, DRAM timing parameters, average voltage level of the bitline, memory access pattern, and spatial variation) using 216 DDR4 and 4 HBM2 chips from three major DRAM manufacturers. Among our 27 key experimental observations, we highlight two major results and their implications.*

1. Introduction

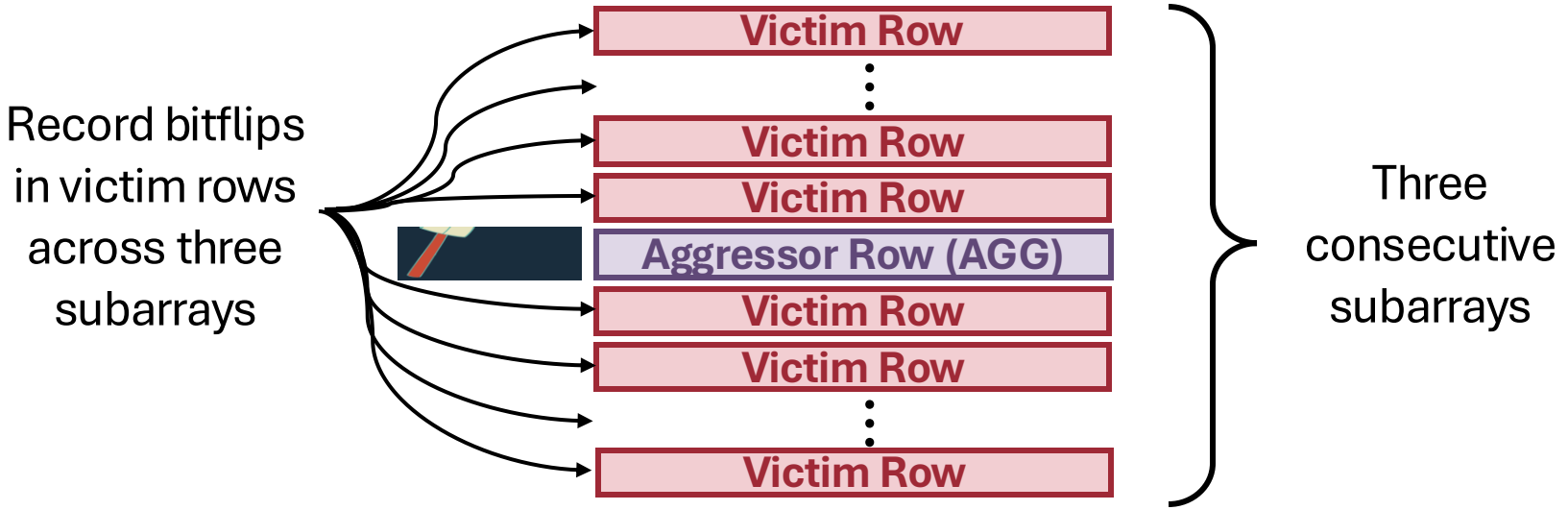
Dynamic random access memory (DRAM) [1] is the dominant main memory technology in nearly all modern computing systems due to its latency and cost characteristics. Continuing to increase DRAM capacity requires increasing the density of DRAM cells by reducing (i.e., scaling) the technology node size of DRAM, which in turn reduces DRAM cell size, cell-to-cell spacing, and cell-to-bitline spacing [2]. Unfortunately, as a result of this aggressive technology node scaling, DRAM suffers from worsening read disturbance problems [3–6].

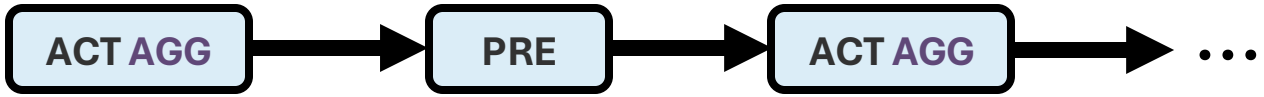
Read disturbance in modern DRAM chips [4–9] is a widespread phenomenon and is reliably used for breaking memory isolation [5,7,9–70], a fundamental building block of robust (i.e., safe, secure, reliable, available) systems. *RowHammer* and *RowPress* are two prominent examples of DRAM read disturbance phenomena where a victim DRAM row experiences bitflips when a nearby aggressor DRAM row is 1) repeatedly activated (i.e., hammered) [5–7] or 2) kept open for a long period (i.e., pressed) [4, 71]. Many prior works [3–5, 7, 9, 41, 42]

https://arxiv.org/pdf/2510.14750

Testing Methodology: Metric & Access Pattern

Minimum time required to induce **the first ColumnDisturb bitflip**	**Lower** is worse
Number of ColumnDisturb bitflips in a subarray	**Higher** is worse
Number of rows with ColumnDisturb bitflips in a subarray	**Higher** is worse





Testing Methodology: Test Parameters

We carefully sweep

- Data Pattern
  - 0x00, 0xAA, 0x33, and 0x77 (and their inverse pattern)
- Temperature (°C)
  - 45, 65, 85, and 95
- Aggressor Row On Time (tAggOn)*
  - 36ns, 7.8us, 70.2us, and 1ms
- DRAM Refresh Interval
  - 64ms, 128ms, 256ms, 512ms, 1s, 2s, 4s, 8s, and 16s
- Aggressor Row Location
  - Aggressor subarray's first, middle, and last rows
- Tested Subarrays (Rows)
  - Up to +49K (+4.5M): *all subarrays in all banks across all DRAM chips*

* The time an aggressor row stays open after each activation during a ColumnDisturb test

Outline

What is ColumnDisturb?

Testing Infrastructure

Real DRAM Chip Results

Implications

Conclusion

Key Takeaways from Major Characterization Experiments

Takeaway 1

ColumnDisturb is a **widespread read disturbance phenomenon** and **becomes worse** as DRAM technology **scales down to smaller** node sizes

Takeaway 2

ColumnDisturb induces bitflips in **thousands of DRAM rows** and is **different** from **RowHammer, RowPress, and Retention** failures

Takeaway 3

The **bitflip directionalities** of ColumnDisturb and RowHammer & RowPress are **significantly different**

Takeaway 4

Significantly more rows are vulnerable to ColumnDisturb **than** retention failures

Takeaway 5

As **temperature increases**, DRAM chips **become more vulnerable** to ColumnDisturb

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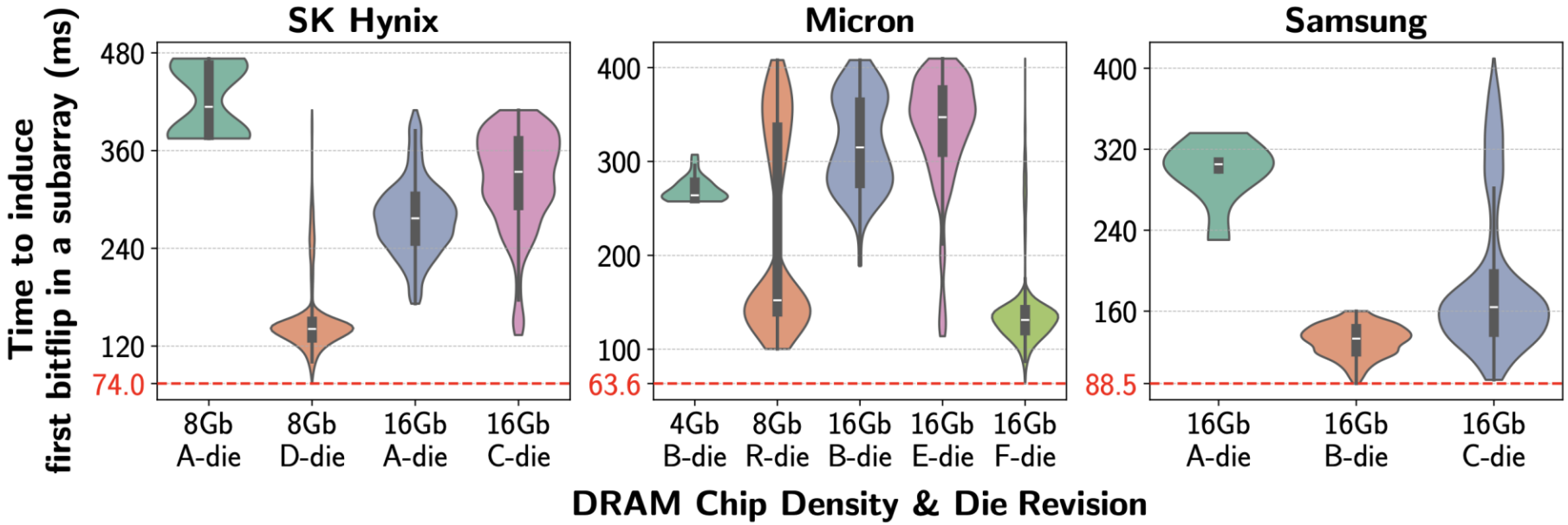
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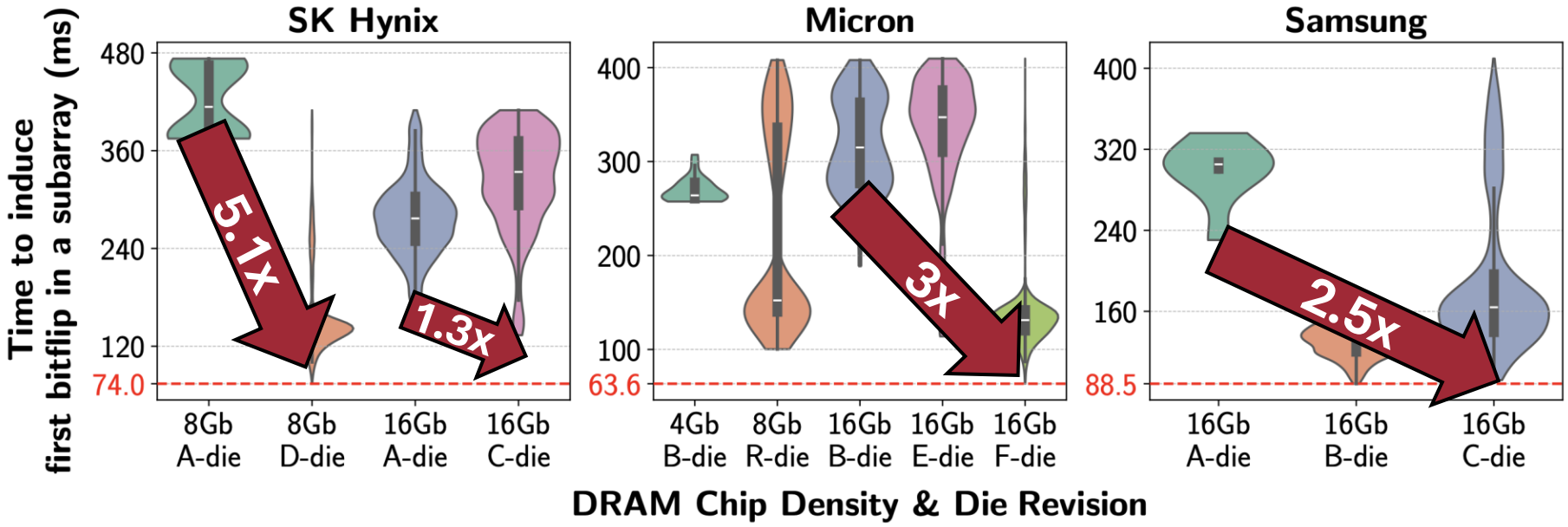
As **temperature increases**, DRAM chips **become more vulnerable** to ColumnDisturb

Minimum Time to Induce First Bitflip



All tested 216 DDR4 chips are **vulnerable** to ColumnDisturb

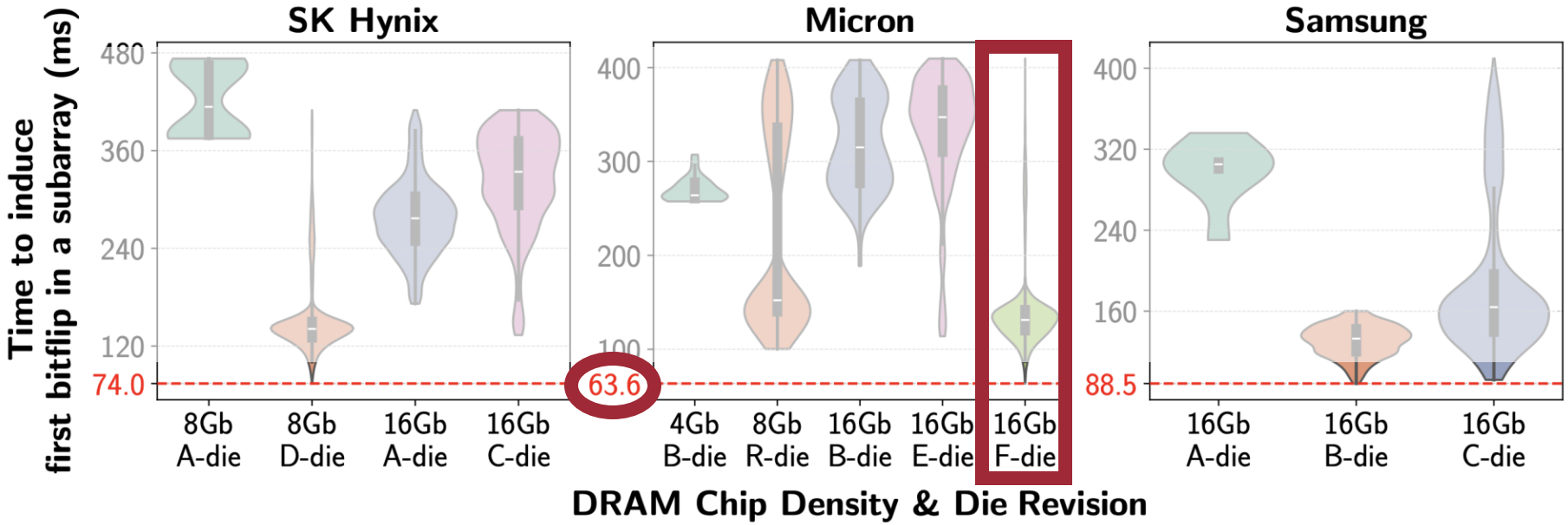
Minimum Time to Induce First Bitflip



Newer chips tend to be
more vulnerable to ColumnDisturb bitflips

This trend is **consistent across all tested manufacturers**

Minimum Time to Induce First Bitflip



Some **real DRAM chips** experience ColumnDisturb bitflips
within the nominal refresh window (64ms)
under nominal operating conditions

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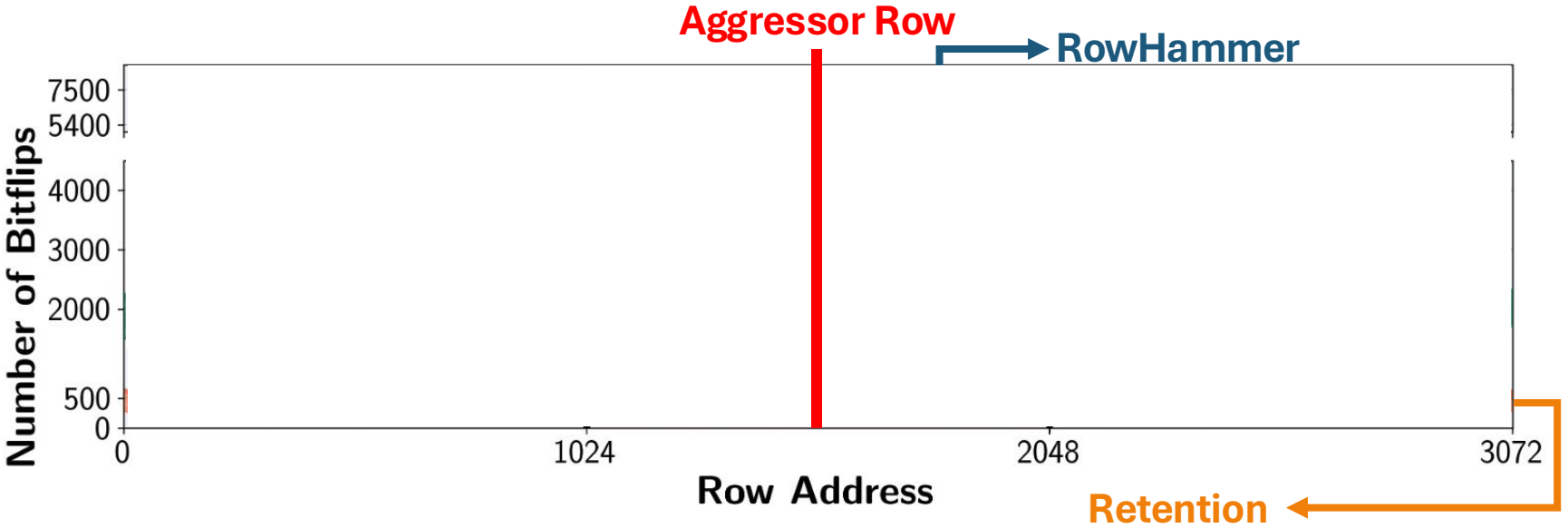
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Significantly more rows are vulnerable to ColumnDisturb **than** retention failures

Takeaway 5

As **temperature increases**, DRAM chips **become more vulnerable** to ColumnDisturb

ColumnDisturb vs. RH/RP & Retention



RowHammer & RowPress induce bitflips **only in the immediate neighboring rows** of the aggressor row

Retention failures are almost **constant** (around 500) for every row

ColumnDisturb induces **many more bitflips than retention failures** and **affects many more rows than RowHammer & RowPress**

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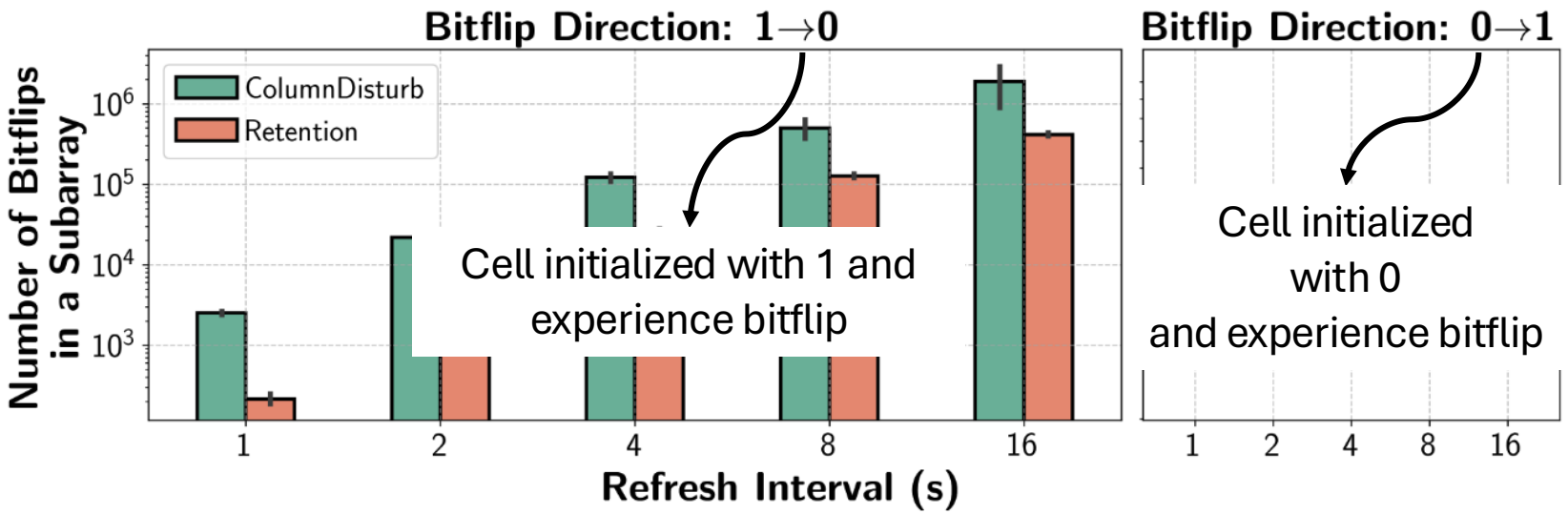
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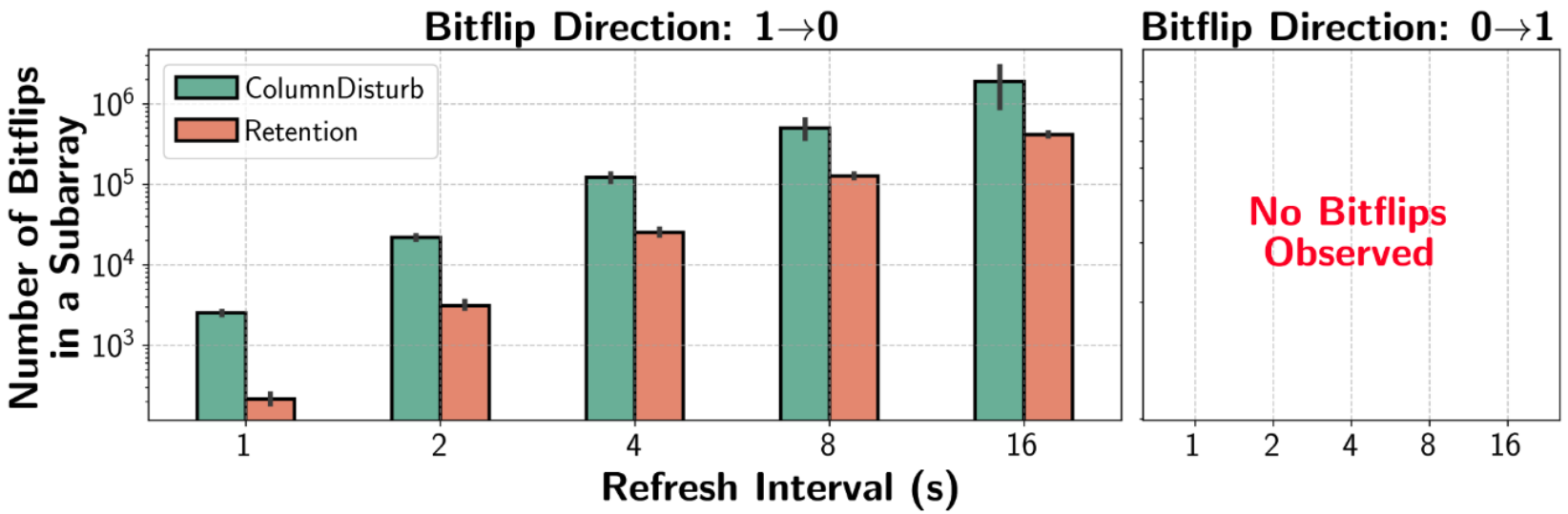
Bitflip Direction



DRAM refreshes all rows within the refresh interval

e.g., 1s means DRAM refreshes all rows within 1 second

Bitflip Direction



ColumnDisturb induces **only 1 to 0 bitflips,** which are **the same direction as retention failures**

This is in stark contrast to **RowHammer and RowPress**, which induce bitflips in **both 1 to 0 and 0 to 1 directions**

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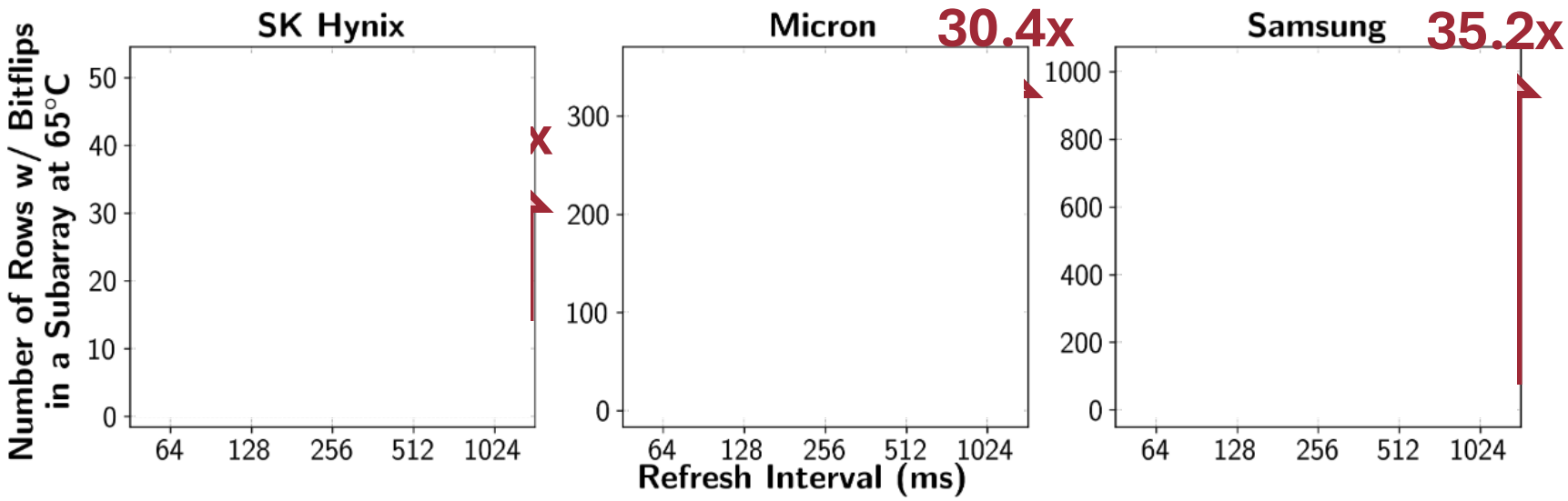
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Significantly more rows are vulnerable to ColumnDisturb **than** retention failures

Takeaway 5

As **temperature increases**, DRAM chips **become more vulnerable** to ColumnDisturb

Number of Rows w/ Bitflips



ColumnDisturb induces bitflips
in significantly more DRAM rows than retention failures

This trend is **consistent** across **all tested manufacturers**

Key Takeaways from Major Characterization Experiments

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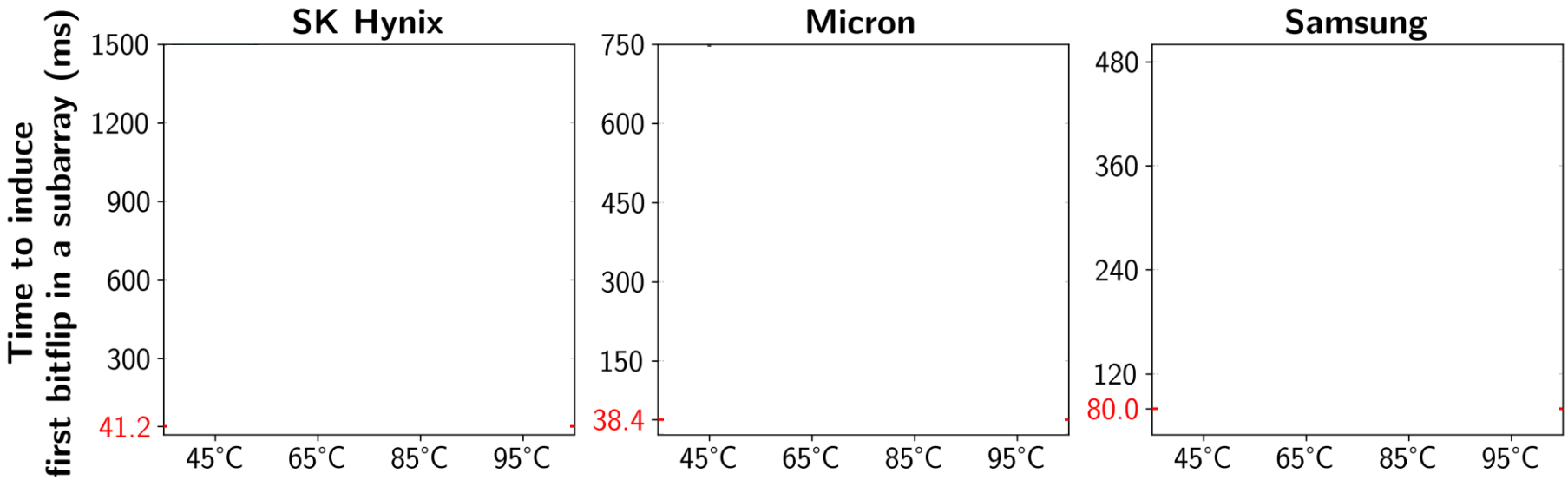
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Significantly more rows are vulnerable to ColumnDisturb **than** retention failures

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As **temperature increases**,
DRAM chips **become more vulnerable** to ColumnDisturb

Effect of Temperature



Time to induce the first ColumnDisturb bitflip **significantly reduces** as **temperature increases**

This trend is **consistent** across **all tested manufacturers**

Outline

What is ColumnDisturb?

Testing Infrastructure

Real DRAM Chip Results

Implications

Conclusion

Implications Summary

1. Implications on Future Computing Systems

ColumnDisturb could jeopardize the robustness of computing systems

- ColumnDisturb bitflips occur in very far away rows from the aggressor row, even **in nominal operating conditions**
- ColumnDisturb is **getting worse**: newer chips are **more vulnerable**

2. Implications on Retention-Aware Refresh Mechanisms

ColumnDisturb dramatically decreases the benefits of these mechanisms

- At long refresh intervals (e.g., >512ms), **a large majority of rows** experience ColumnDisturb bitflips
- ColumnDisturb induces **significantly more bitflips** than retention failures

Implications for System Robustness

- **ColumnDisturb bitflips** occur in **very far away rows** from the aggressor row
  - E.g., one of the bitflips we observe in nominal refresh window is **374 rows away** from the aggressor row

- Most of the read disturbance mitigations refresh only **a few neighboring rows** as RowHammer and RowPress induces bitflips **in close proximity**

- **Even refreshing only a few rows** is shown to be **a difficult problem**

- **We describe and analytically evaluate two solutions**
  - **Increasing DRAM Refresh Rate**
    - Reduces DRAM **throughput** by 42.1%
  - **Proactively Refreshing Victim Rows**
    - Reduces DRAM **throughput** by 12.4%

Implications for System Robustness

ColumnDisturb: Understanding Column-based Read Disturbance in Real DRAM Chips and Implications for Future Systems

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1. Introduction

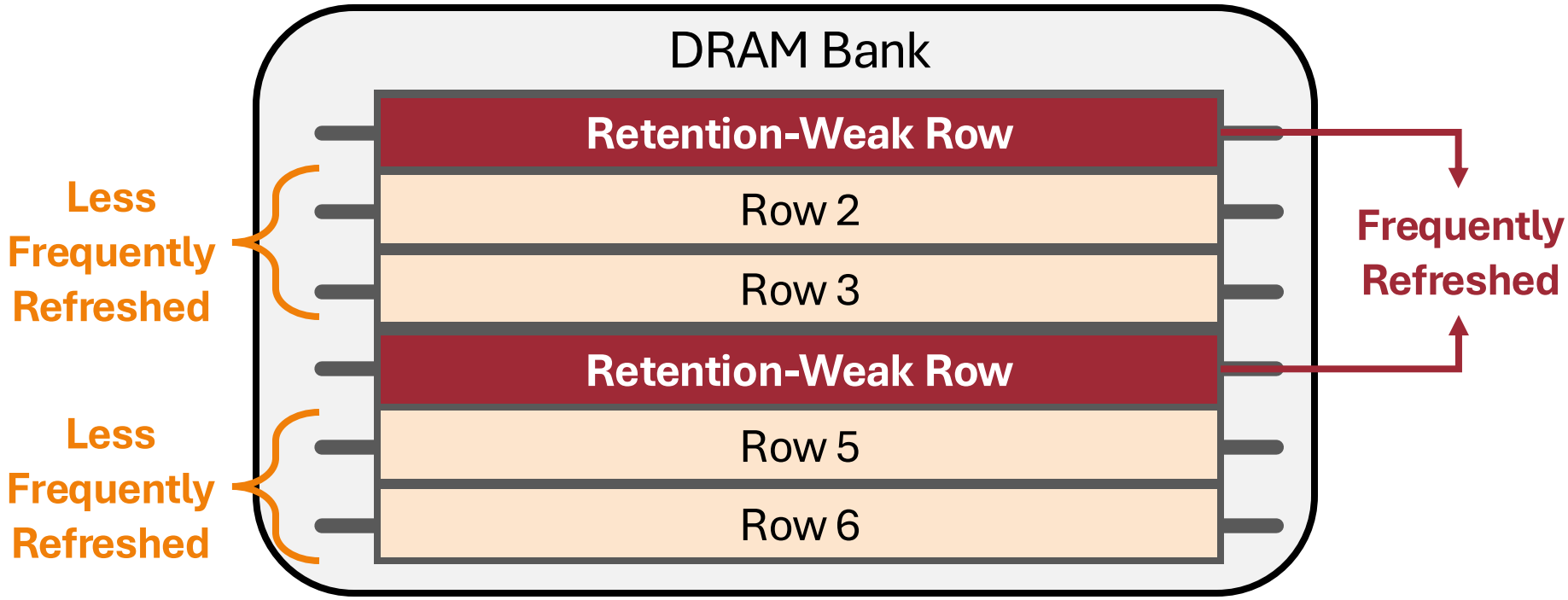
Dynamic random access memory (DRAM) [1] is the dominant main memory technology in nearly all modern computing systems due to its latency and cost characteristics. Continuing to increase DRAM capacity requires increasing the density of DRAM cells by reducing (i.e., scaling) the technology node size of DRAM, which in turn reduces DRAM cell size, cell-to-cell spacing, and cell-to-bitline spacing [2]. Unfortunately, as a result of this aggressive technology node scaling, DRAM suffers from worsening read disturbance problems [3–6].

Read disturbance in modern DRAM chips [4–9] is a widespread phenomenon and is reliably used for breaking memory isolation [5,7,9–70], a fundamental building block of robust (i.e., safe, secure, reliable, available) systems. *RowHammer* and *RowPress* are two prominent examples of DRAM read disturbance phenomena where a victim DRAM row experiences bitflips when a nearby aggressor DRAM row is 1) repeatedly activated (i.e., hammered) [5–7] or 2) kept open for a long period (i.e., pressed) [4, 71]. Many prior works [3–5, 7, 9, 41, 42]

https://arxiv.org/pdf/2510.14750

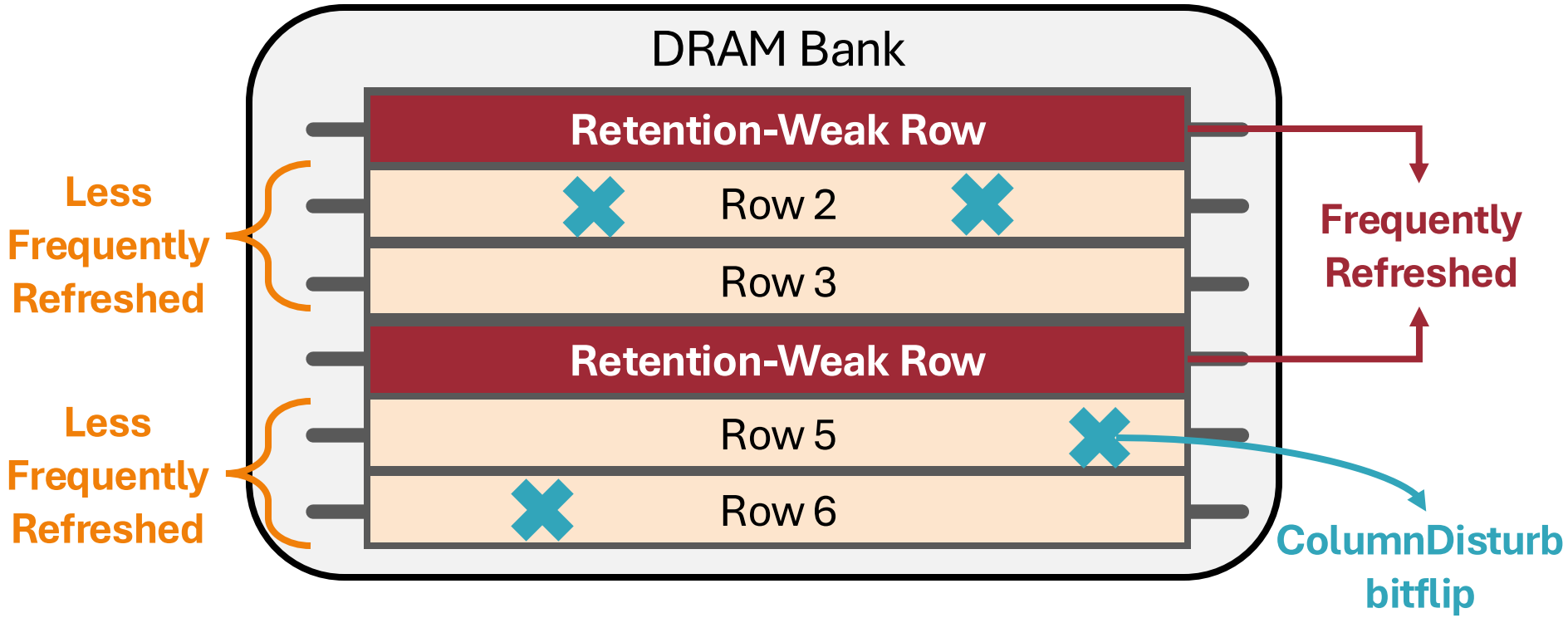
Retention-Aware Refresh Mechanisms

- Leverage the heterogeneity in DRAM cell data retention time
  - Most of the rows **do not need to be frequently refreshed**



Implications on Retention-Aware Refresh Mechanisms

- However, in the presence of **ColumnDisturb**, **less frequently refreshed rows become vulnerable** to ColumnDisturb



Benefits of retention-aware refresh mechanisms significantly reduces in the presence of ColumnDisturb

Implications on Retention-Aware Refresh Mechanisms

ColumnDisturb: Understanding Column-based Read Disturbance in Real DRAM Chips and Implications for Future Systems

İsmail Emir Yüksel[1] Ataberk Olgun[1] F. Nisa Bostancı[1]
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https://arxiv.org/pdf/2510.14750

More in the Paper

- **Many more characterization results**
  - Average voltage level on the perturbed columns
  - Temperature
  - Spatial variation
  - Access pattern
  - Data pattern
  - Aggressor row on time
- **Details and analyses on Implications**
  - Effectiveness of System-Level and on-Die ECC
  - More details on ColumnDisturb mitigation mechanisms
  - Implications on a retention-aware refresh mechanism: RAIDR
    - How ColumnDisturb affects energy-efficiency and the speedup of such mechanisms
  - ..

Available on arXiv

ColumnDisturb: Understanding Column-based Read Disturbance in Real DRAM Chips and Implications for Future Systems

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Outline

What is ColumnDisturb?

Testing Infrastructure

Real DRAM Chip Results

Implications for Future Systems & Mechanisms

Conclusion

Conclusion

We demonstrate and extensively study a new widespread read disturbance phenomenon in real DRAM chips: ColumnDisturb

We characterize 216 DDR4 and 4 HBM2 chips from three major manufacturers

- ColumnDisturb affects **all tested chips**
- ColumnDisturb is **getting worse** as DRAM technology node size **reduces**
- ColumnDisturb is **different** from **RowHammer, RowPress, and Retention** failures

We **describe and evaluate two major implications of ColumnDisturb**

- ColumnDisturb affects **robustness of future systems**
- ColumnDisturb makes **retention-aware refresh mechanisms much less effective**

We hope ColumnDisturb inspires system-level, architectural-level, and device-level solutions and understanding to enable ColumnDisturb-resilient future systems

ColumnDisturb

Understanding Column-based Read Disturbance in Real DRAM Chips and Implications for Future Systems

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A. Giray Yağlıkçı Onur Mutlu

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PuDHammer

Experimental Analysis of Read Disturbance Effects of Processing-using-DRAM in Real DRAM Chips

İsmail Emir Yüksel

Akash Sood Ataberk Olgun Oğuzhan Canpolat Haocong Luo

Nisa Bostancı Mohammad Sadrosadati Giray Yağlıkçı Onur Mutlu

SAFARI

ETH zürich

Executive Summary

Motivation: Processing-using-DRAM (PuD) alleviates data movement bottlenecks

- DRAM can perform many PuD operations by activating multiple DRAM rows in quick succession or simultaneously (i.e., multiple-row activation)
- Modern DRAM is subject to read disturbance (e.g., RowHammer)
  - Repeatedly activating even a single row induces bitflips in unaccessed rows

Problem: No prior work studies the read disturbance effects of multiple-row activation

Goal: Understand and analyze read disturbance effects of multiple-row activation

Experimental Study: 316 COTS DDR4 chips from four major manufacturers to study read disturbance effects of multiple-row activation, which we call PuDHammer

- PuDHammer significantly exacerbates DRAM read disturbance, causing up to 158.58x reduction in min. hammer count to induce first bitflip
- PuDHammer bypasses in-DRAM RowHammer mitigation in DDR4 and induces 11340x more bitflips than RowHammer

Mitigation: We adapt Per Row Activation Counting (PRAC) for PuDHammer

- Adapted PRAC solution incurs an average system performance overhead of 48.26%

Outline

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Problem & Goal

Testing Infrastructure

Read Disturbance Effect of CoMRA

Read Disturbance Effect of SiMRA

Mitigating PuDHammer

Conclusion

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Read Disturbance Effect of CoMRA

Read Disturbance Effect of SiMRA

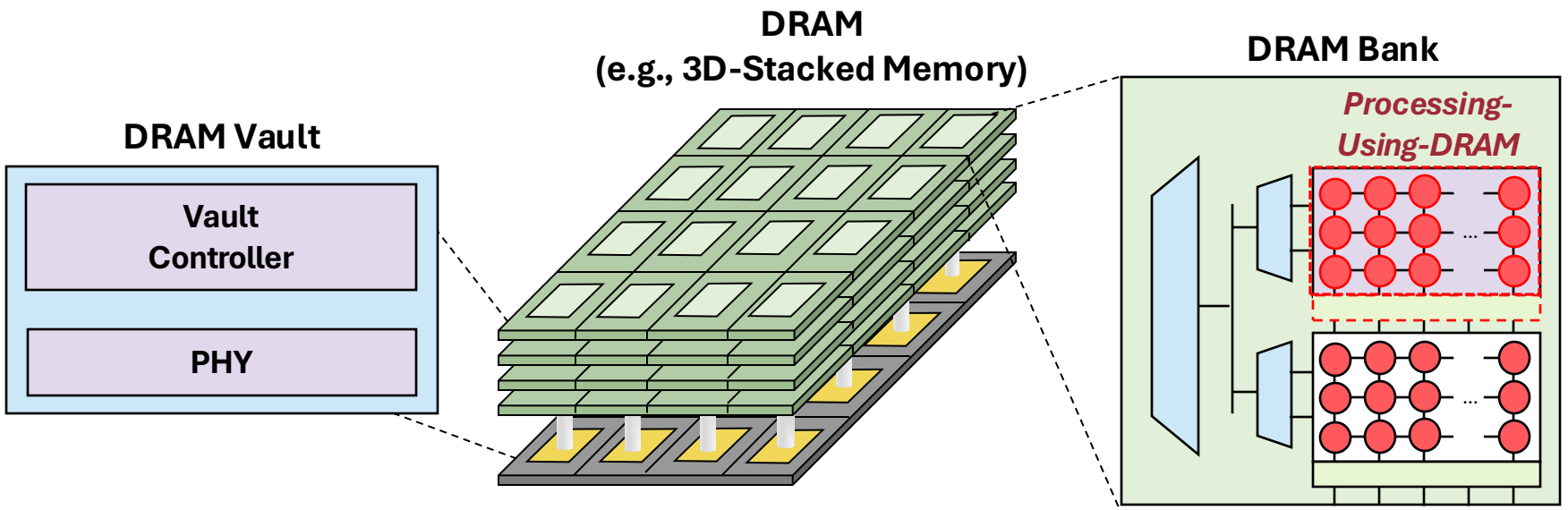
Mitigating PuDHammer

Conclusion

Processing-using-DRAM (PuD)

Processing-using-DRAM uses the analog operational principles of DRAM cells to perform computation

Multiple-row activation is a key technique to realize many PuD operations (e.g., AND and OR)

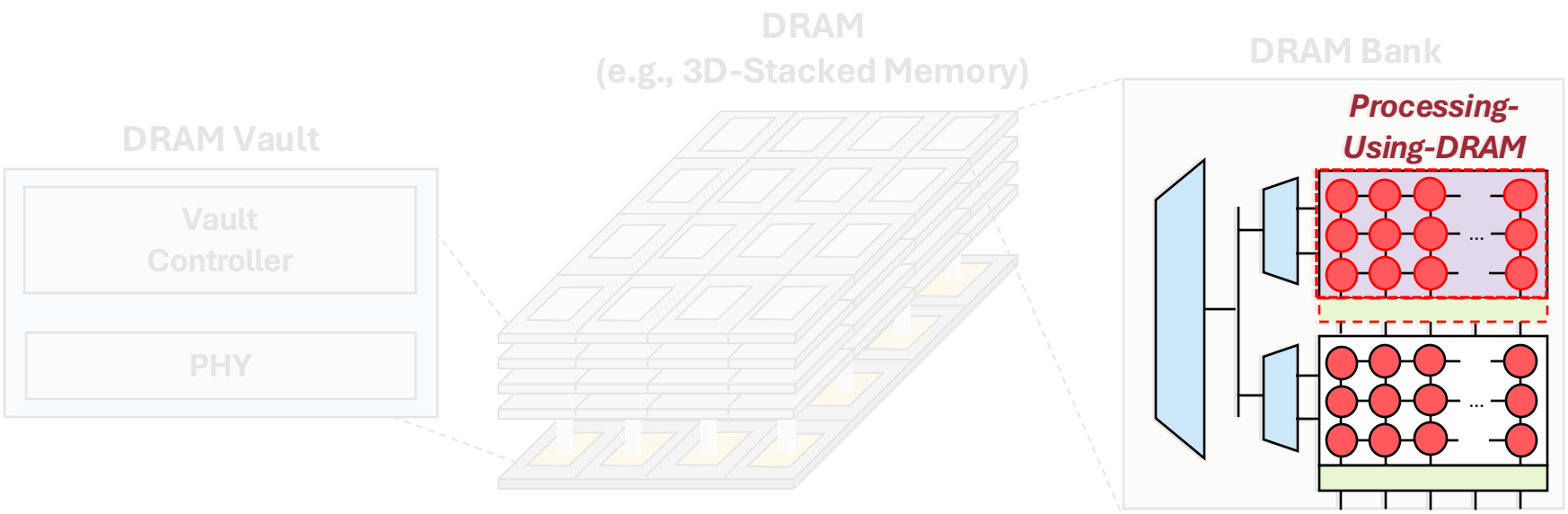


Processing-using-DRAM (PuD)

Multiple-row activation is a key technique to realize many PuD operations

1. **Consecutive Multiple-Row Activation (CoMRA)**

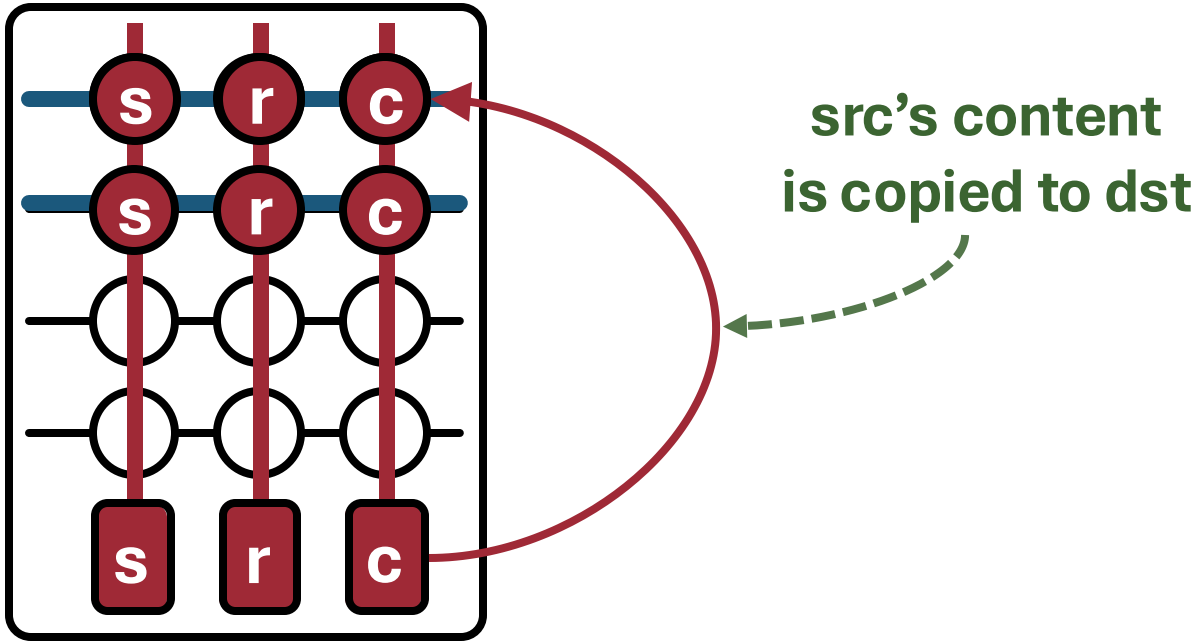
2. **Simultaneous Multiple-Row Activation (SiMRA)**



Consecutive Multiple-Row Activation

Violating timing of PRE command **greatly**
in **COTS** DRAM chips
activates two rows **consecutively**

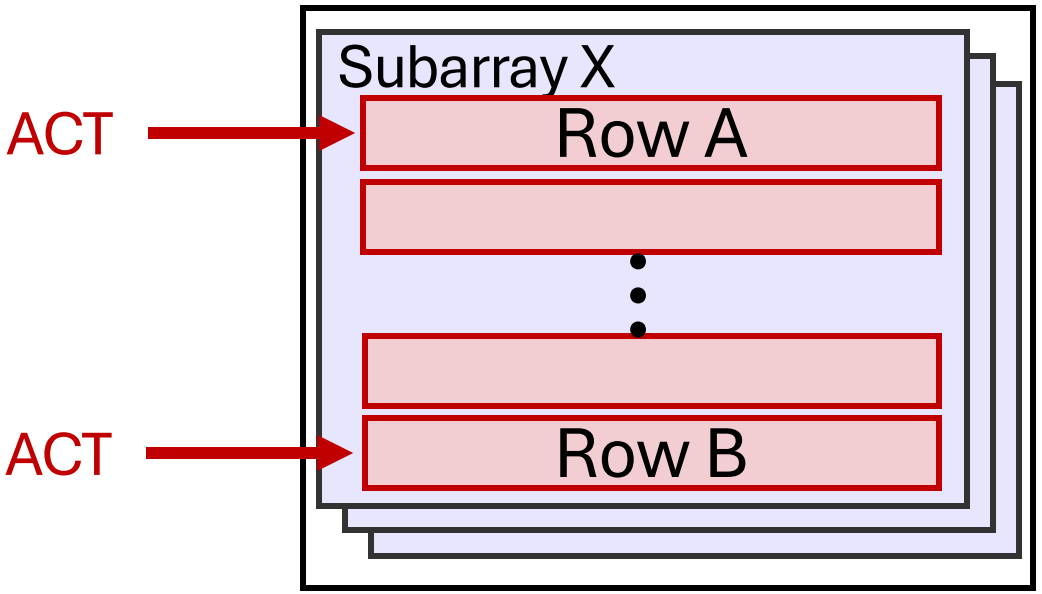




Simultaneous Multiple-Row Activation

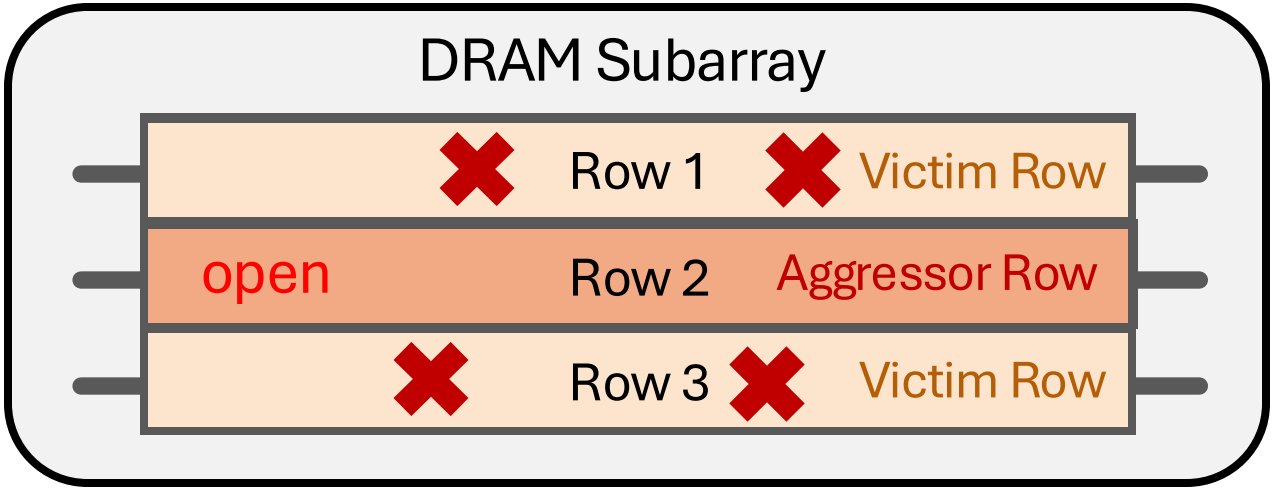
Activating two rows in **quick succession** in **COTS** DRAM chips **simultaneously** activates **multiple rows** in a subarray





Read Disturbance in DRAM

- Unfortunately, even activating **a single DRAM row** **disturbs** the data integrity of other **unaccessed DRAM rows**
- Prominent example: **RowHammer**



Repeatedly **opening (activating)** and **closing** a DRAM row **many times** causes **RowHammer bitflips** in adjacent rows

[Kim+, ISCA'14]

Outline

Background

Problem & Goal

Testing Infrastructure

Read Disturbance Effect of CoMRA

Read Disturbance Effect of SiMRA

Combined Effect of CoMRA, SiMRA, and RowHammer

Conclusion

Problem & Goal

Problem

*No prior work **investigates** the **read disturbance effects of multiple-row activation***

Goal

*Understand how **multiple-row activation affects read disturbance vulnerability** in DRAM*

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Background

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Real DRAM Chip Testing Infrastructure

Read Disturbance Effect of CoMRA

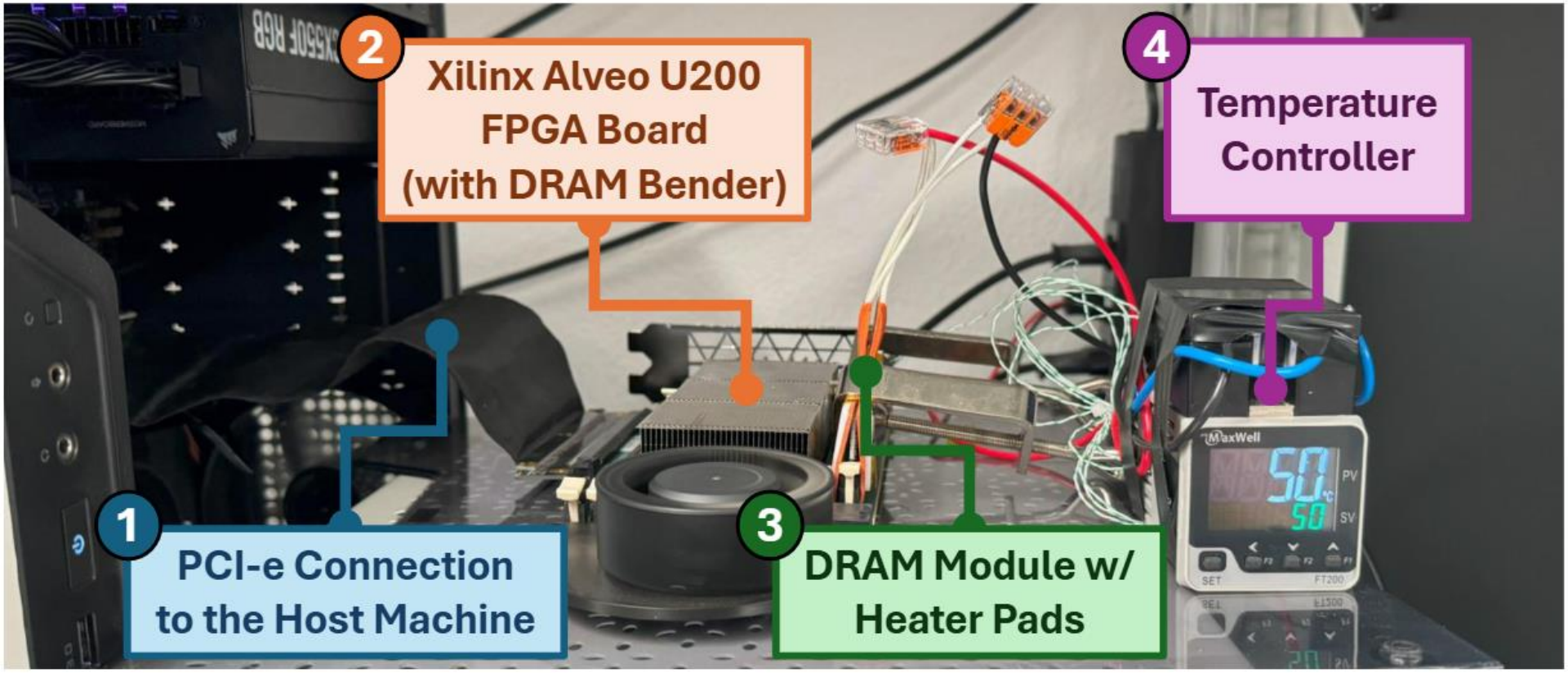
Read Disturbance Effect of SiMRA

Mitigating PuDHammer

Conclusion

DRAM Bender: DRAM Testing Infrastructure

Fine-grained control over DRAM commands, timings, temperature, and voltage



Olgun et al., "DRAM Bender: An Extensible and Versatile FPGA-based Infrastructure to Easily Test State-of-the-art DRAM Chips," TCAD, 2023.

DRAM Chips Tested

- 316 DDR4 chips from four major DRAM manufacturers
- Covers different die revisions and chip densities

Chip Mfr.	#Modules	#Chips	Die Rev.	Density	Org.
SK Hynix	1	8	A	4Gb	x8
	8	64	A	8Gb	x8
	2	16	C	16Gb	x8
	6	48	D	8Gb	x8
Micron	1	8	B	4Gb	x8
	4	32	E	16Gb	x16
	4	32	F	16Gb	x8
	2	16	R	8Gb	x8
Samsung	1	8	A	16Gb	x8
	5	40	B	16Gb	x8
	1	4	C	4Gb	x16
	1	8	C	16Gb	x8
	1	8	E	4Gb	x8
Nanya	3	24	C	8Gb	x8

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Read Disturbance Effect of CoMRA

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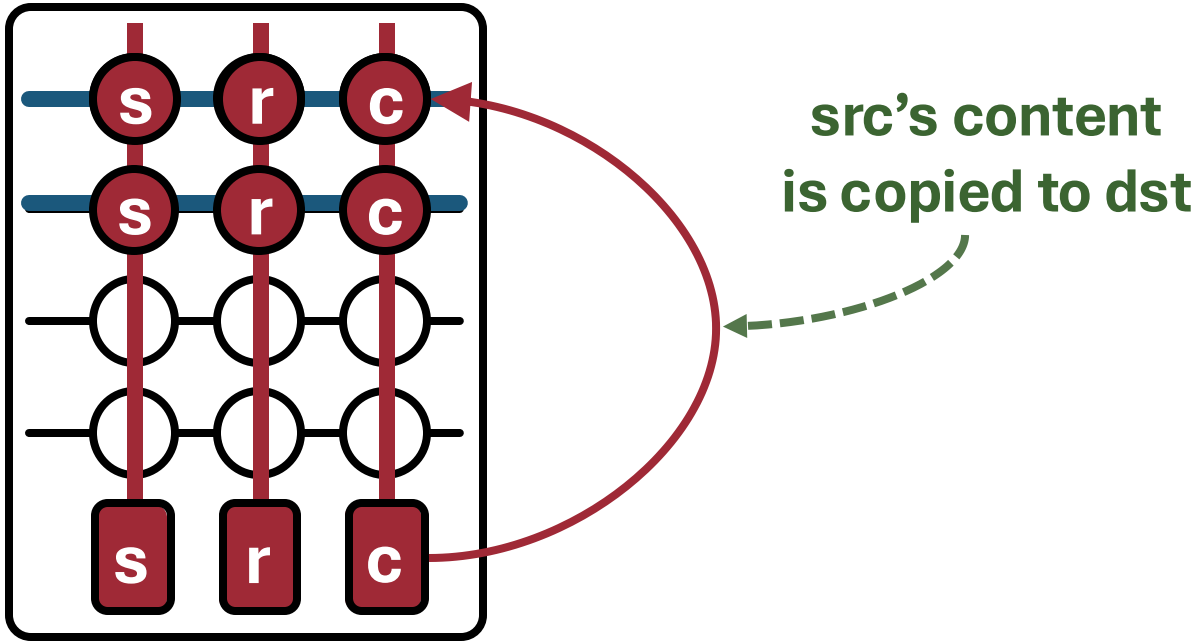
Mitigating PuDHammer

Conclusion

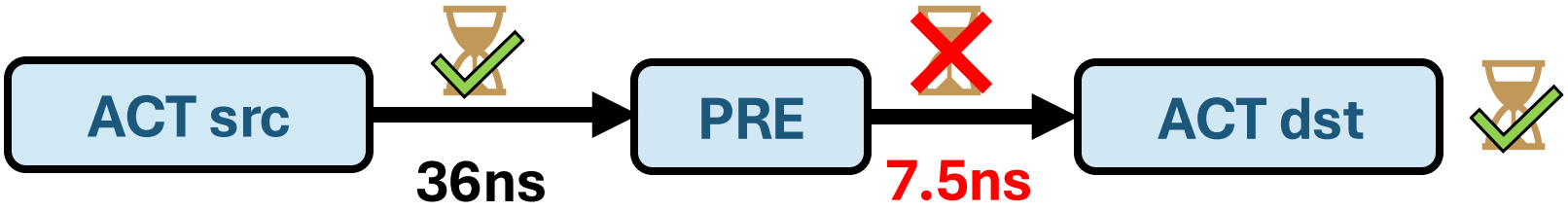
Consecutive Multiple-Row Activation

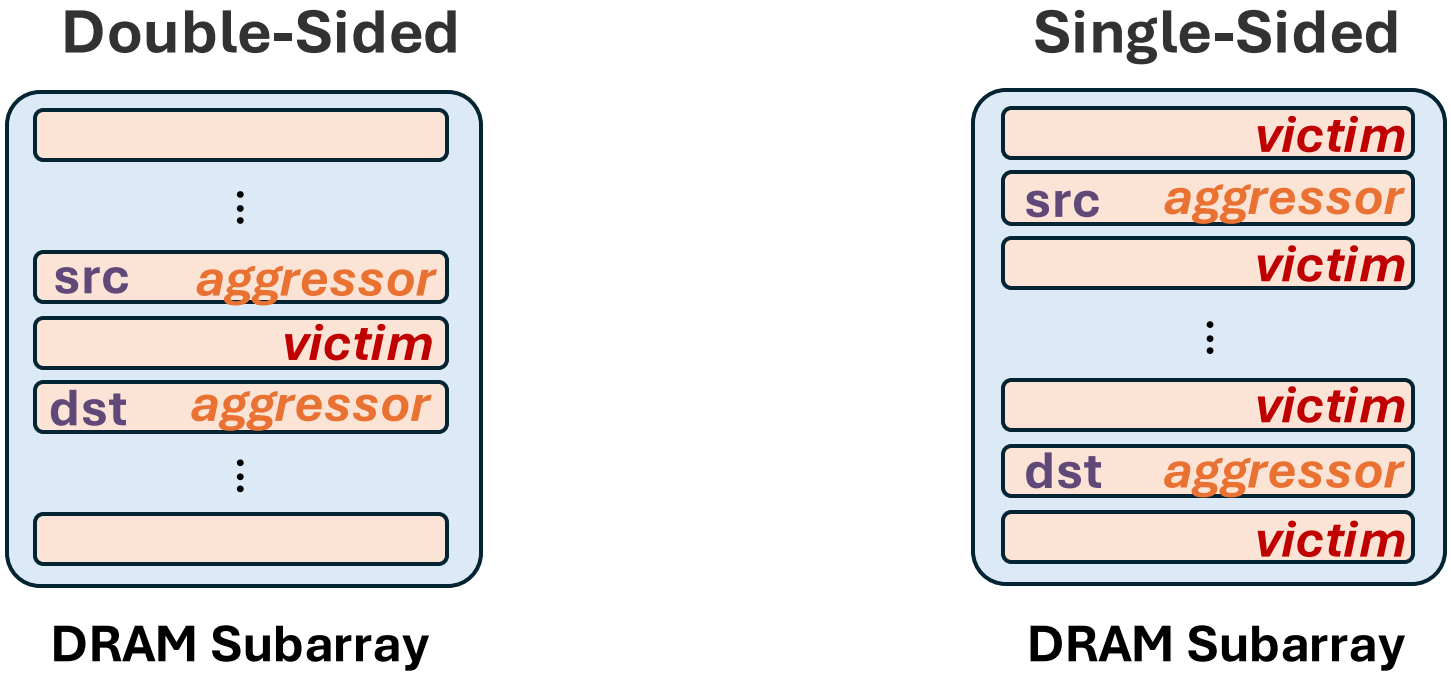
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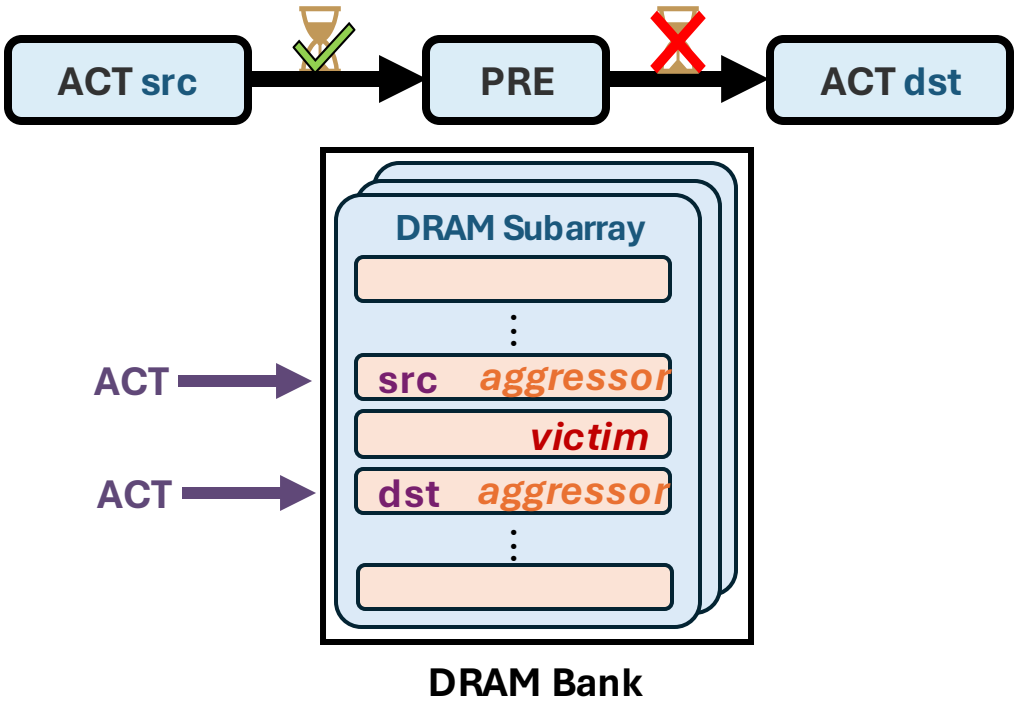
Hammering with CoMRA





Characterization Methodology

- Carefully sweep
  - Row addresses: src and dst
  - Timing parameters: Between PRE → ACT
  - Data Pattern: 0x00, 0xFF, 0xAA, and 0x55
  - Temperature (°C): 50, 60, 70, and 80

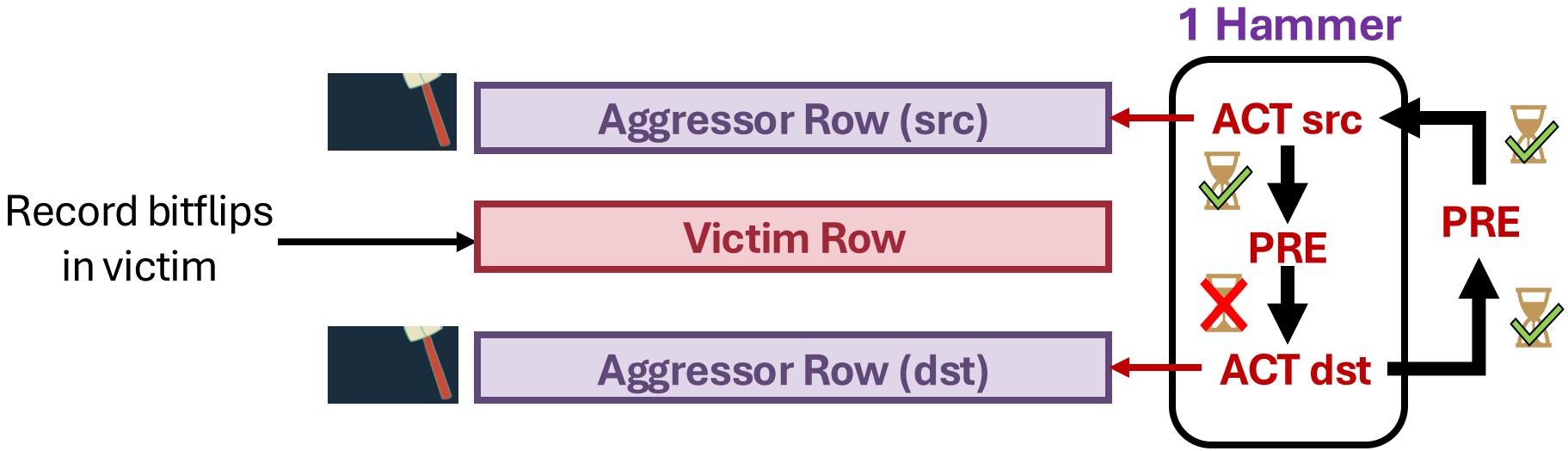




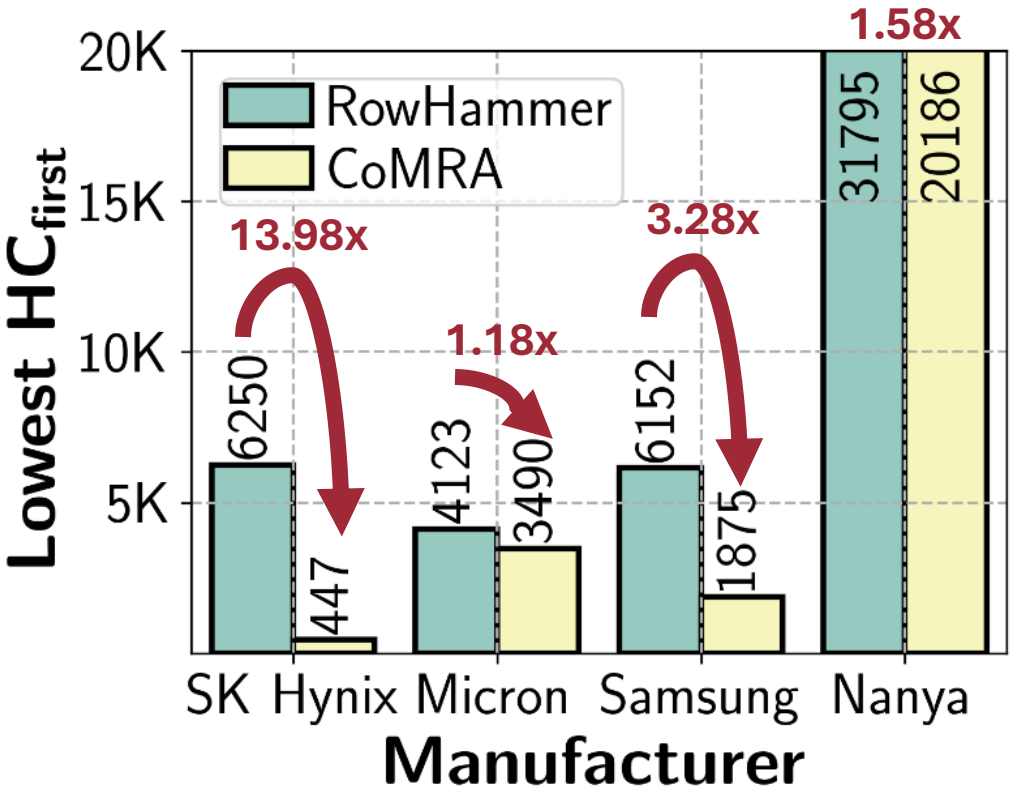
Read Disturbance Vulnerability Metric: HC_{first}

The minimum hammer count
required to induce the **first bitflip**

Lower is worse



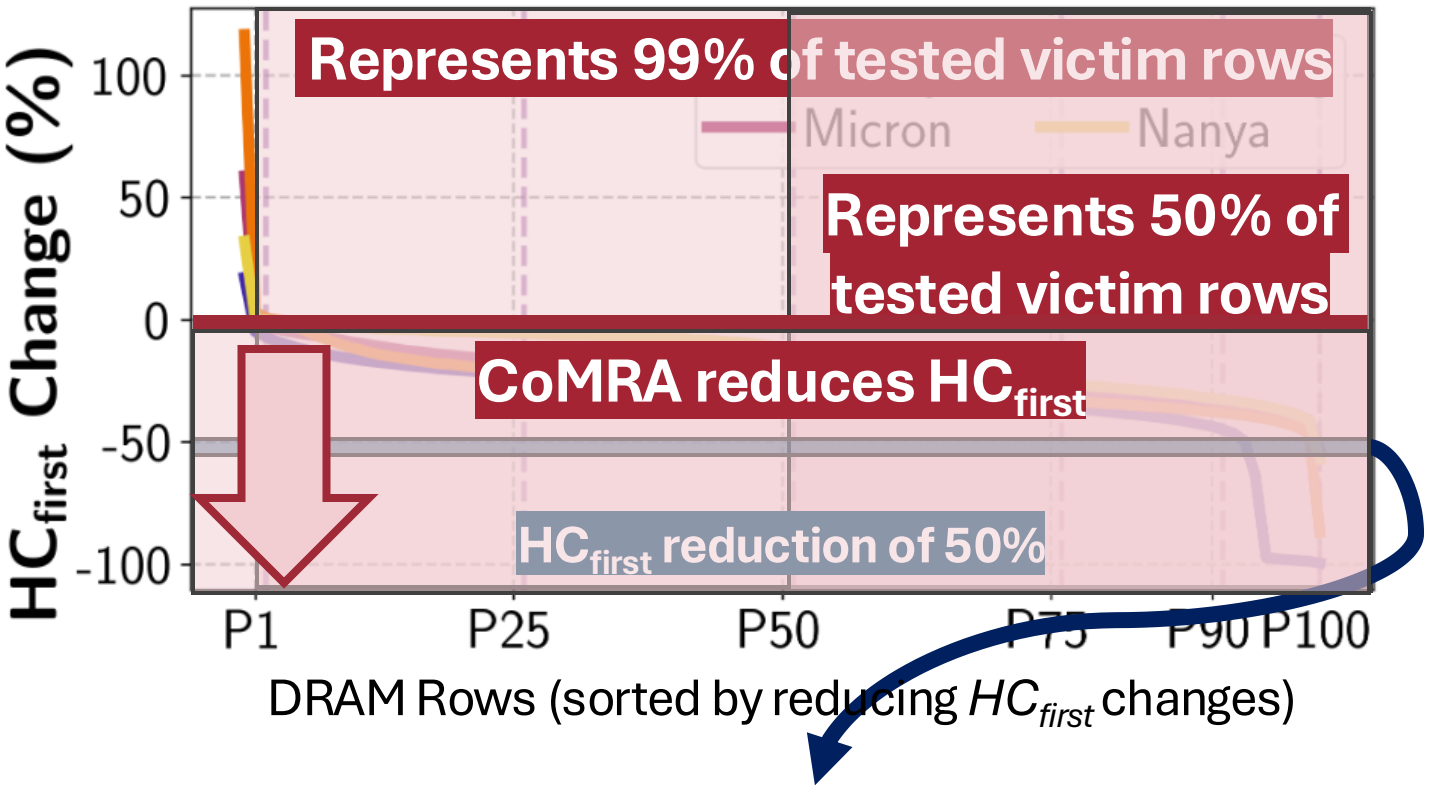
Double-Sided CoMRA vs. RowHammer (I)



Double-sided CoMRA decreases lowest HC$_{first}$ by 13.98x compared to double-sided RowHammer

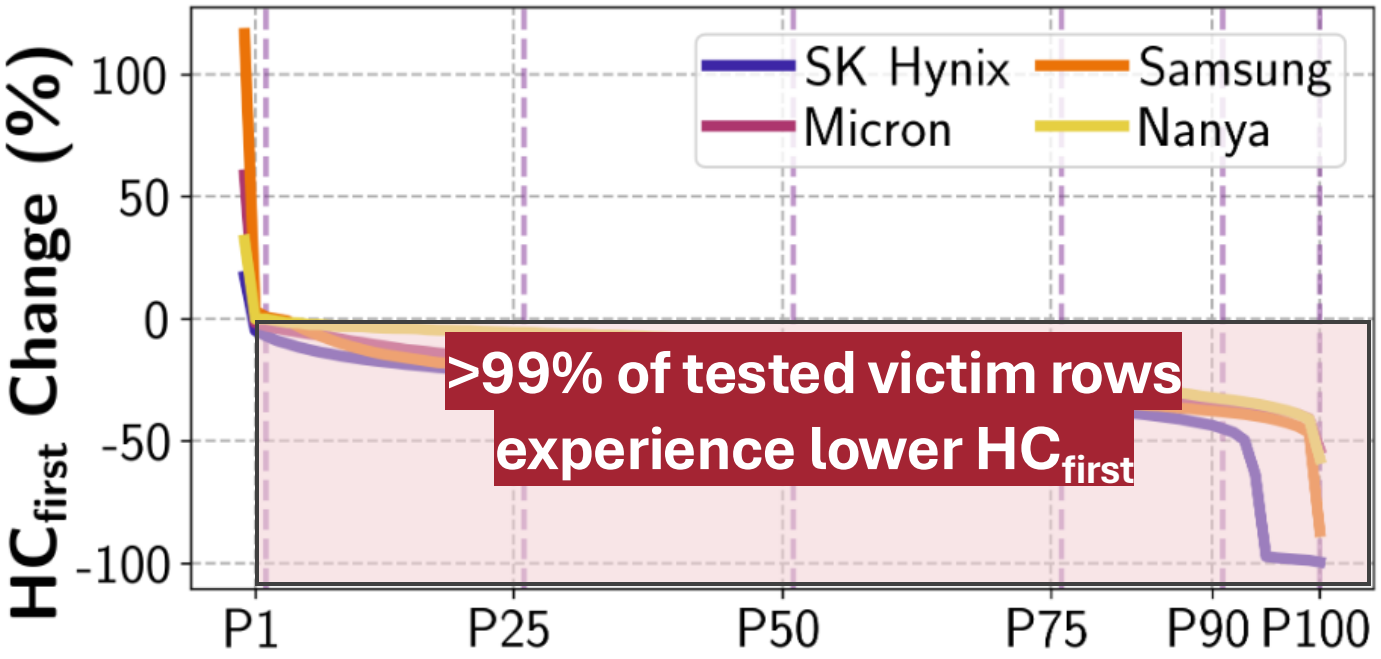
Lowest HC$_{first}$ reduction trend consistent across all four manufacturer

Double-Sided CoMRA vs. RowHammer (II)



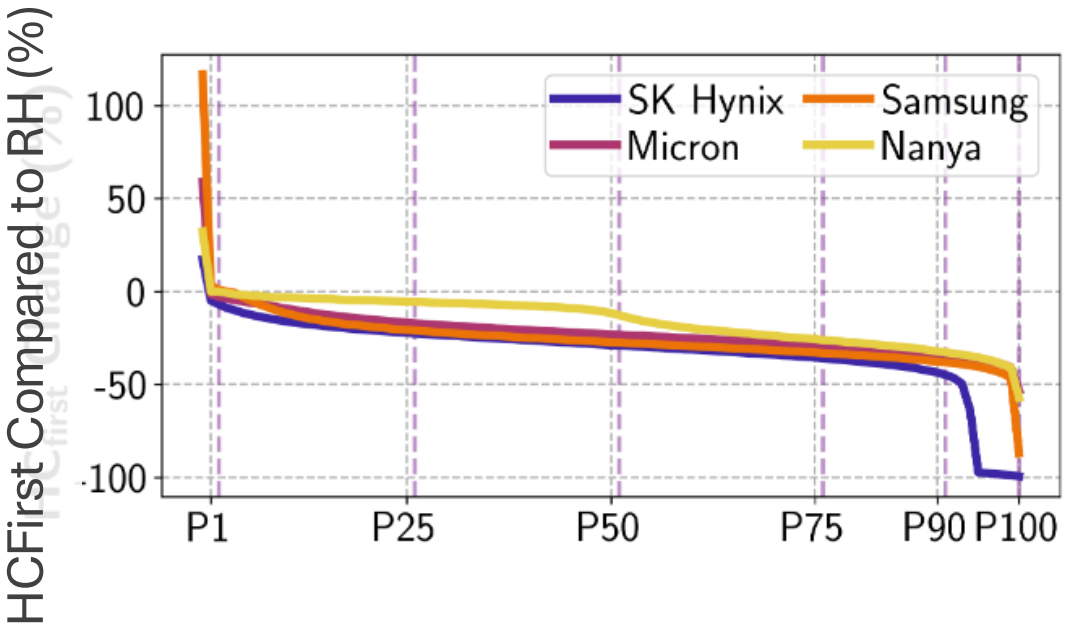
RowHammer HC_{first} = 100
CoMRA HC_{first} = 50

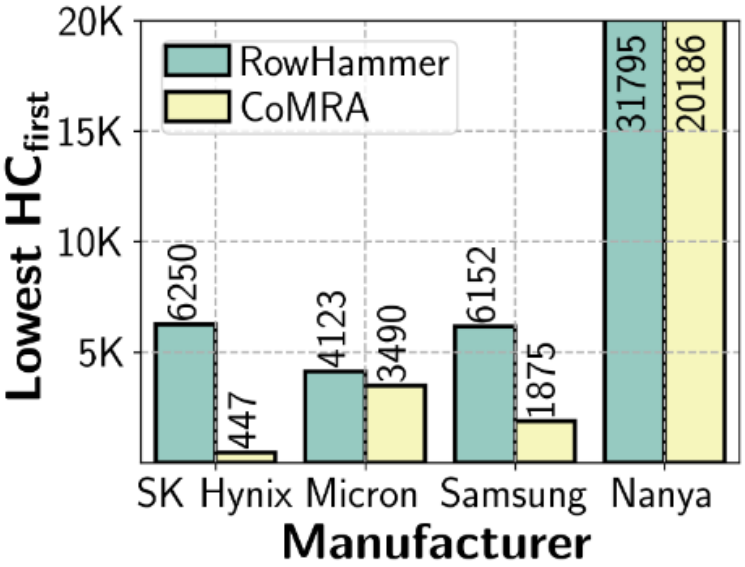
Double-Sided CoMRA vs. RowHammer (II)



Double-sided CoMRA decreases HC_{first} for a large fraction of rows compared to double-sided RowHammer

Double-Sided CoMRA vs. RowHammer





Key Takeaway

CoMRA exacerbates DRAM's vulnerability to read disturbance in all four major manufacturers

Outline

Background

Problem & Goal

Testing Infrastructure

Read Disturbance Effect of CoMRA

Read Disturbance Effect of SiMRA

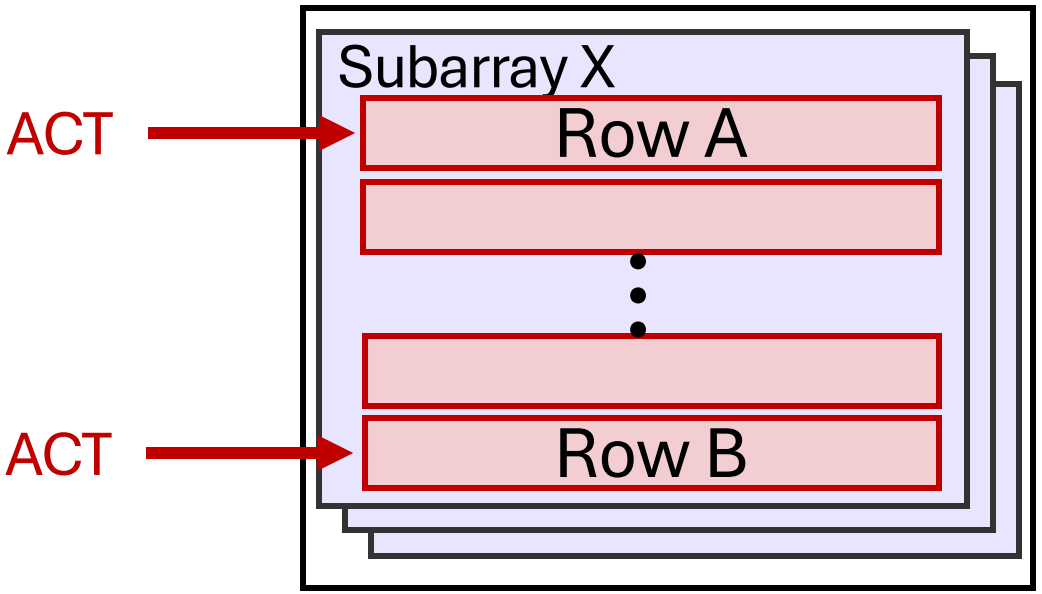
Mitigating PuDHammer

Conclusion

Recall: SiMRA

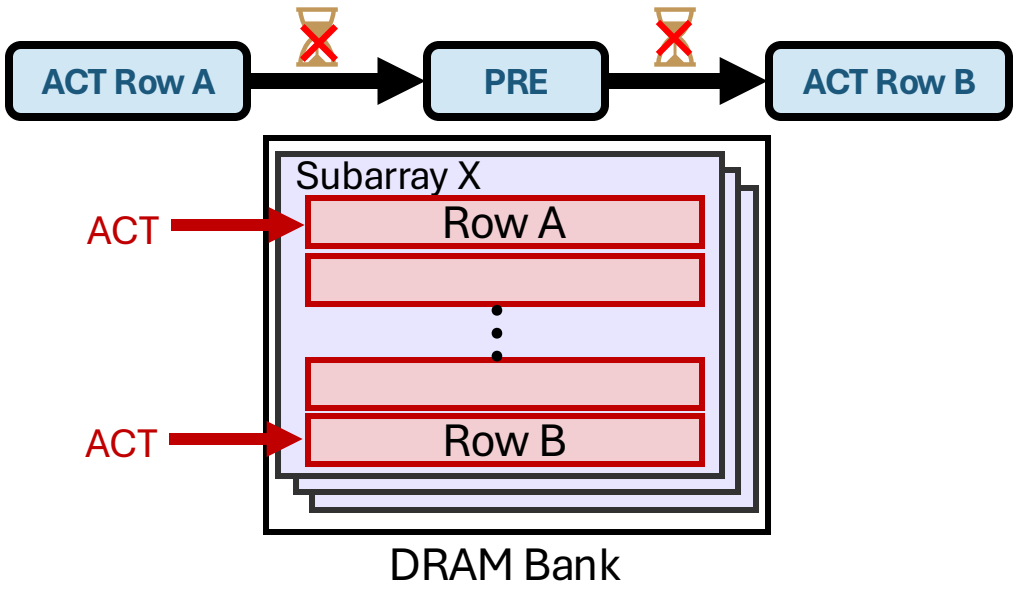
Activating two rows in **quick succession** in **COTS** DRAM chips **simultaneously** activates **multiple rows** in a subarray





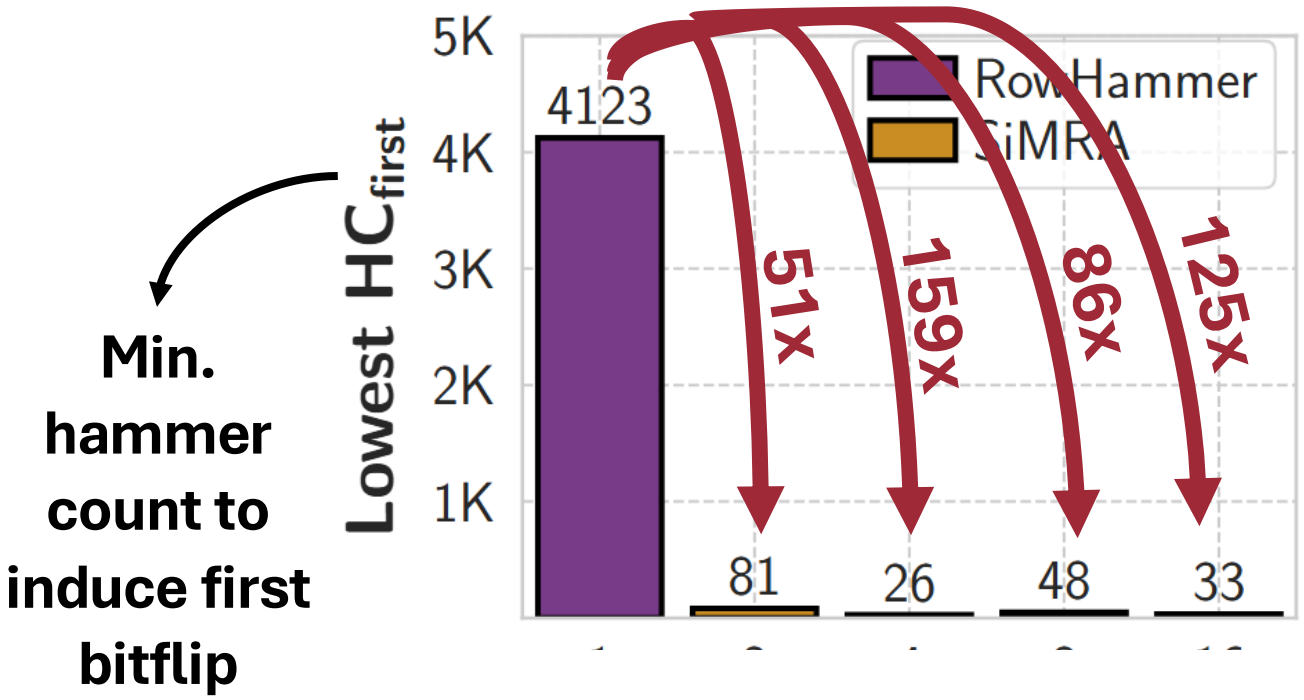
Characterization Methodology

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  - Temperature (°C): 50, 60, 70, and 80





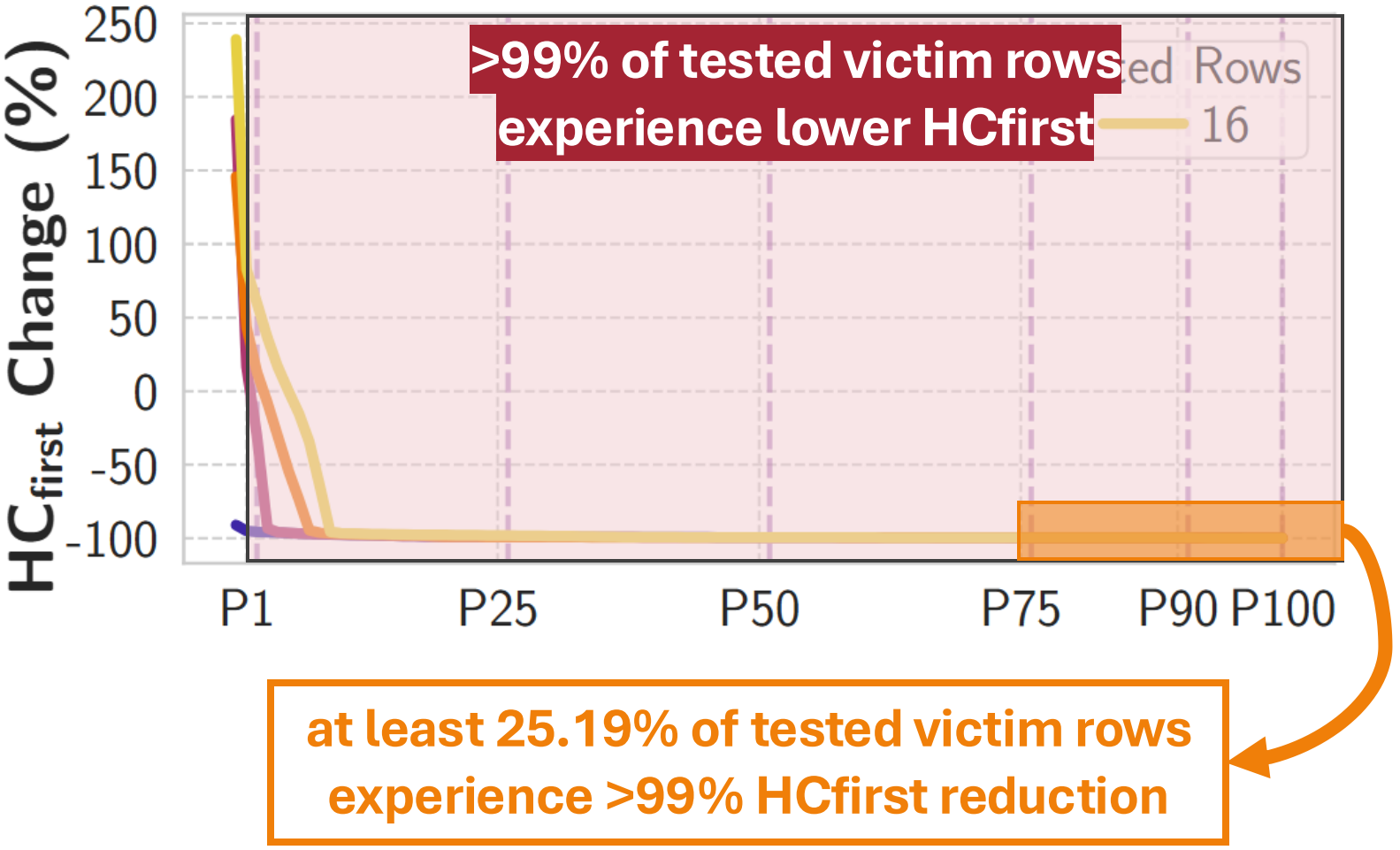
Double-Sided SiMRA vs. RowHammer (I)



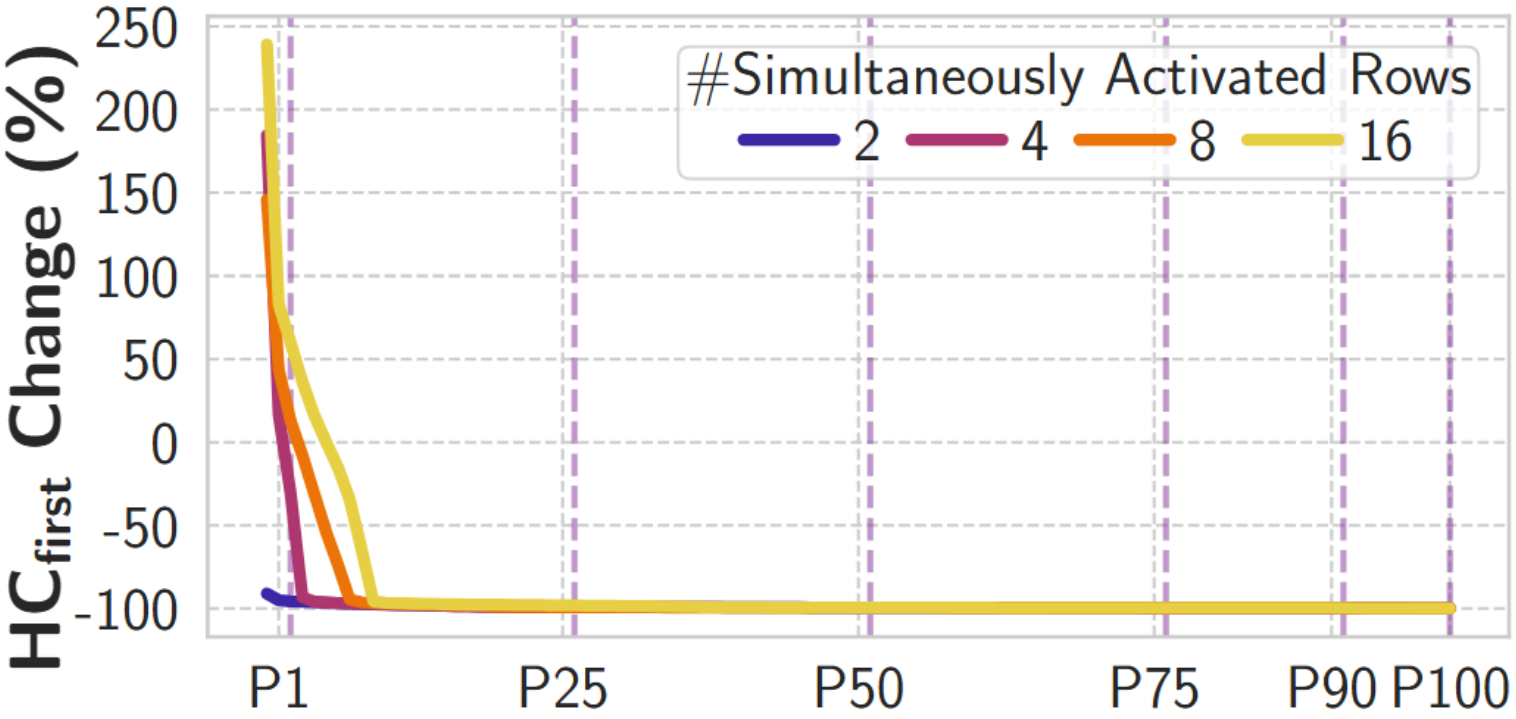
Double-sided SiMRA decreases lowest HC_{first} by 159x compared to double-sided RowHammer

Lowest HC_{first} reduction trend consistent across all tested number of simultaneously activated rows

Double-Sided SiMRA vs. RowHammer (II)

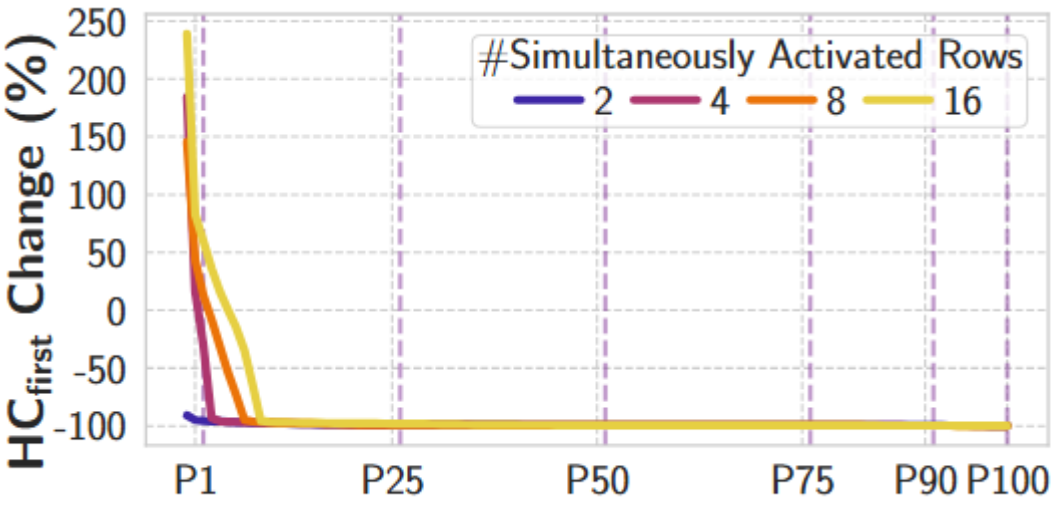


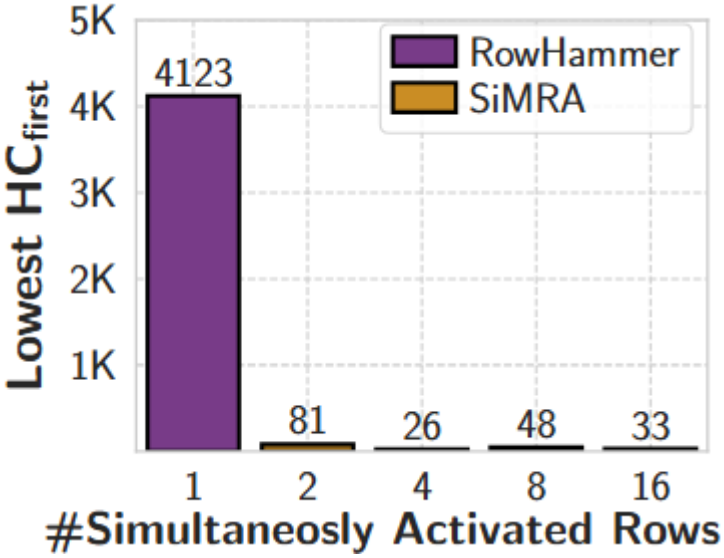
Double-Sided SiMRA vs. RowHammer (II)



Double-sided SiMRA significantly decreases HC_{first} for a majority of rows compared to double-sided RowHammer

Double-sided SiMRA vs. RowHammer





Key Takeaway

SiMRA drastically exacerbates DRAM's vulnerability to read disturbance

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Mitigating PuDHammer

1 Analyze the effectiveness of existing in-DRAM RowHammer mitigation against PuDHammer

2 Adapt RowHammer mitigations to account for PuDHammer

Mitigating PuDHammer

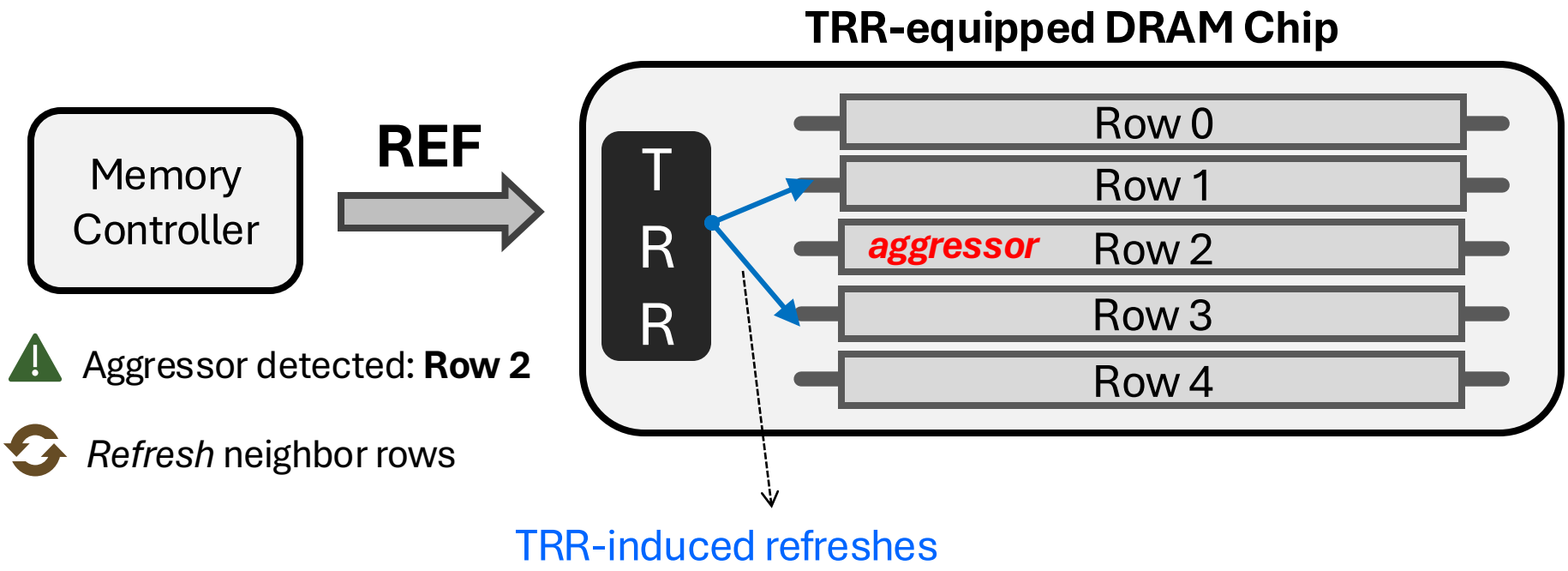
1 **Analyze the effectiveness of existing in-DRAM RowHammer mitigation against PuDHammer**

2 **Adapt RowHammer mitigations to account for PuDHammer**

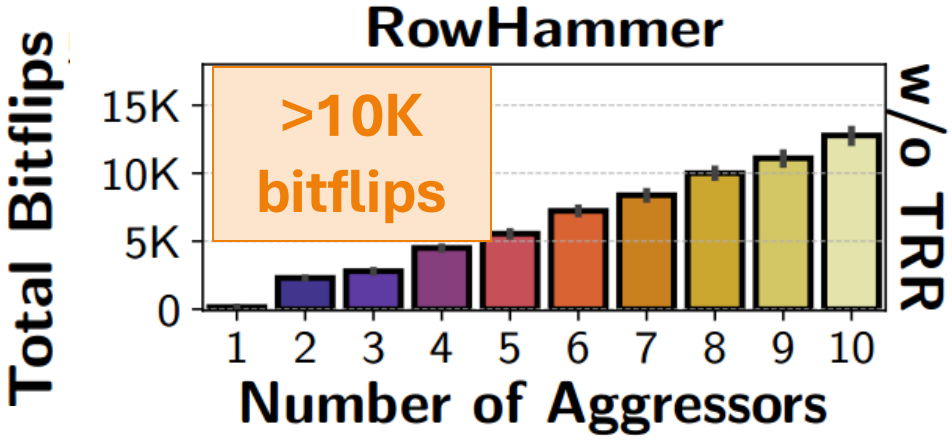
Target Row Refresh (TRR)

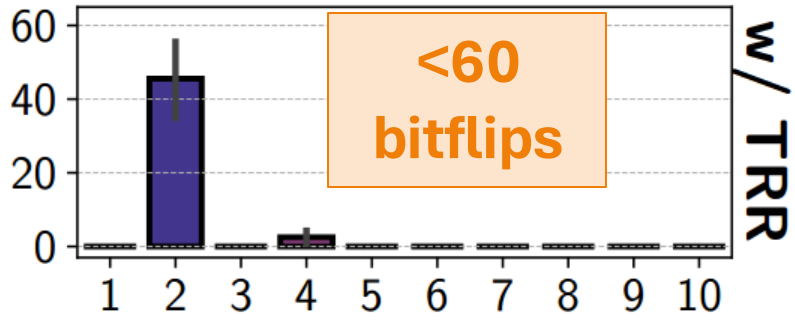
DRAM vendors equip their DRAM chips with a *proprietary* mitigation mechanisms known as **Target Row Refresh (TRR)**

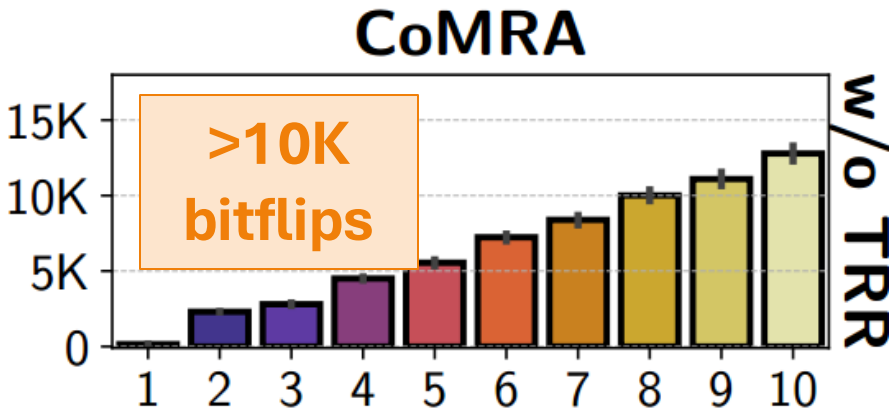
Key Idea: TRR refreshes nearby rows upon detecting an aggressor row

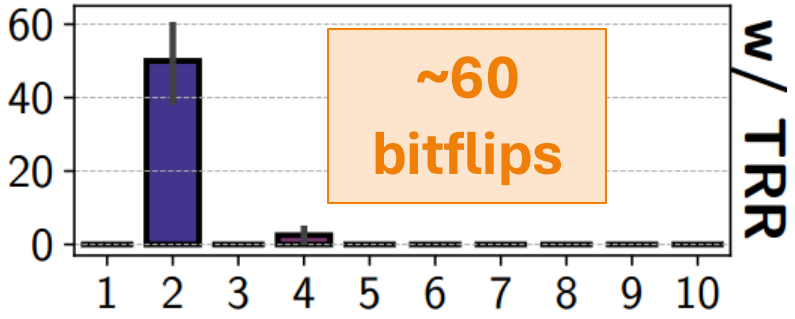


Real DRAM Chip Results



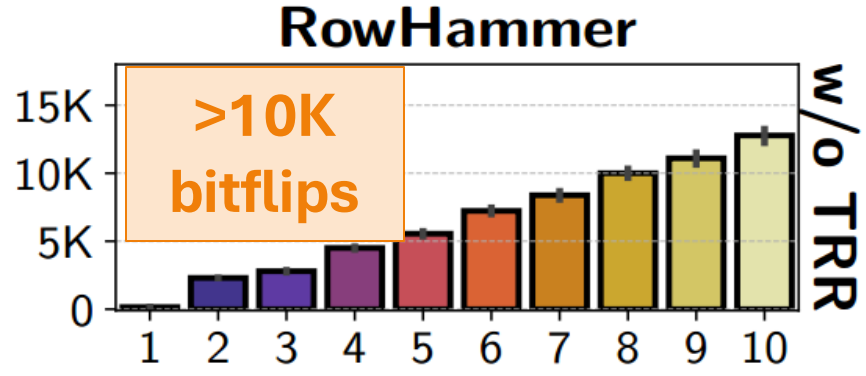


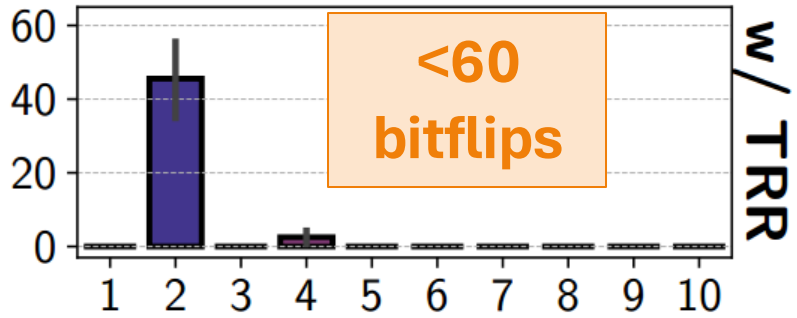


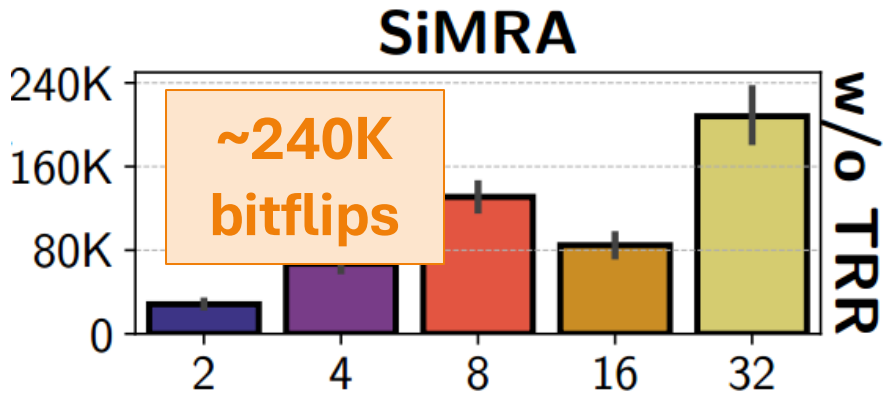


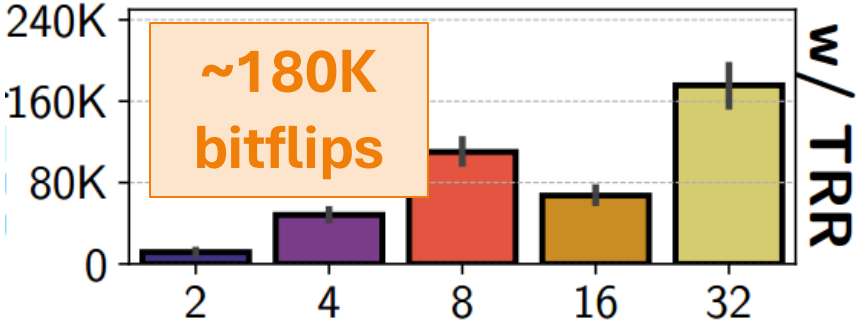
CoMRA induces 1.10x more bitflips than RowHammer on average in the presence of TRR

Real DRAM Chip Results









SiMRA induces 11340x more bitflips than RowHammer on average in the presence of TRR

Mitigating PuDHammer

1 Analyze the effectiveness of existing in-DRAM RowHammer mitigation against PuDHammer

2 **Adapt RowHammer mitigations to account for PuDHammer**

Adapting PRAC for PuDHammer

A Naïve Approach (PRAC-Naïve)

Reducing RowHammer threshold to SiMRA's lowest observed HCfirst

Weighted Counting (PRAC-WC)

Increase counters based on the vulnerability caused by the operation (RH, CoMRA, and SiMRA)

Methodology

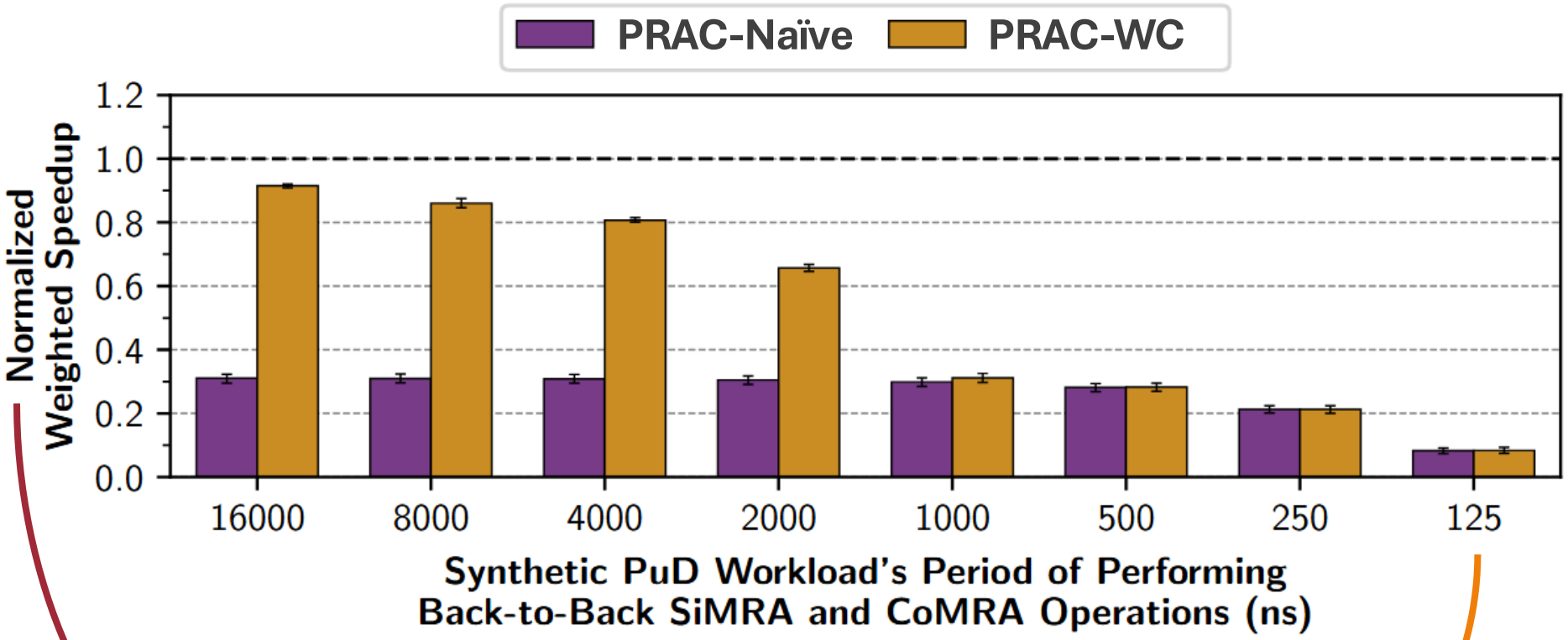
- **Environment:** Cycle-level DRAM simulator Ramulator2 [Kim+ CAL'15, Luo+ CAL'23]

- **System Configuration:**

Processor	4-core, out-of-order, 4.2GHz clock frequency
DRAM	DDR5, 1 channel, 2 rank/channel, 8 bank groups, 4 banks/bank group, 128K rows/bank
Memory Ctrl.	64-entry read and write requests queues, Scheduling policy: FR-FCFS with a column cap of 4
PRAC	4 RFMs per Back-off
PIM Unit	1-core, issues CoMRA & SiMRA operations with a fixed invertal

- **Workloads:**
  - 60 five-core multiprogrammed workload mixes.
  - Each mix is composed of
    - four workloads from five major benchmark suites and
    - one synthetic workload that periodically performs back-to-back one SiMRA with 32-row activation and one CoMRA operation

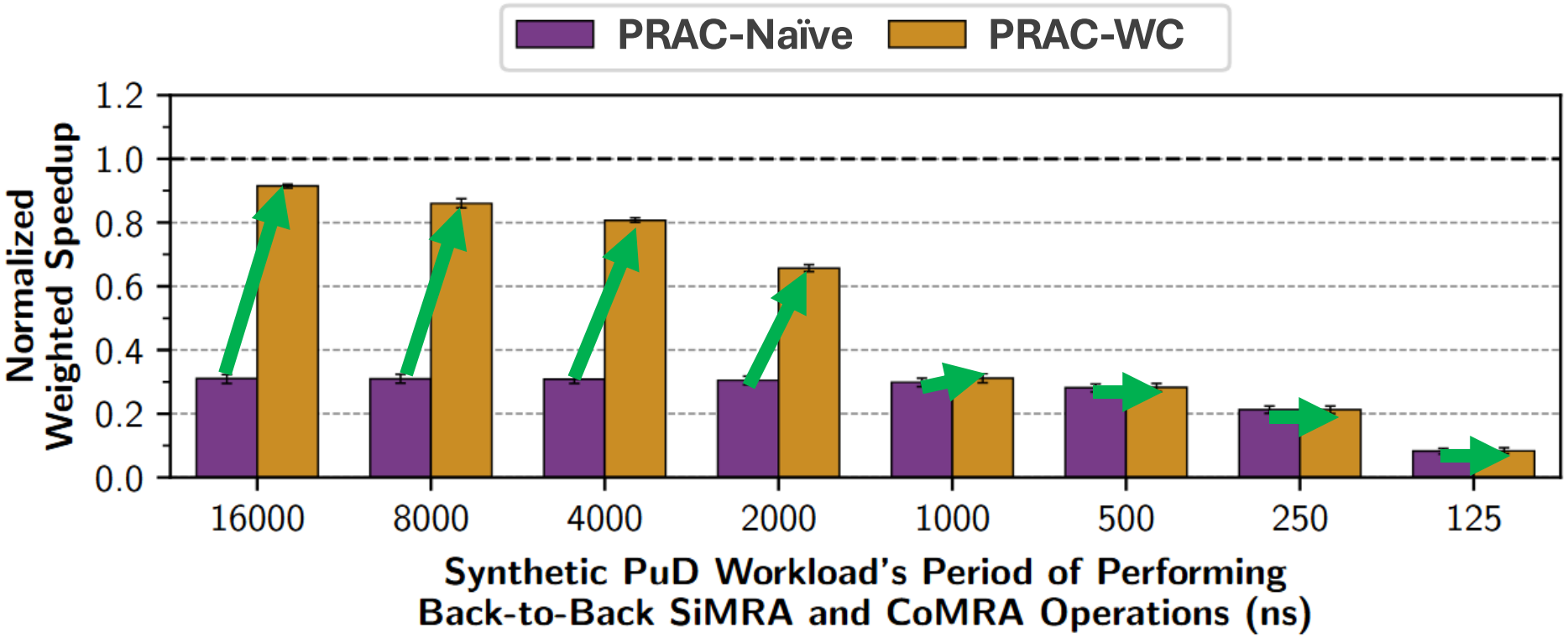
Weighted Speedup Results



Normalized to no RH mitigation

DRAM performs PuD operations in every 125ns

Weighted Speedup Results



PRAC-WC greatly outperforms PRAC-Naïve until 2us

PRAC-WC incurs an average (maximum) performance overhead of 48.26% (98.83%)

More in the Paper

- **More characterization results**
  - Temperature, spatial variation, timing parameters, access pattern, data pattern, aggressor row on time analyses for CoMRA and SiMRA
  - Combined Read Disturbance Effect of RowHammer with PuD
    - Combined pattern is more effective than RowHammer alone (e.g., 1.66x reduction in HCfirst on average)

- **Details and analyses on mitigating PuDHammer**
  - Discussion on how modifying DRAM chips could help mitigating PuDHammer
  - More details on PRAC adaptation
    - Challenges on adapting for PuDHammer
    - An area-efficient PRAC solution
    - ..

Available on arXiv

PuDHammer: Experimental Analysis of Read Disturbance Effects of Processing-using-DRAM in Real DRAM Chips

İsmail Emir Yüksel Akash Sood Ataberk Olgun Oğuzhan Canpolat Haocong Luo
F. Nisa Bostancı Mohammad Sadrosadati A. Giray Yağlıkçı Onur Mutlu

ETH Zürich

Processing-using-DRAM (PuD) is a promising paradigm for alleviating the data movement bottleneck using a DRAM array's massive internal parallelism and bandwidth to execute very wide data-parallel operations. Performing a PuD operation involves activating multiple DRAM rows in quick succession or simultaneously, i.e., multiple-row activation. Multiple-row activation is fundamentally different from conventional memory access patterns that activate one DRAM row at a time. However, repeatedly activating even one *DRAM row (e.g., RowHammer) can induce bitflips in unaccessed DRAM rows because modern DRAM is subject to read disturbance, a worsening safety, security, and reliability issue. Unfortunately, no prior work investigates the effects of multiple-row activation, as commonly used by PuD operations, on DRAM read disturbance.*

In this paper, we present the first characterization study of read disturbance effects of multiple-row activation-based PuD (which we call PuDHammer) using 316 real DDR4 DRAM chips from four major DRAM manufacturers. Our detailed characterization results covering various operational conditions and parameters (i.e., temperature, data patterns, access patterns, timing parameters,

CPU and GPU) [1, 2]. Unfortunately, this data movement is a major bottleneck that consumes a large fraction of execution time and energy in many modern applications [1–28]. Processing-using-DRAM (PuD) [29–34] is a promising paradigm that can alleviate the data movement bottleneck. PuD uses the analog operational properties of the DRAM array circuitry to enable massively parallel in-DRAM computation (i.e., PuD operations), which can be used to accelerate important applications including databases and web search [29, 30, 32, 35–43], data analytics [29, 44–48], graph processing [32, 48–51], genome analysis [52–57], cryptography [58, 59], hyper-dimensional computing [60–62], and generative AI [63–72].

A wide variety of PuD operations (e.g., in-DRAM data copy and bulk bitwise operations) rely on a key PuD technique called *multiple-row activation*, which accesses (activates) multiple DRAM rows in quick succession or simultaneously [29–32, 40, 73–84]. Multiple-row activation is fundamentally different from conventional DRAM operations that access only a *single* DRAM row at a time.

https://arxiv.org/pdf/2506.12947

Outline

Background

Problem & Goal

Testing Infrastructure

Read Disturbance Effect of CoMRA

Read Disturbance Effect of SiMRA

Mitigating PuDHammer

Conclusion

Conclusion

We extensively study the interaction between read disturbance and multiple-row activation-based Processing-using-DRAM operations

We characterize 316 DDR4 chips from four major manufacturers

- PuDHammer **significantly exacerbates** the **read disturbance vulnerability** (e.g., HCfirst reduces from 4K to 26)
- PuDHammer **bypasses in-DRAM RowHammer mitigation** (TRR) and induces **11340x more bitflips** than RowHammer

We **adapt Per Row Activation Counting (PRAC) for PuDHammer**

- Incurs **48.26% performance overhead** on average across 60 multiprogrammed workload mixes

We hope our findings guide system-level and architectural solutions to enable read-disturbance-resilient future PuD systems

PuDHammer

Experimental Analysis of Read Disturbance Effects of Processing-using-DRAM in Real DRAM Chips

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SAFARI

ETH zürich

ColumnDisturb

Understanding Column-based Read Disturbance in Real DRAM Chips and Implications for Future Systems

Backup Slides

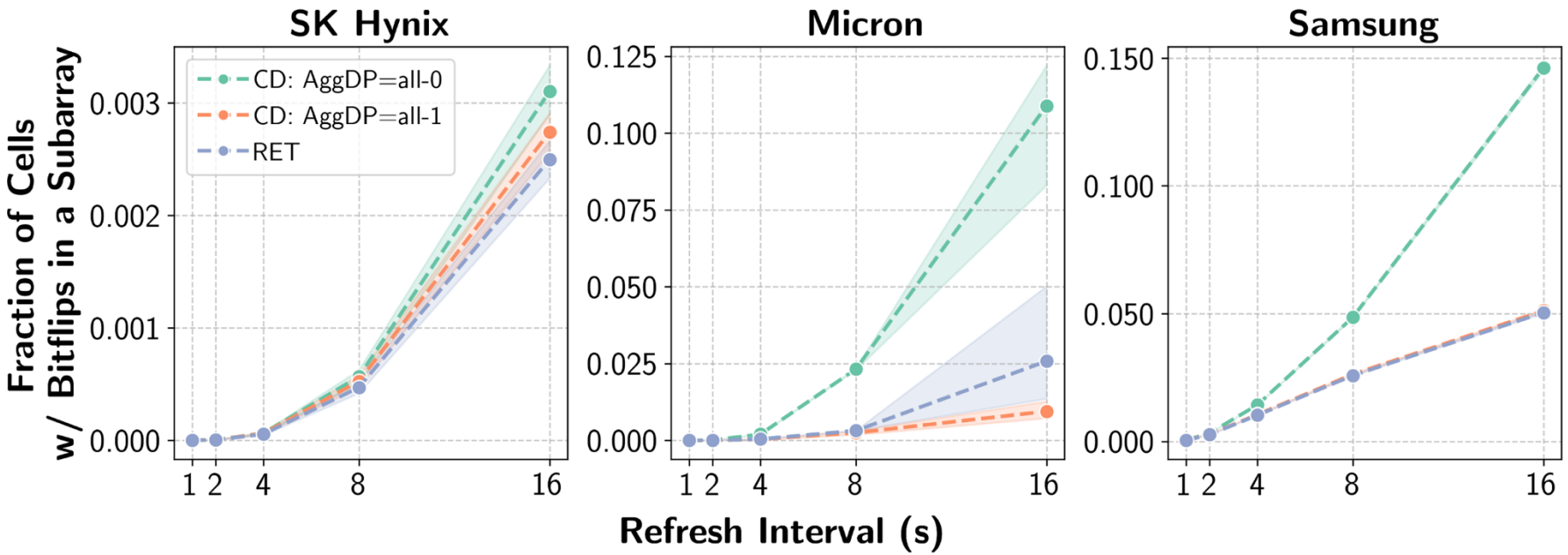
İsmail Emir Yüksel

Ataberk Olgun F. Nisa Bostancı Haocong Luo

A. Giray Yağlıkçı Onur Mutlu

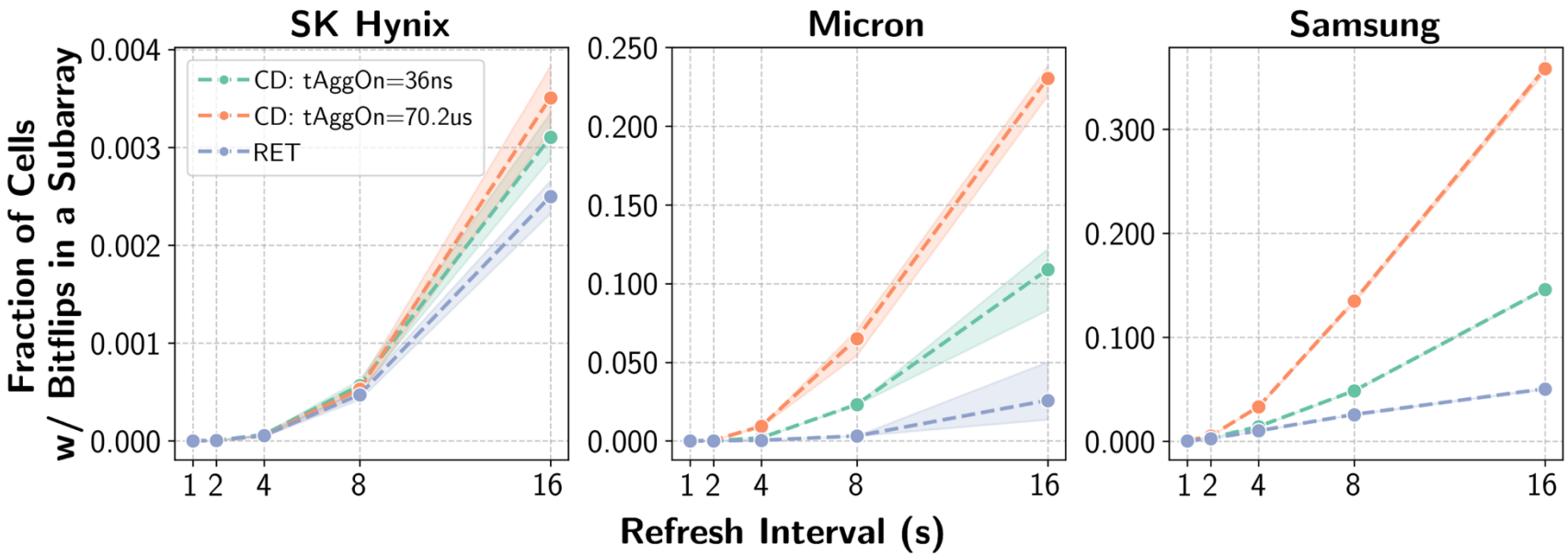
ETH zürich SAFARI CISPA

Foundational Results: Data Pattern



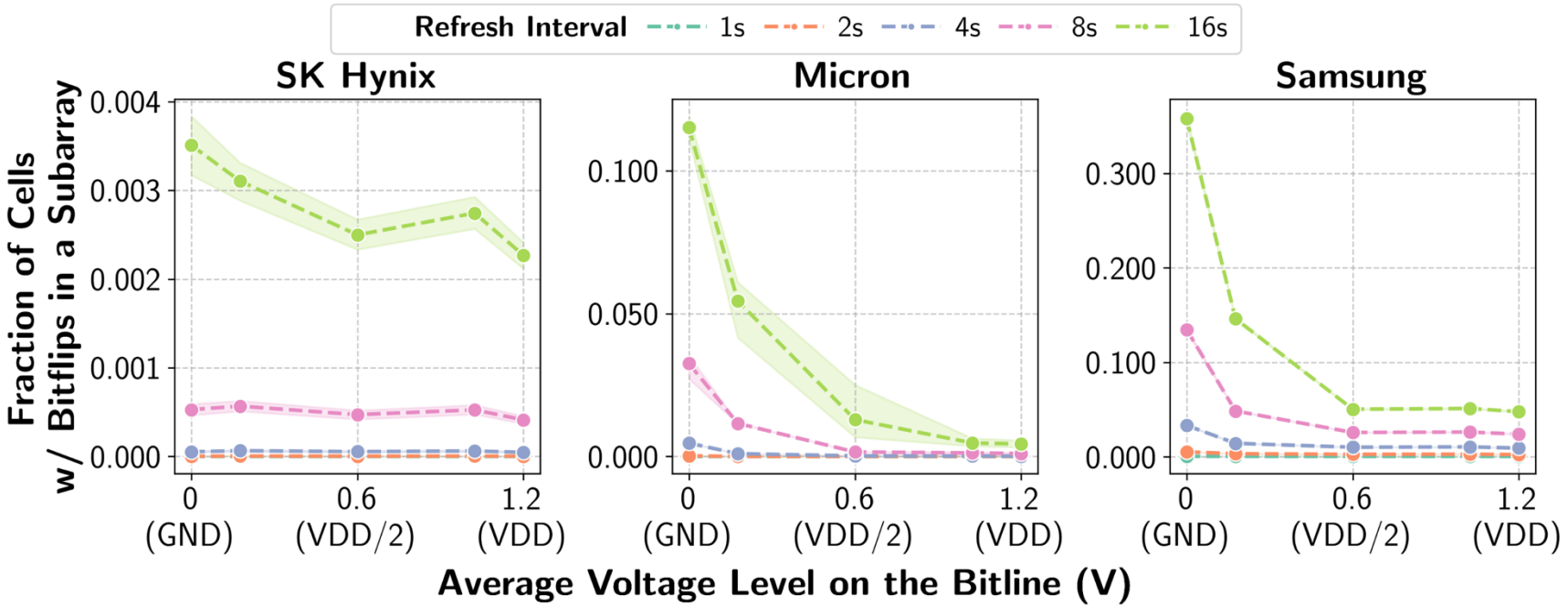
ColumnDisturb induces many more bitflips when the aggressor row data pattern is all-0 versus all-1

Foundational Results: tAggOn



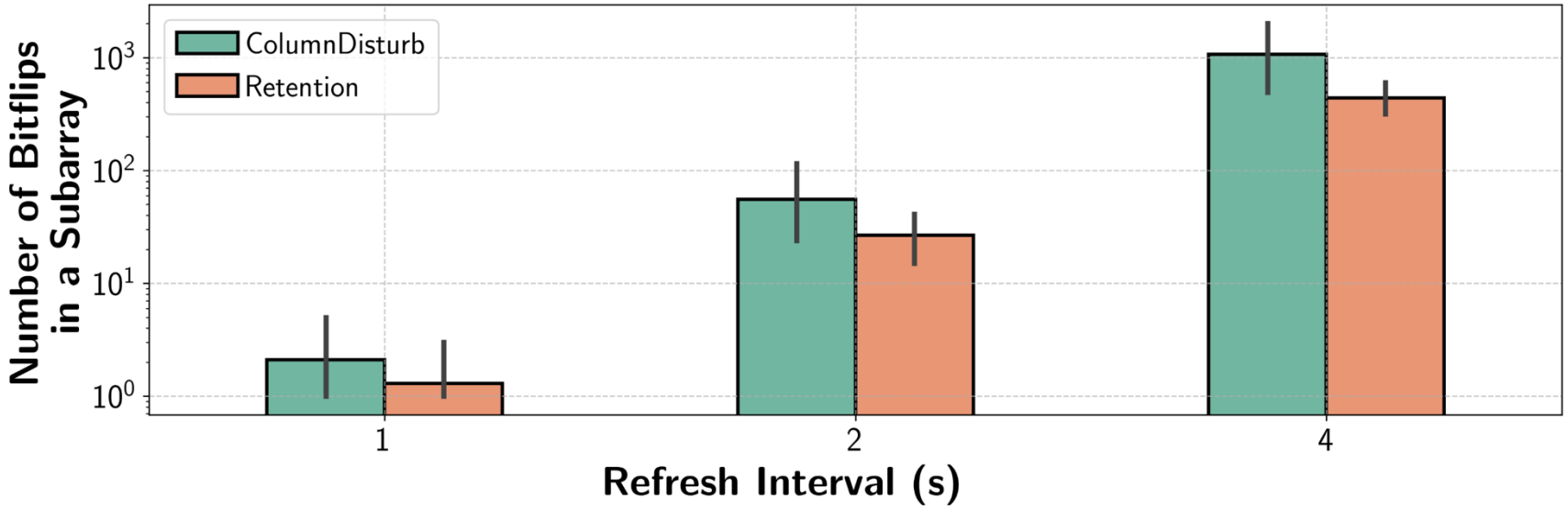
Fraction of cells with ColumnDisturb bitflips in a subarray significantly increases as tAggOn increases

Foundational Results: AVG(V_{BL})



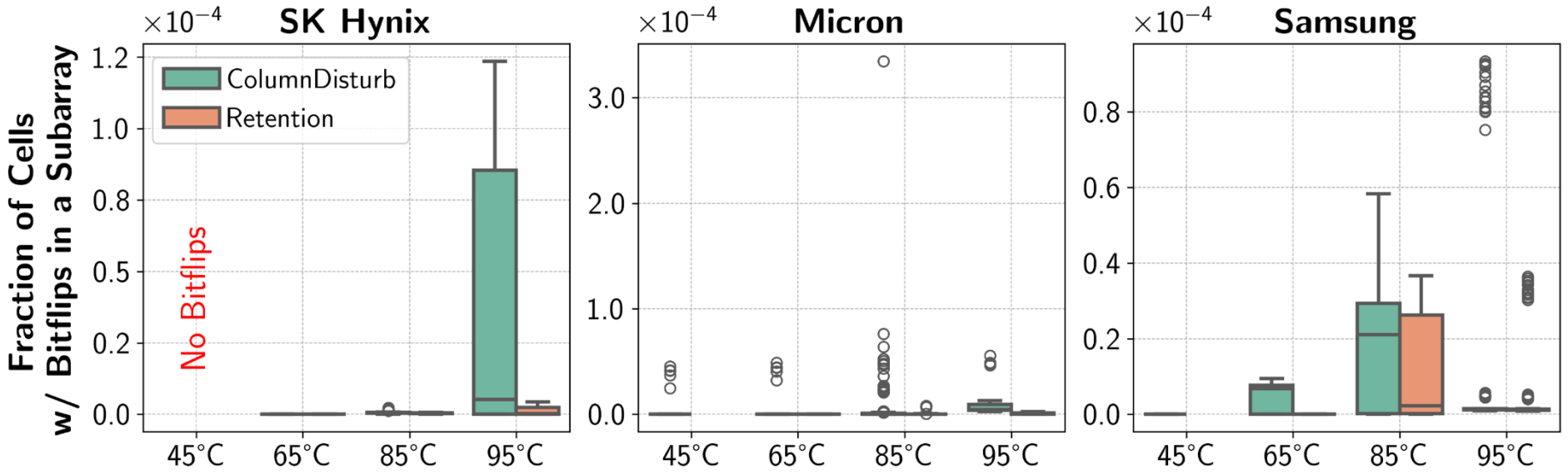
A DRAM chip becomes
significantly morev ulnerable to ColumnDisturb
as the average voltage level on the perturbed columns decreases

Foundational Results: CD on HBM



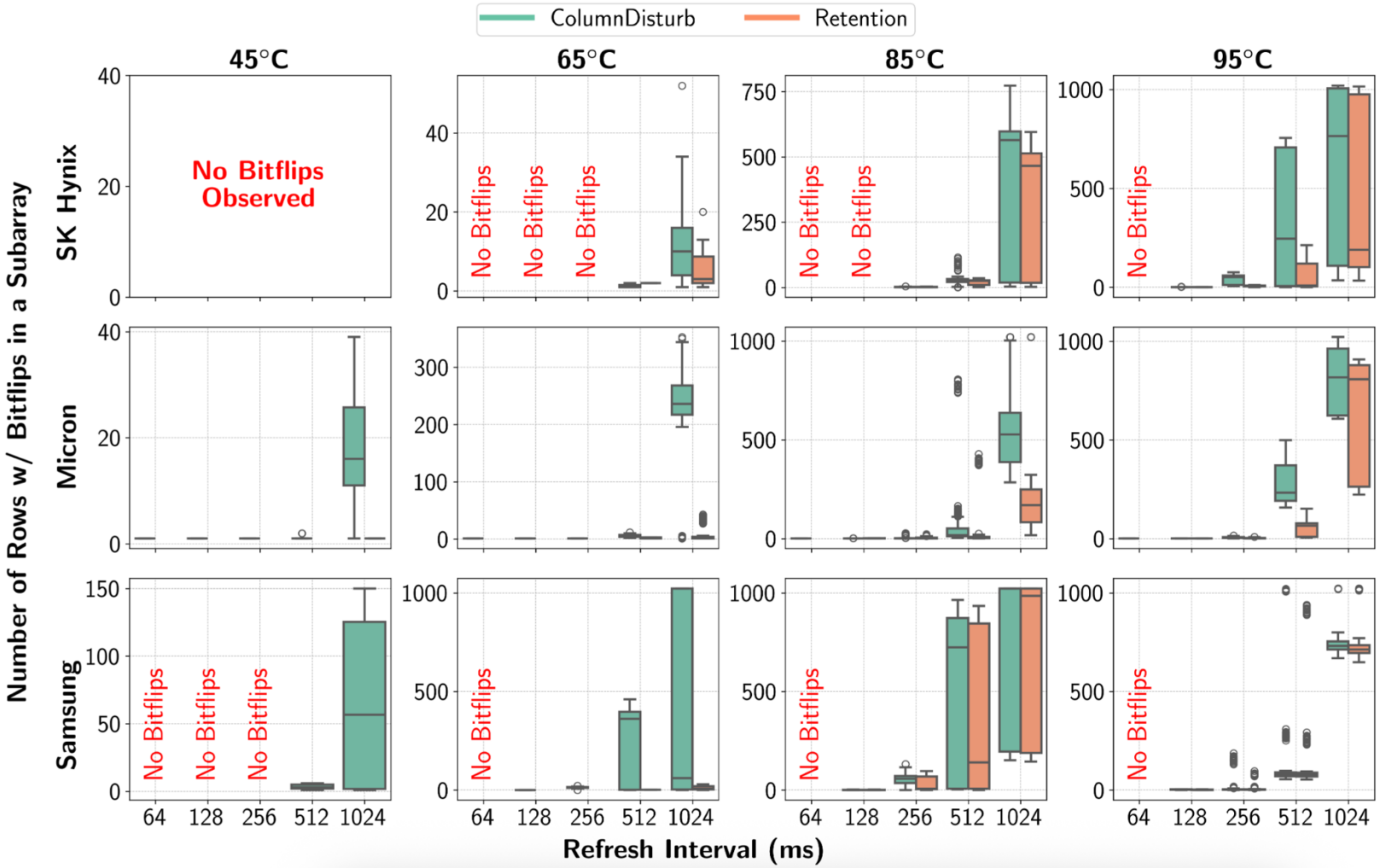
HBM2 chips are vulnerable to ColumnDisturb.
ColumnDisturb induces many more bitflips than retention failures

Temperature: Fraction of Cells w/ Bitflips

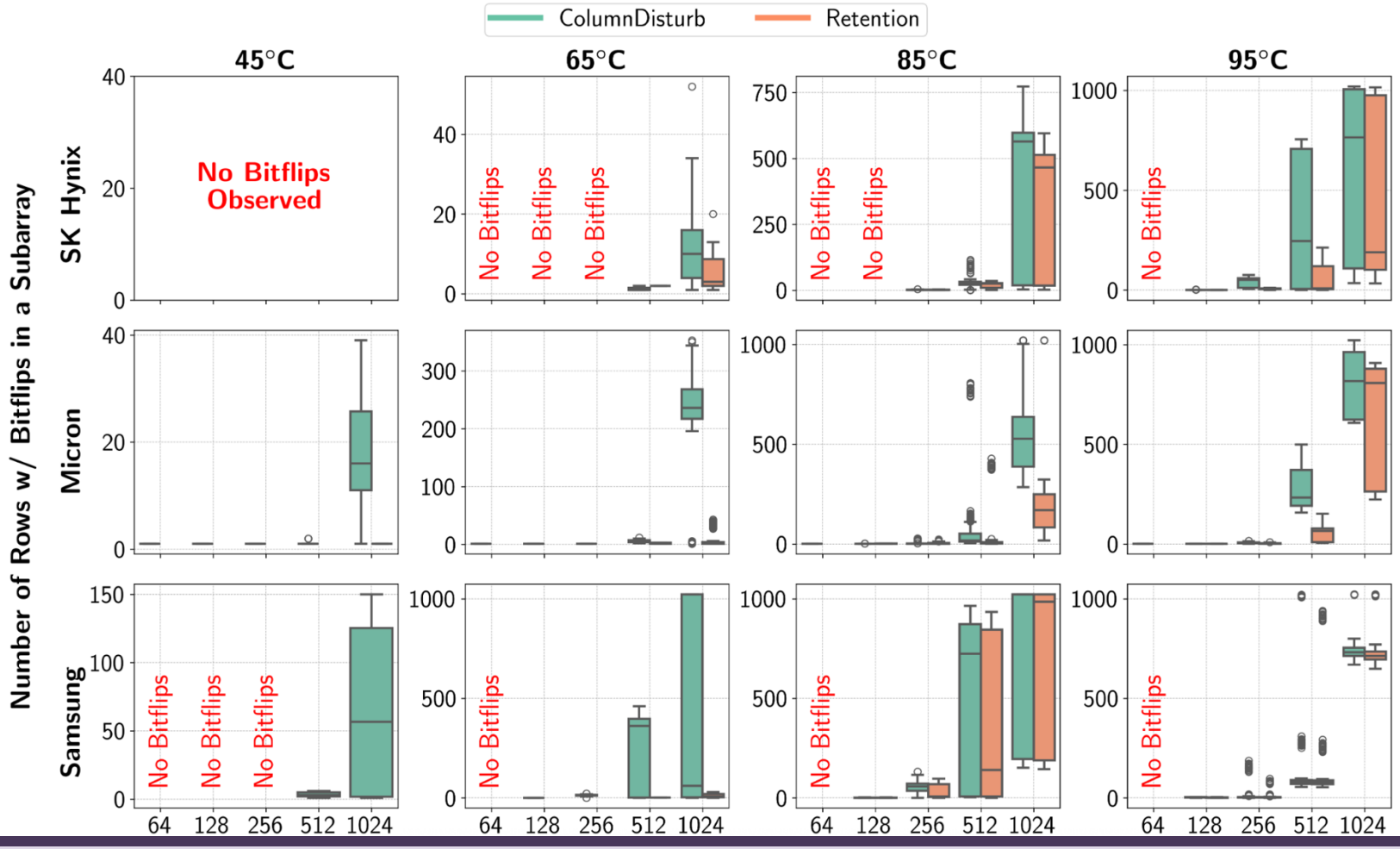


ColumnDisturb is more sensitive to temperature than retention failures

Temperature: Number of Rows w/ Bitflips

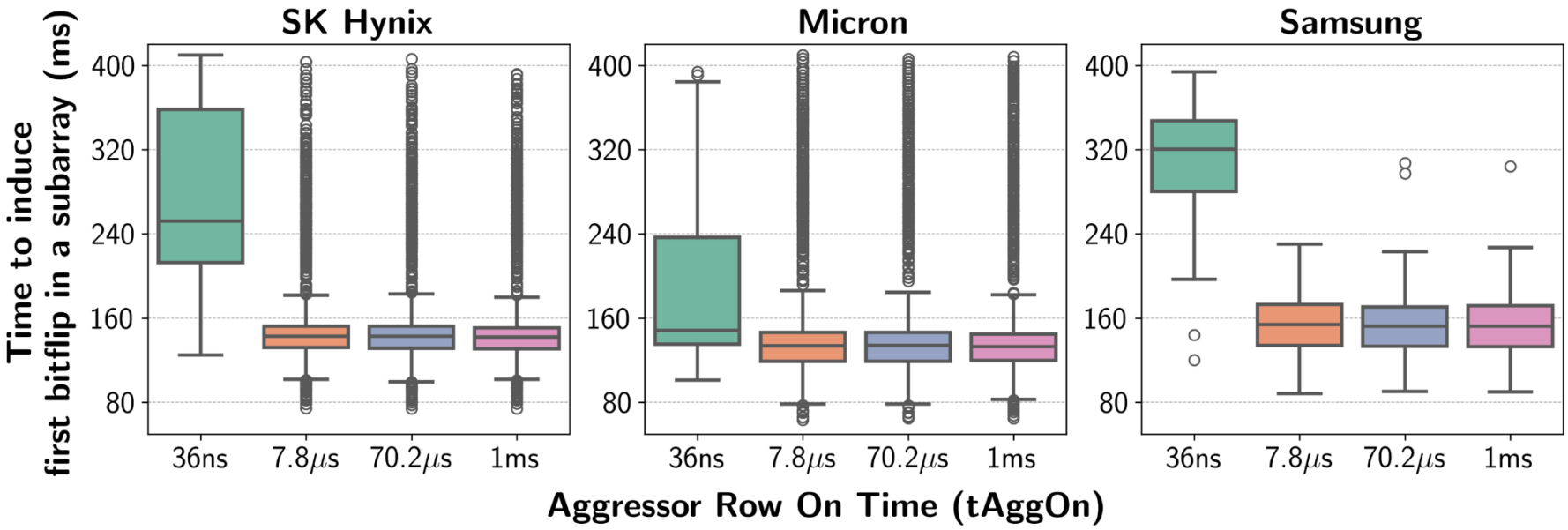


Temperature: Number of Rows w/ Bitflips



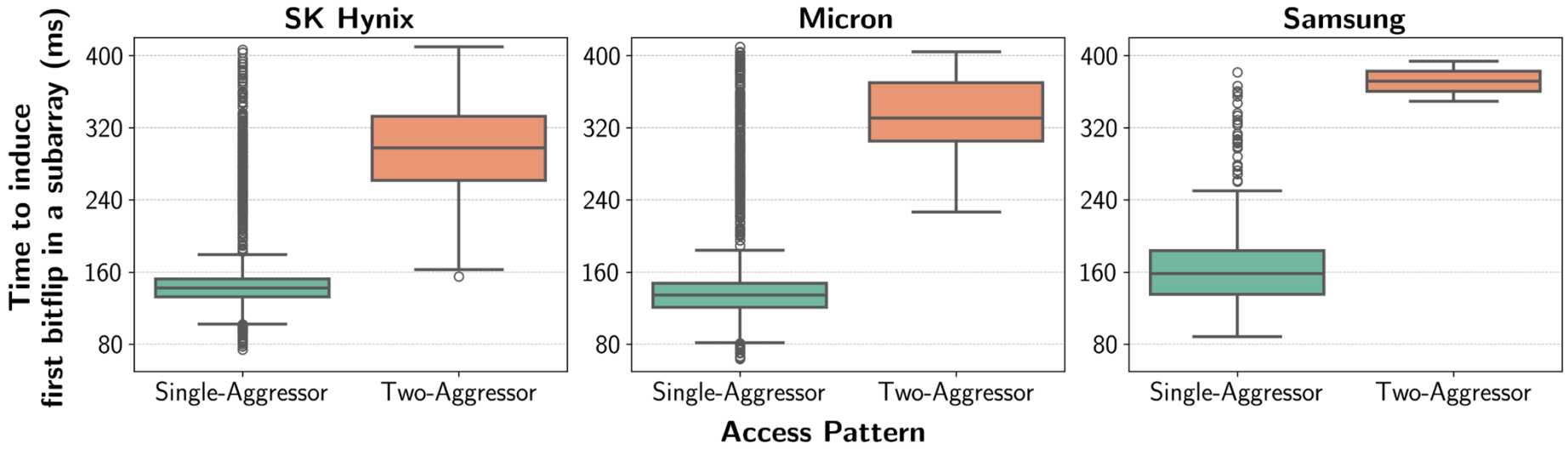
The blast radius of both ColumnDisturb and retention failures increase with higher temperature

In-Depth: Aggressor Row On Time



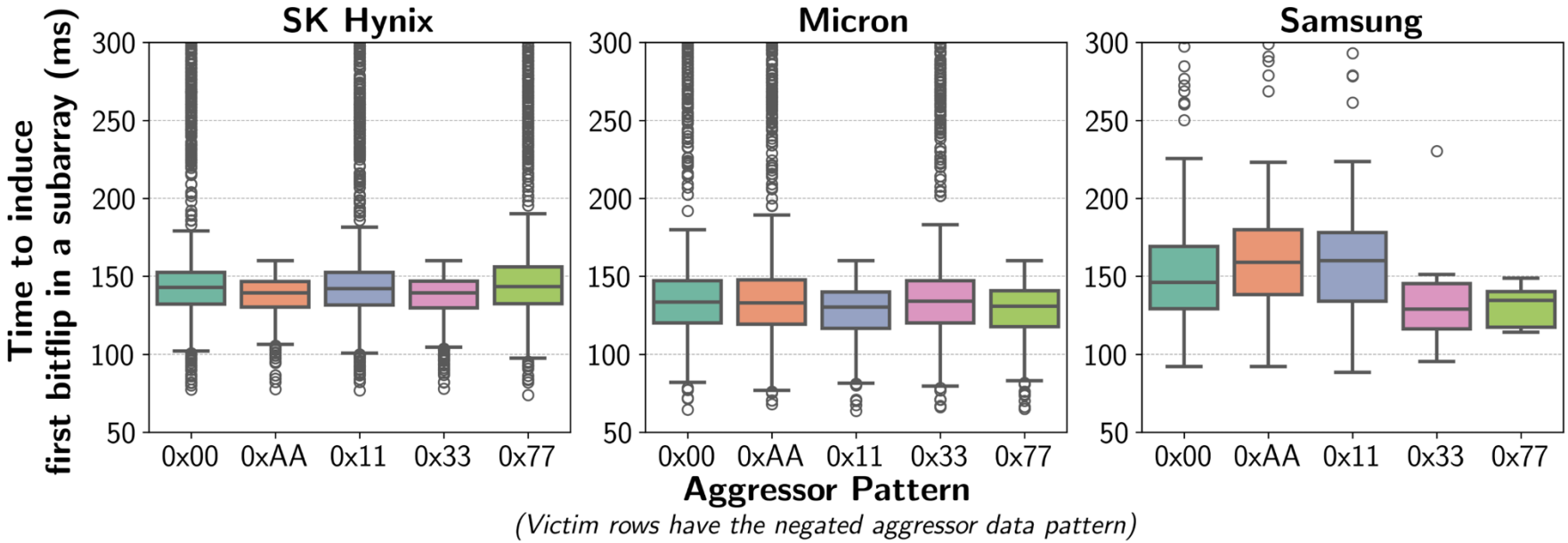
Keeping a row open (tAggOn >> tRAS) is significantly more effective than immediately closing it (tAggOn = tRAS) in inducing the first ColumnDisturb bitflip in a subarray

In-Depth: Access Pattern



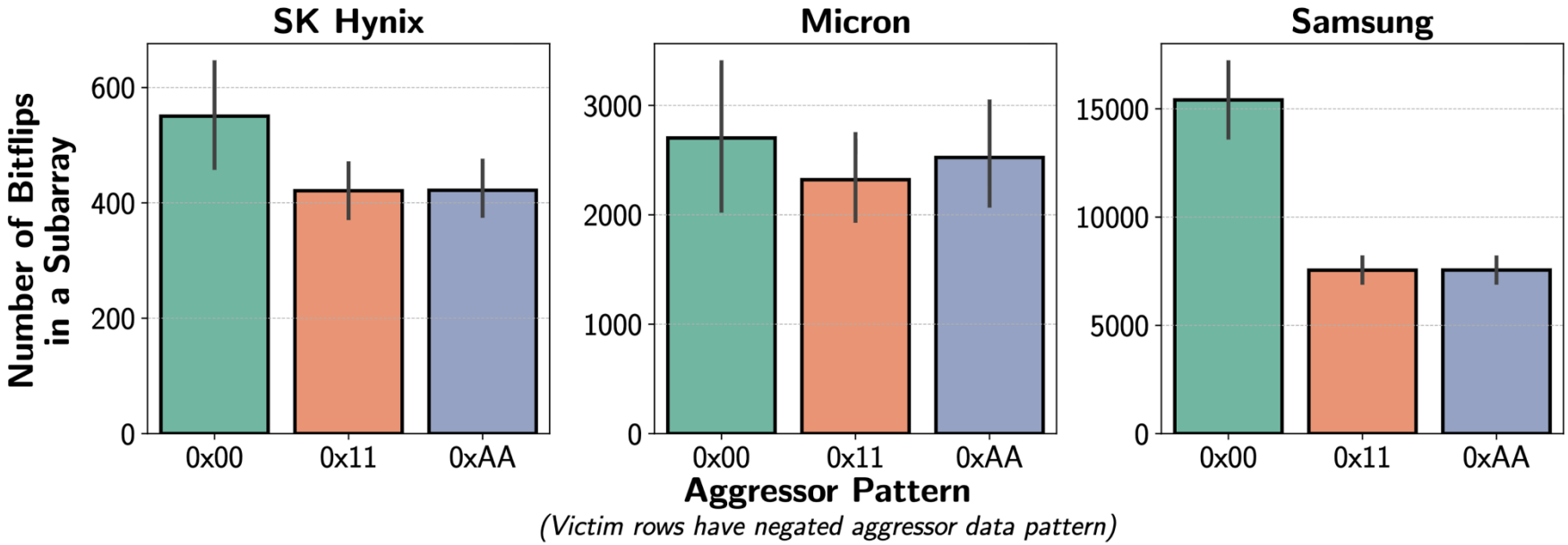
Single-aggressor access pattern induces the first ColumnDisturb bitflip in a subarray significantly faster than the two-aggressor access pattern

In-Depth: Data Pattern: TTIFB



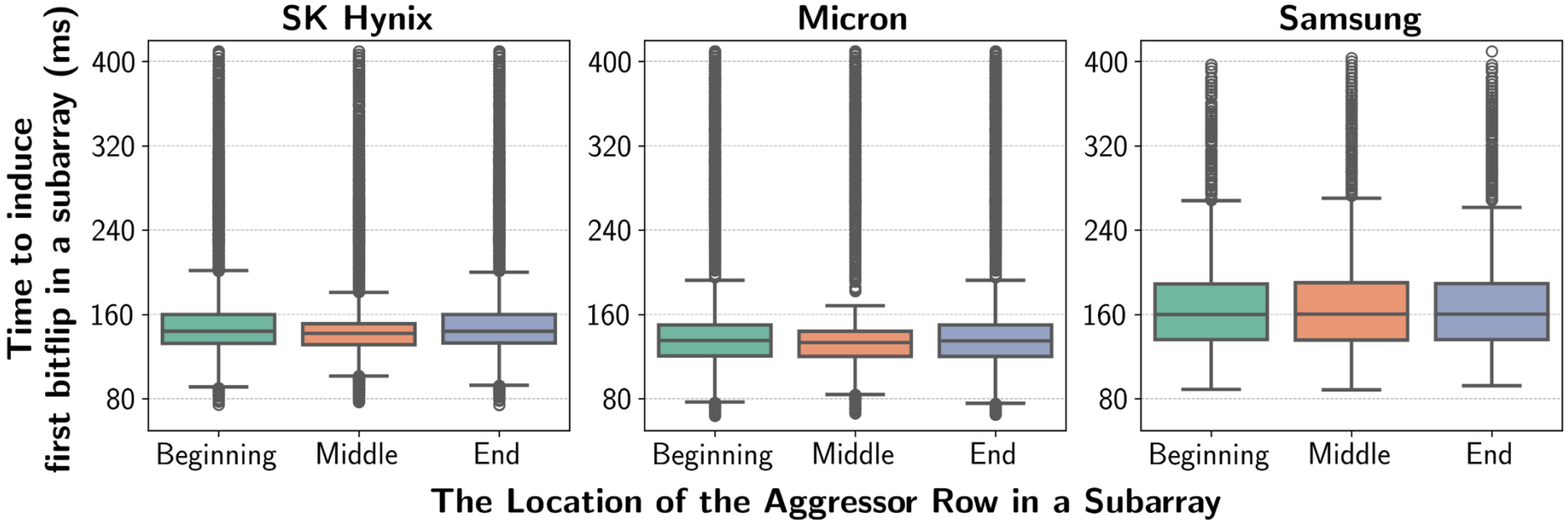
Data pattern has a small effect on the time to induce the first ColumnDisturb bitflip in a subarray

In-Depth: Data Pattern: # of Bitflips



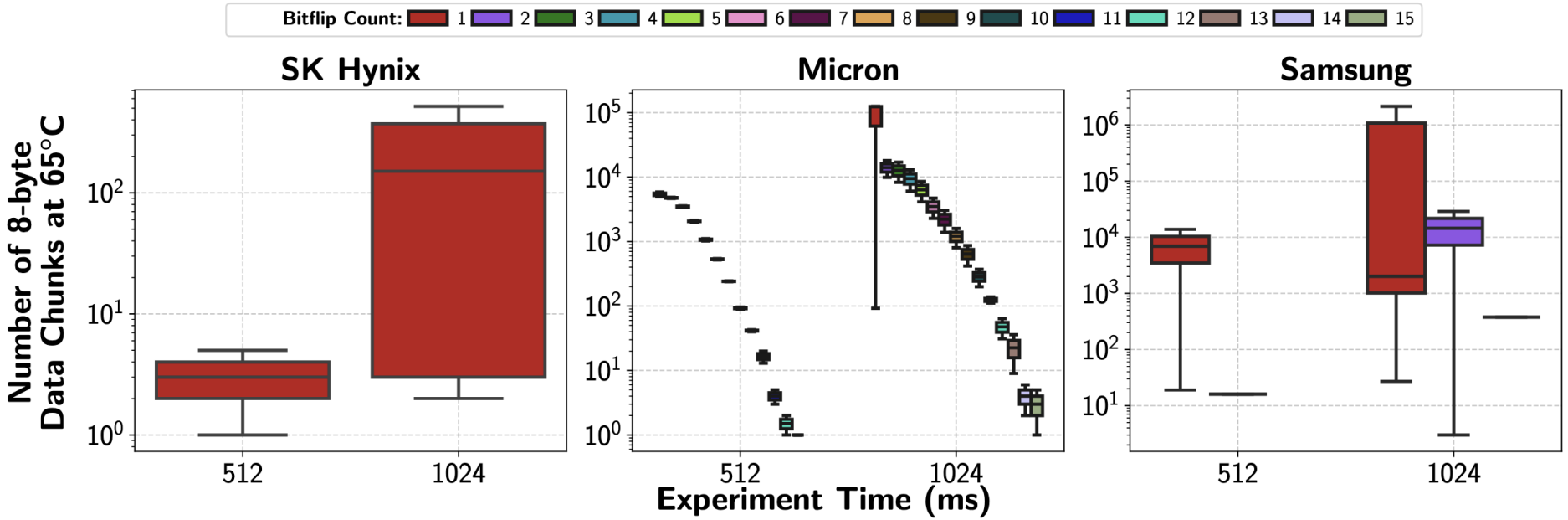
Data pattern greatly affects total ColumnDisturb:
having more logic-0 values
In the perturbed columns generates more bitflips

Aggressor Row Location in a Subarray



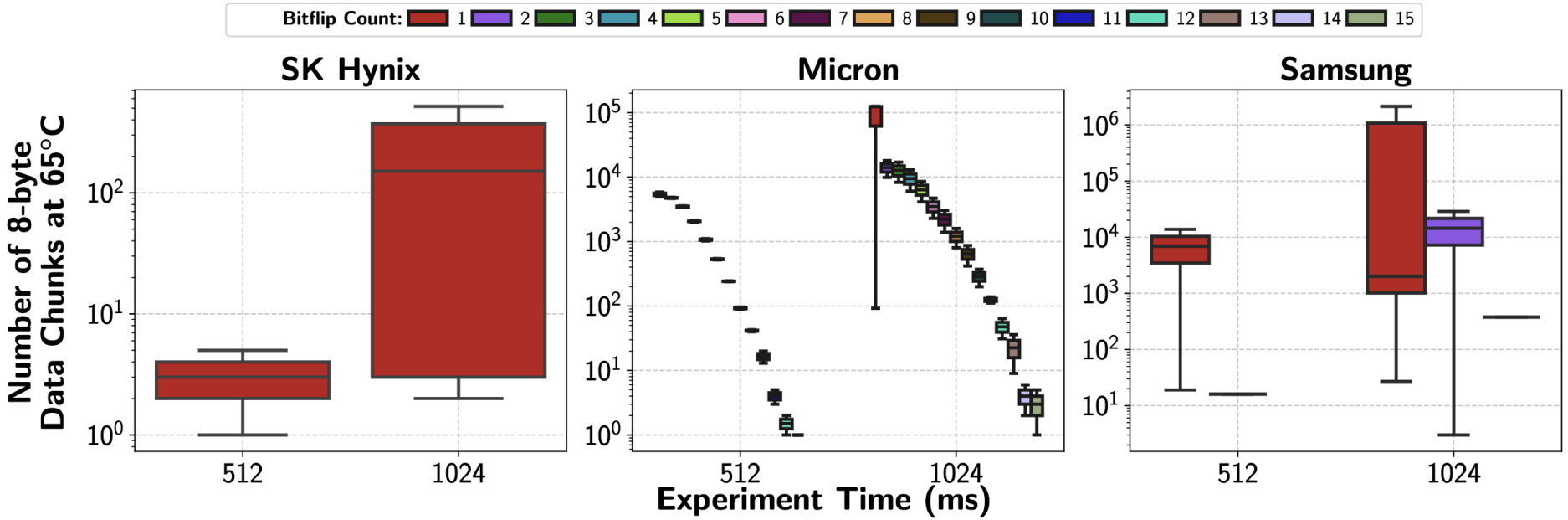
The location of the aggressor row only slightly affects the time to induce the first ColumnDisturb bitflip

Effectiveness of ECC



Many 8-byte data chunks experience 3 (up to 15) ColumnDisturb bitflips, which the SECDED ECC cannot correct or detect

Effectiveness of ECC



Since ColumnDisturb can induce 15 bitflips in multiple 8-byte data chunks, a (7,4) Hamming code could correct such bitflips with very large DRAM storage overheads (75%, i.e., three parity bits for every four data bits)

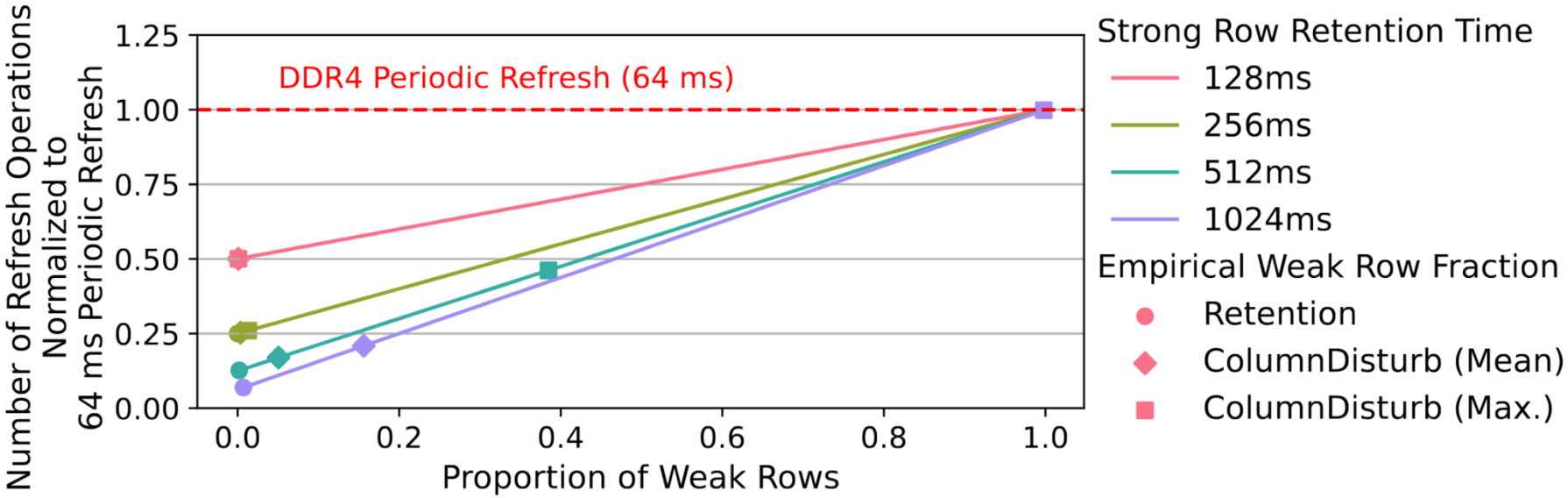
Implications for System Robustness

Increasing DRAM Refresh Rate

- **Increasing DRAM refresh rate** in future DRAM chips to prevent ColumnDisturb-induced bitflips
- DRAM chips perform periodic refreshes to avoid retention failures
  - E.g., Each row in DDR5 chips is refreshed **in every 32ms**

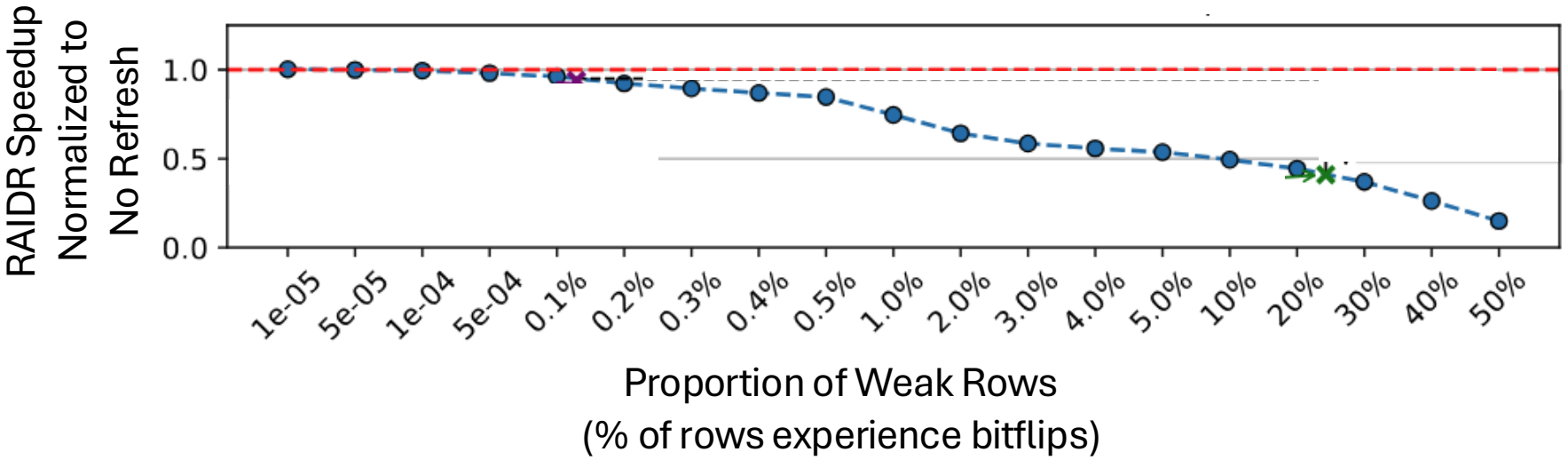
- The benefit
  - It can be applied **immediately in today's systems**
- Downside
  - It **degrades** system performance and energy efficiency **drastically**
- For a 32Gb DDR5 chip, 4x higher refresh rate
  - **Reduces system throughput** by 42.1%
  - **Increases energy consumption** by 67.5%

Number of Refresh Operations Needed



**In the presence of ColumnDisturb,
for a strong row retention time of 1024 ms,
the number of refresh operations increases by 3.02x on average**

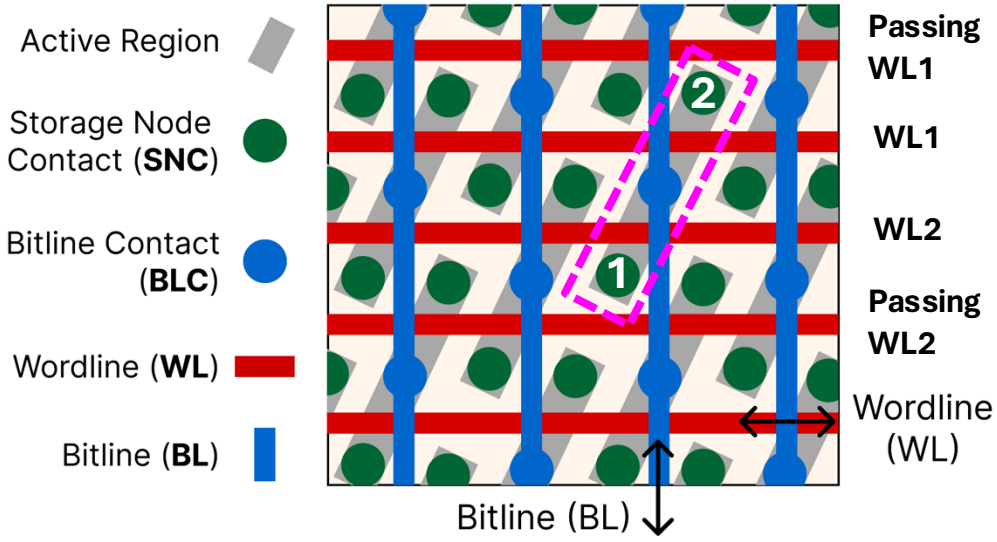
Implications on Retention-Aware Refresh



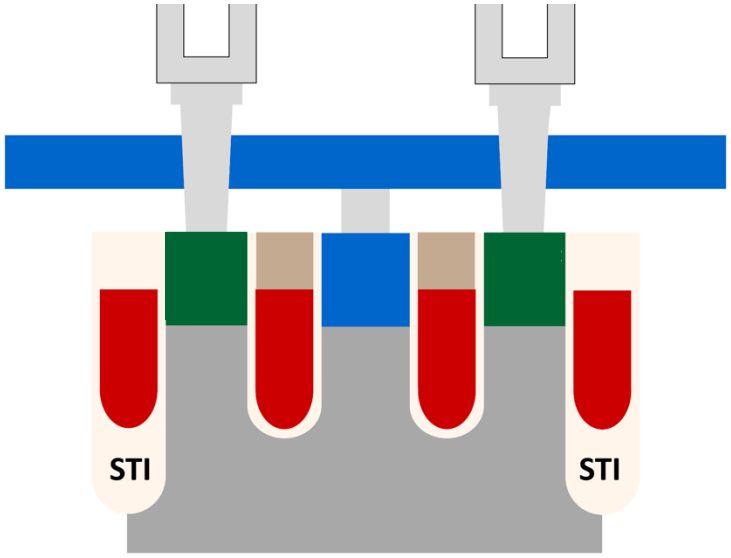
The speedup benefits of RAIDR **reduce significantly in the presence of ColumnDisturb**
compared to a baseline where there is no ColumnDisturb bitflips

Device-Level Mechanisms – Physical Layout

- **Modern $6F^2$ DRAM cell array layout**



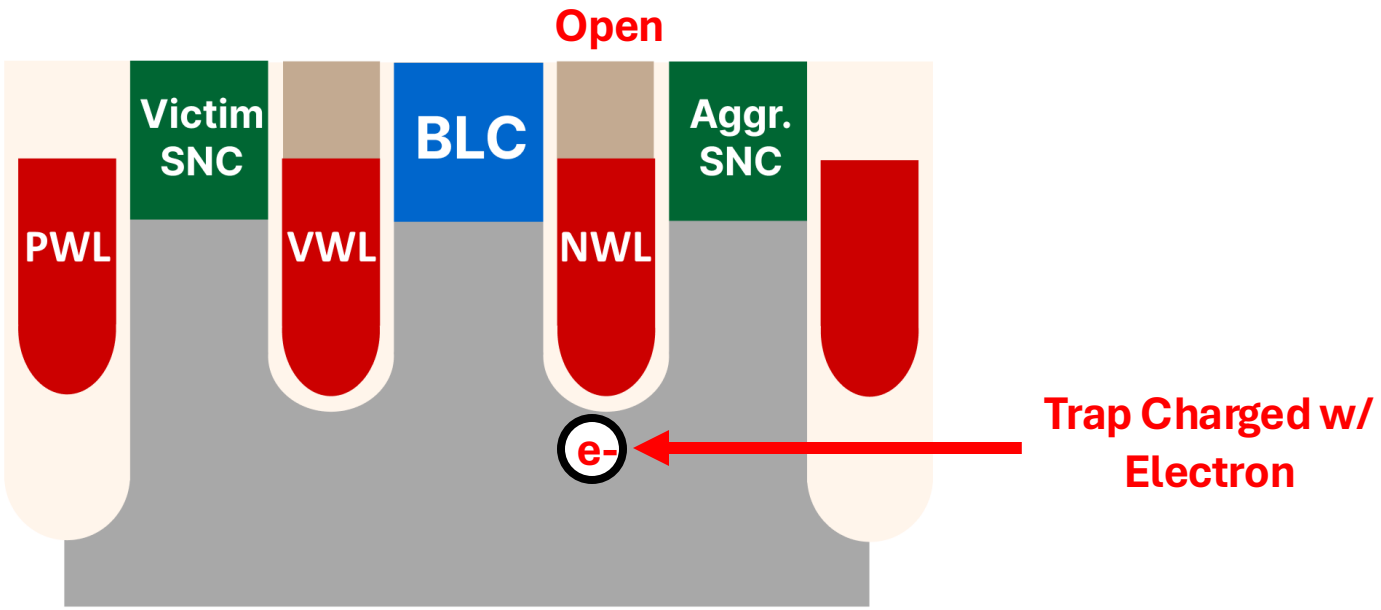
a) Physical Layout of $6F^2$ DRAM (Top View)



b) Cross-section of an Active Region (Side View, 2 Cells)

Device-Level Mechanism – RowHammer

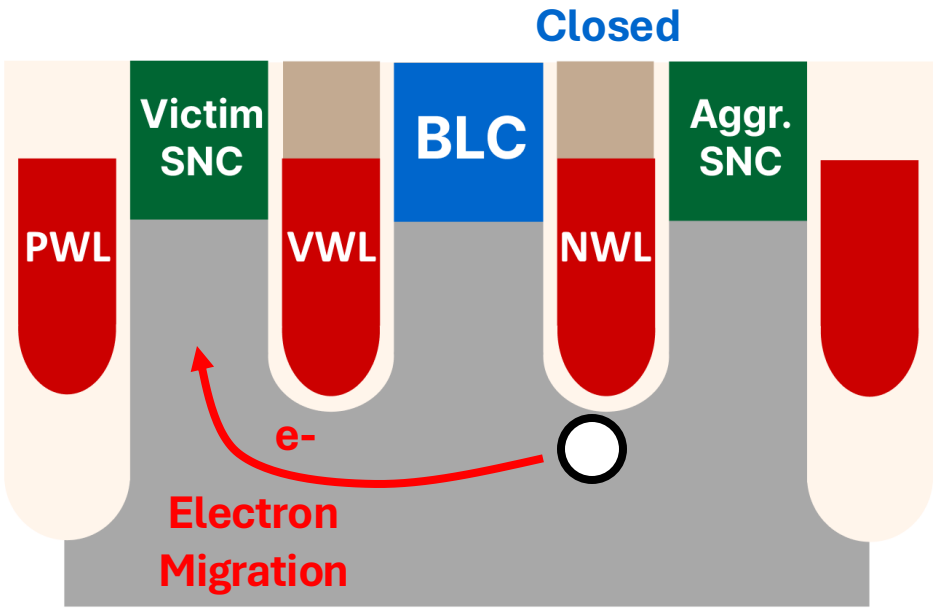
- **Key Error Mechanisms of RowHammer**
  - Trap-assisted Electron Migration [Yang+, EDL'19] [Walker+, TED'21] [Zhou+, IRPS'23]
    1. When NWL (aggressor) is open, acceptor-like traps are charged with electrons



PWL: Passing Wordline **VWL:** Victim Wordline
NWL: Neighboring Wordline (Aggressor)

Device-Level Mechanism – RowHammer

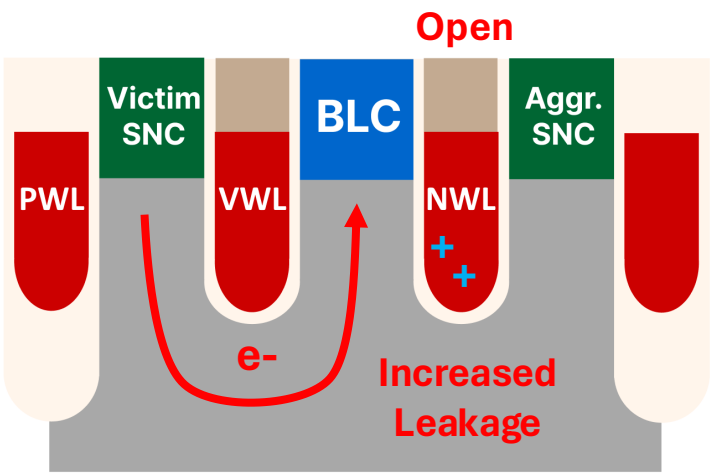
- **Key Error Mechanisms of RowHammer**
  - Trap-assisted Electron Migration [Yang+, EDL'19] [Walker+, TED'21] [Zhou+, IRPS'23]
    1. When NWL (aggressor) is open, acceptor-like traps are charged with electrons
    2. When NWL (aggressor) is closed, electrons are emitted from traps and migrate towards the victim cell



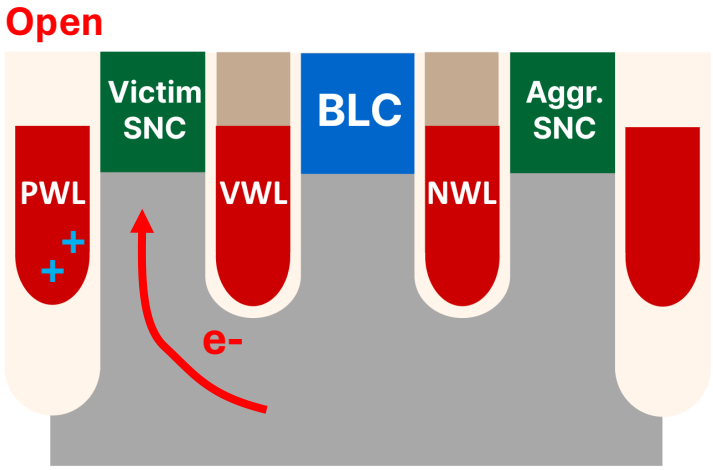
PWL: Passing Wordline **VWL:** Victim Wordline
NWL: Neighboring Wordline (Aggressor)

Device-Level Mechanism – RowPress

- **Key Error Mechanisms of RowPress**
  - **NWL RowPress:** When NWL is kept open for a long period, its strong electric field increases the leakage from the victim to the BLC,
    causing 0→1 bitflips [Zhou+, TED'24] [Zhou+, IRPS'24]
  - **PWL RowPress:** When PWL is kept open for a long period, its strong electric field draws electrons towards the victim,
    causing 1→0 bitflips [Zhou+, TED'24] [Zhou+, IRPS'24]



NWL RowPress



PWL RowPress

Device-Level ColumnDisturb Mechanism

- ColumnDisturb happens due to columns are perturbed and then driven. There is a direct contact between the cell and the column

- This contact spans and affects thousands of rows

- We have preliminary hypotheses on what might be happening
  - *"ColumnDisturb exacerbates subthreshold leakage of the access transistor [112] and/or the dielectric leakage between the capacitor and the bitline [2]."*

We need device-level studies
to fundamentally understand ColumnDisturb, and
our study can inspire such future works and be a good starting point

Access Pattern Difference

- For **RowHammer and RowPress**,
  - one must need to hammer/press **the same aggressor row(s)** to observe bitflips.

- However, **for ColumnDisturb,**
  - one can **distribute the number of activations** across many rows
  - As long as **the activated rows perturbs the same set of columns**, **it does not matter** whether **one aggressor row or thousands of aggressor rows are used**

- For example (assume one cell requires 3K activations for a bitflip)
  - **Activating a single row 3K times = activating 1K rows 3 times**
  - **It is not easy to *detect and refresh cells***

PuDHammer

Experimental Analysis of Read Disturbance Effects of Processing-using-DRAM in Real DRAM Chips

Backup Slides

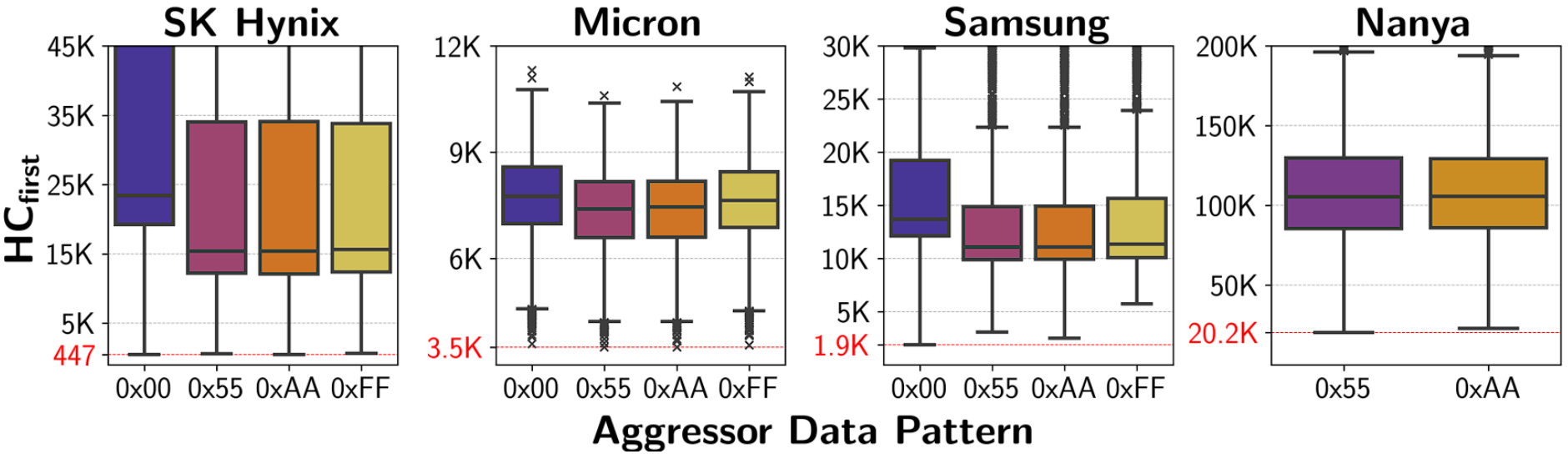
İsmail Emir Yüksel

Akash Sood Ataberk Olgun Oğuzhan Canpolat Haocong Luo

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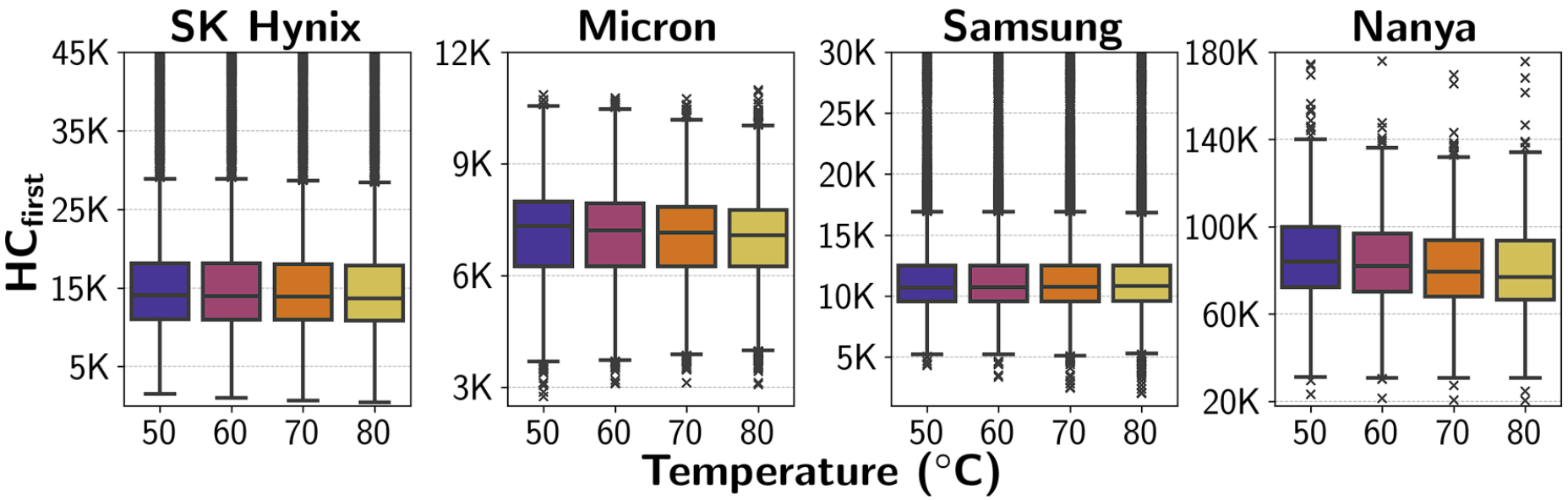
SAFARI ETH zürich

CoMRA – Data Pattern



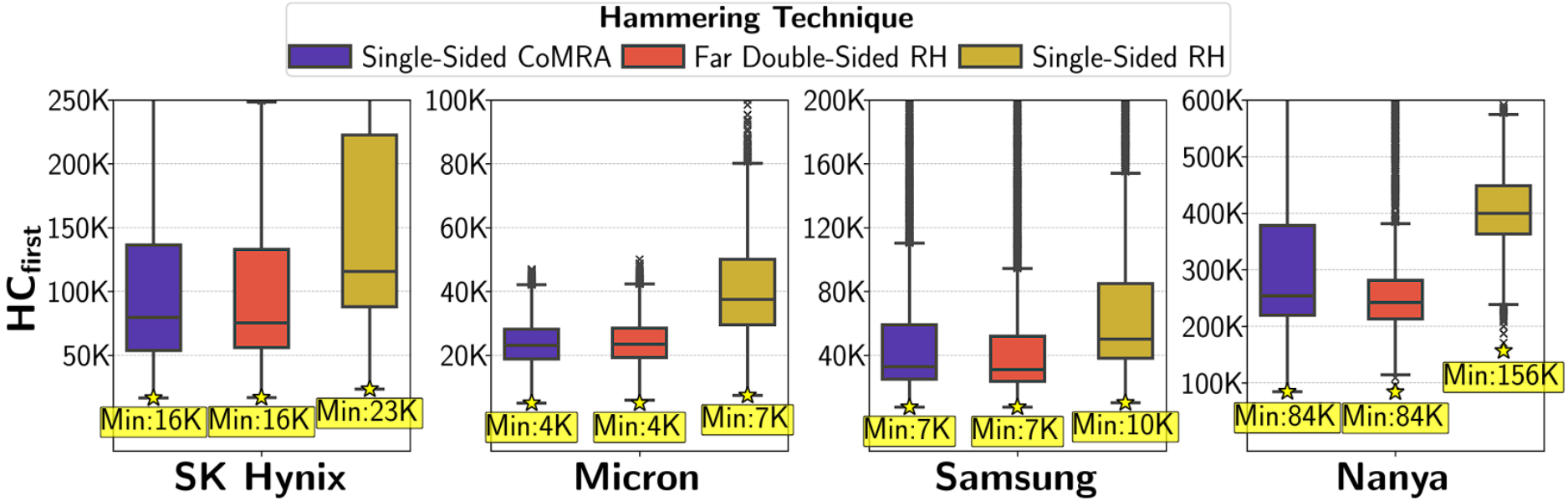
CheckerBoard pattern (i.e., 0x55/0xAA) is, in general, the most effective data pattern among the ones tested.

CoMRA – Temperature



Read disturbance effects of hammering with double-sided CoMRA tend to get worse as temperature increases.

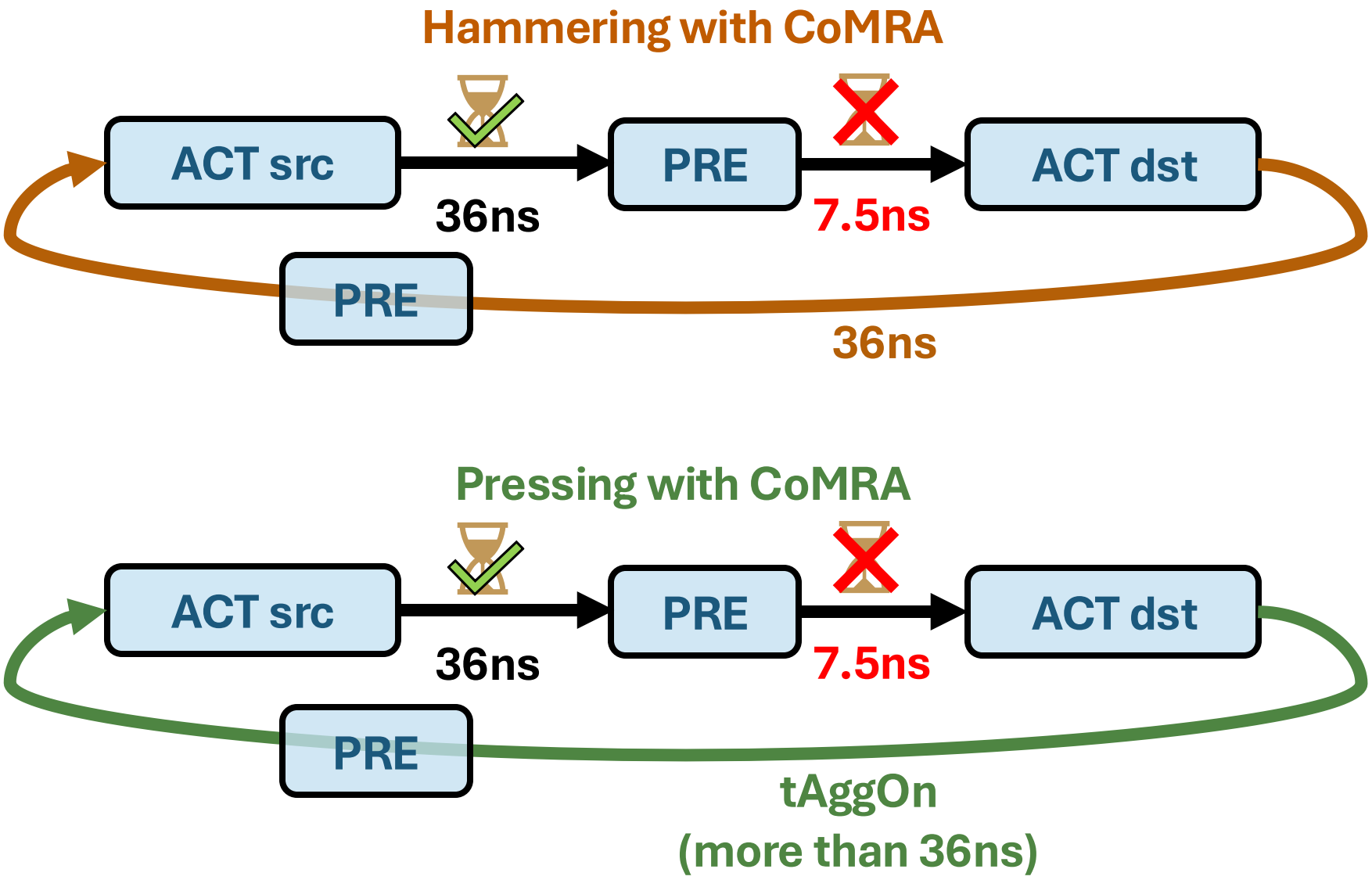
Single-Sided CoMRA vs. RowHammer



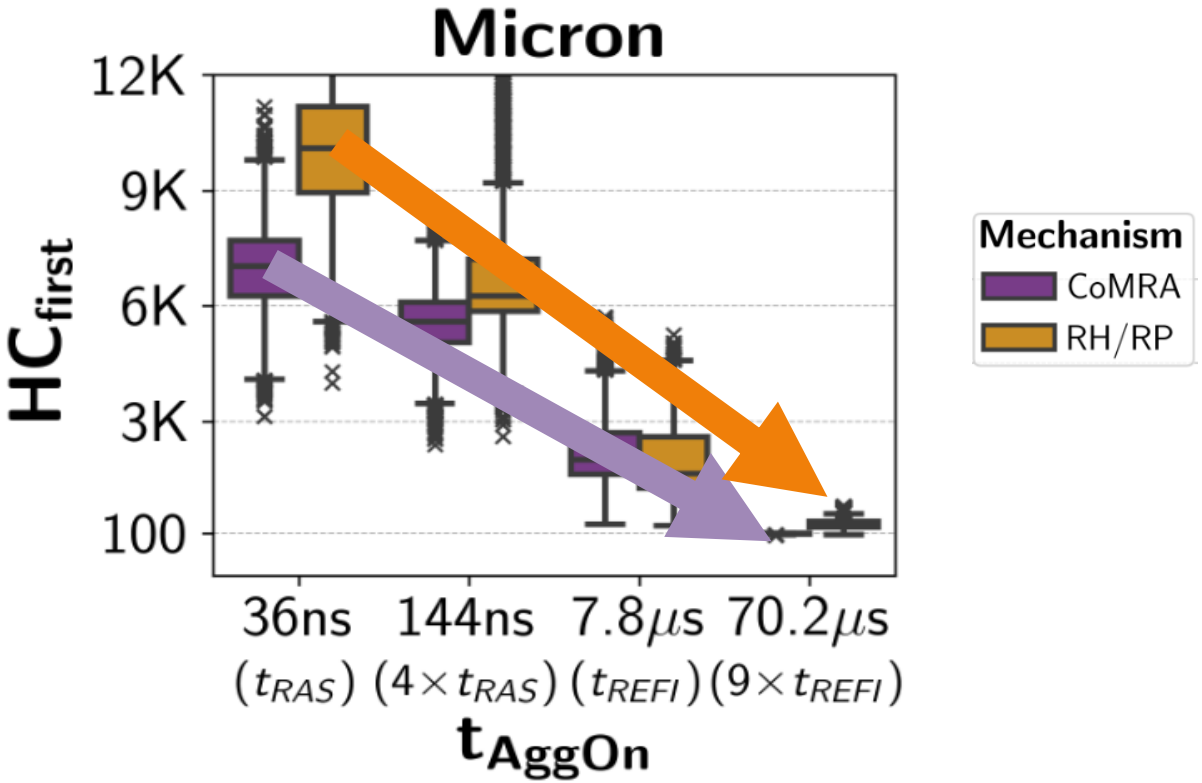
Single-sided CoMRA decreases HCfirst compared to single-sided RowHammer

Single-sided CoMRA exhibits similar HCfirst distribution with far double-sided RowHammer.

CoMRA (w/ increased tAggOn)



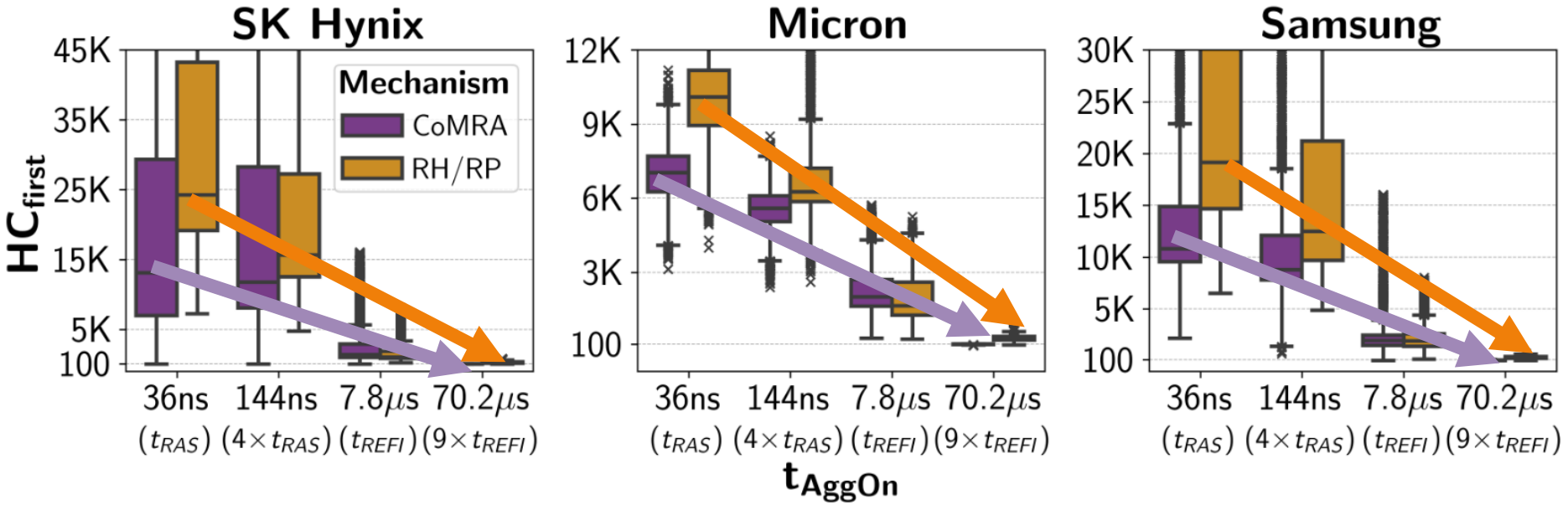
CoMRA (w/ Aggressor Row On Time) vs RowPress



For CoMRA
From 36ns to 70.2us
79x reduction

For RH/RP
From 36ns to 70.2us
31x reduction

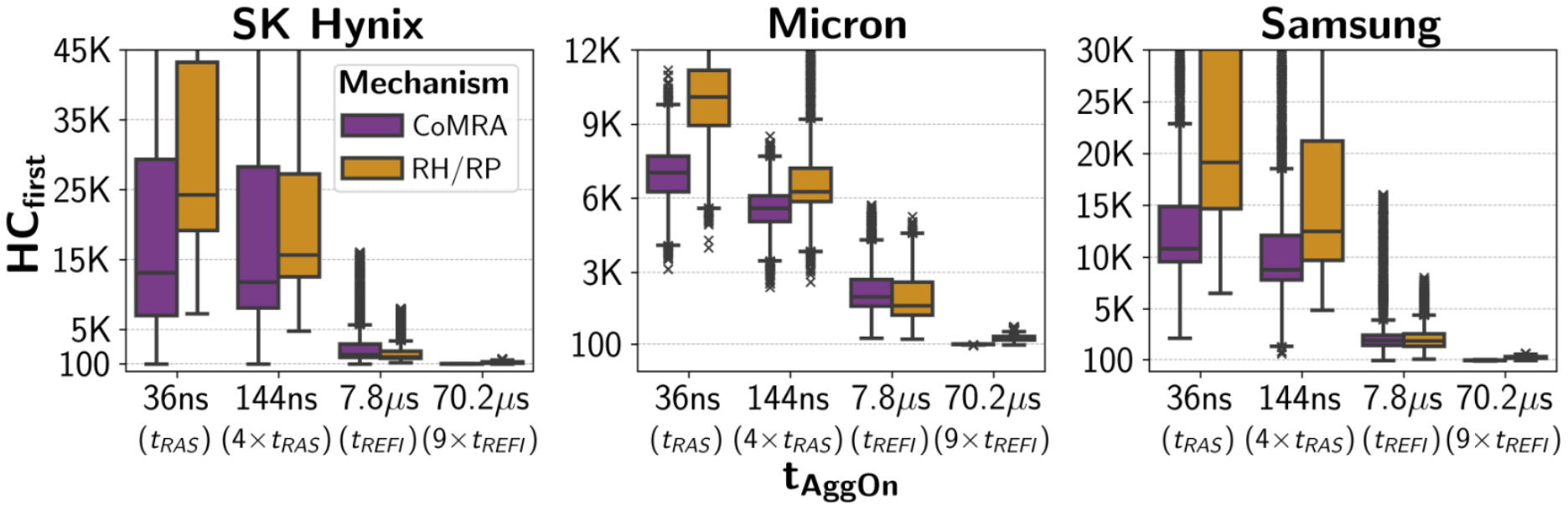
CoMRA (w/ Aggressor Row On Time) vs RowPress



Trend consistent across all four manufacturer:

Increasing tAggOn significantly reduces HCfirst for both CoMRA and RH/RP

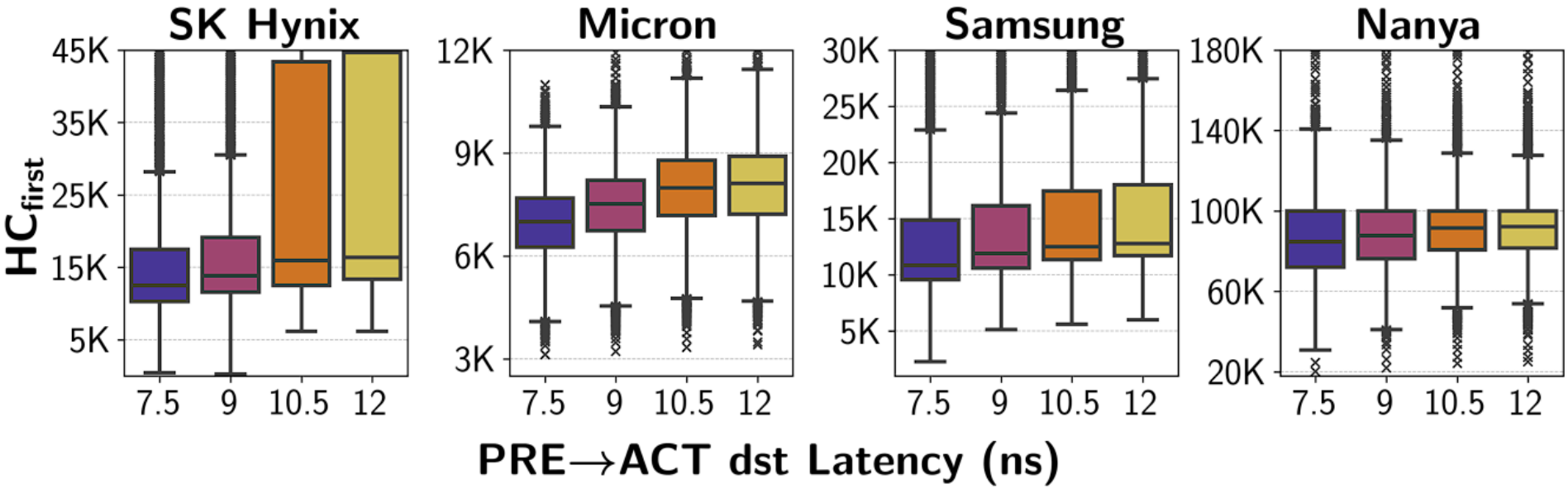
CoMRA (w/ Aggressor Row On Time) vs RowPress



Key Takeaway 2

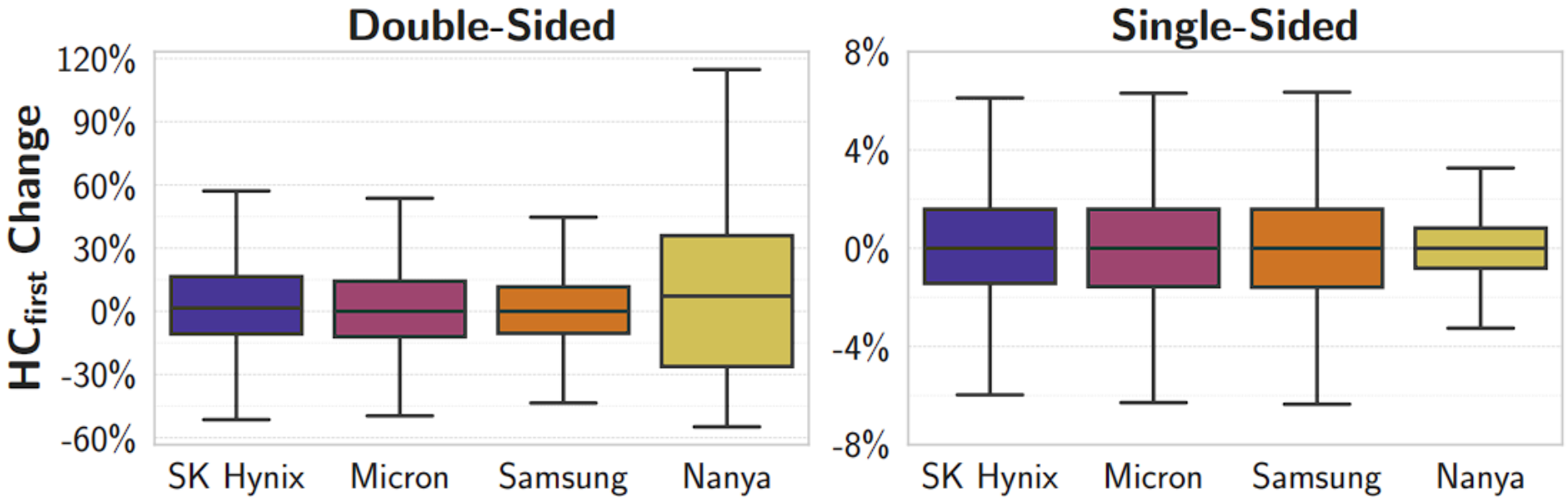
Pressing with CoMRA is more effective than hammering with CoMRA

CoMRA – Timing Delay



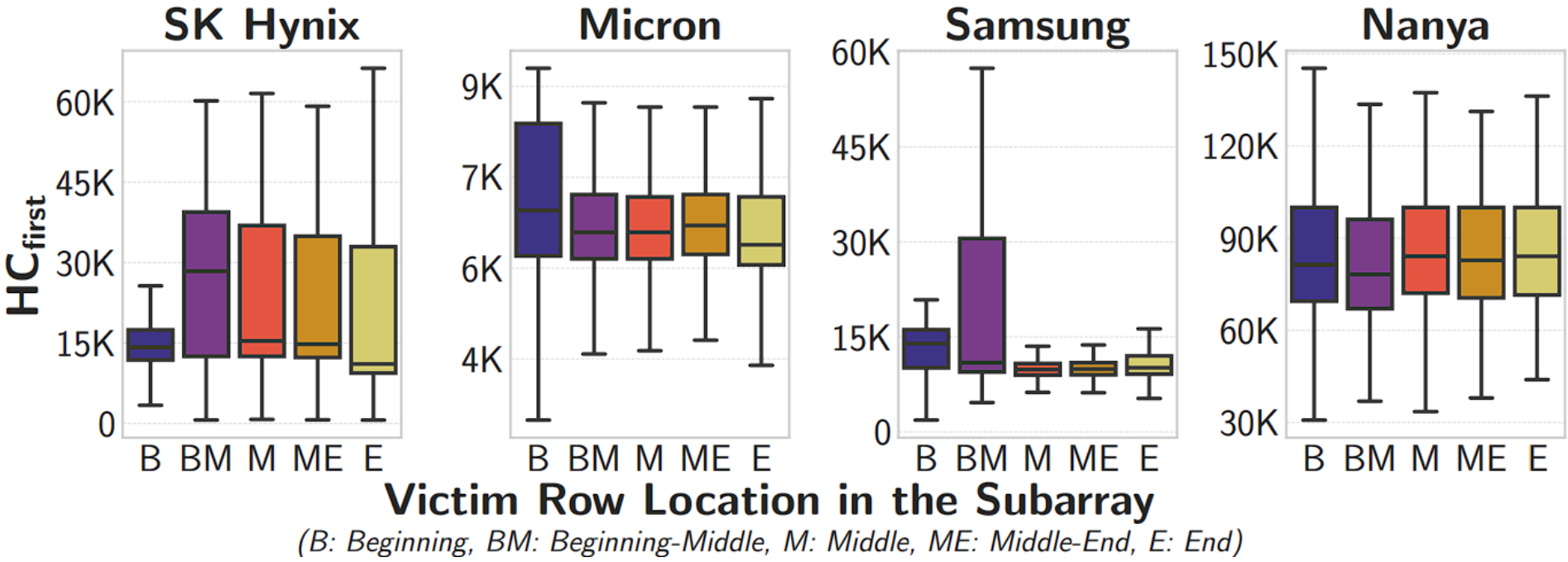
HCfirst increases as the latency of PRE→ACT dst increases

CoMRA – Copy Direction



In most cases, copy direction has a small effect on HCfirst.

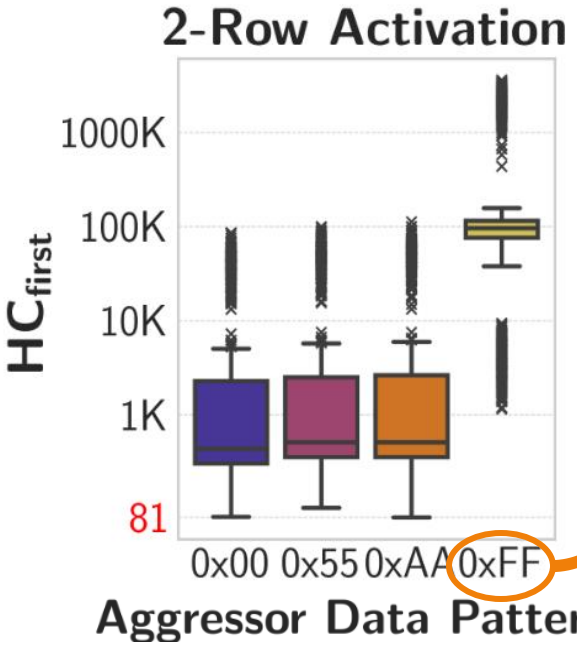
CoMRA – Spatial Variation



HCfirst varies based on
the victim row's location in a subarray

SiMRA Results

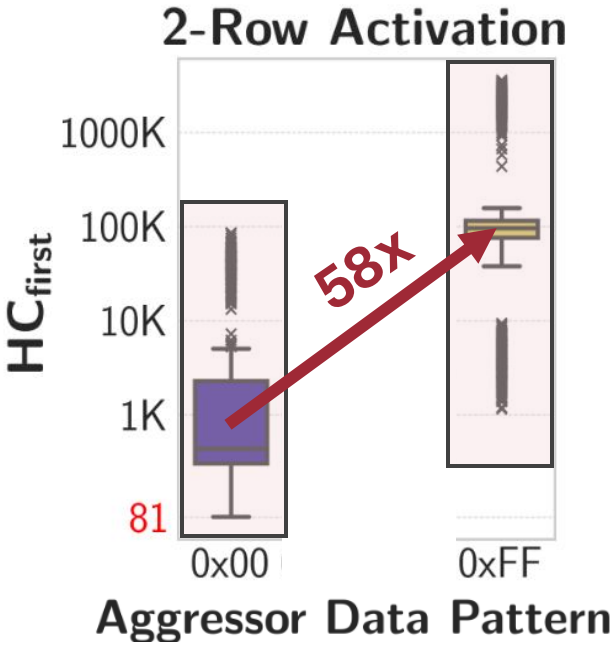
SiMRA -- Impact of the Data Pattern



Aggressor Rows = 0xFF..FF
Victim Rows = 0x00..00

(Victim has negated data pattern)

SiMRA -- Impact of the Data Pattern



(Victim has negated data pattern)

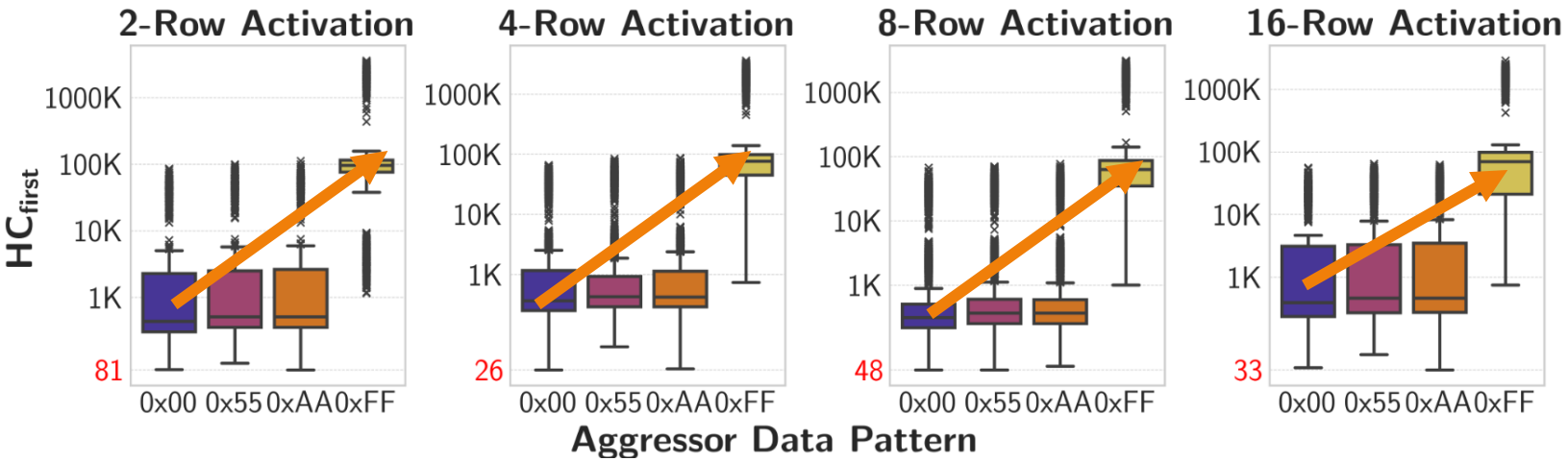
Initializing victim rows with 0x00 increases avg. HCfirst by 58x

Dominant bitflip direction for SiMRA 1->0

Dominant bitflip direction for RH* 0->1

*not shown in the figure

SiMRA -- Impact of the Data Pattern

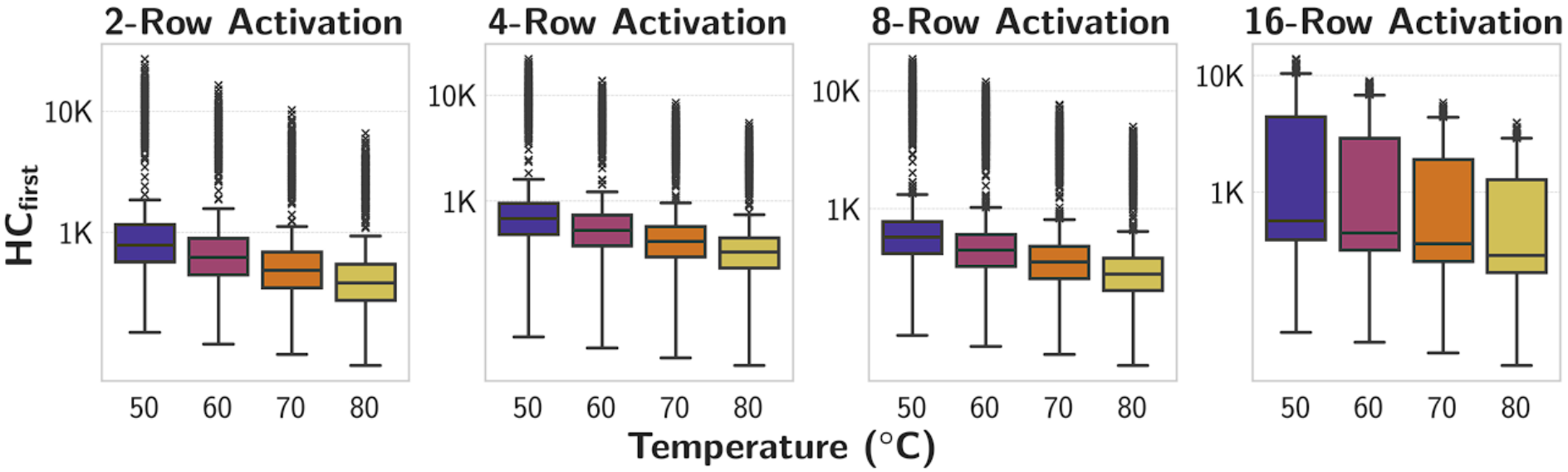


Trend consistent across all tested number of simultaneously activated rows

Key Takeaway

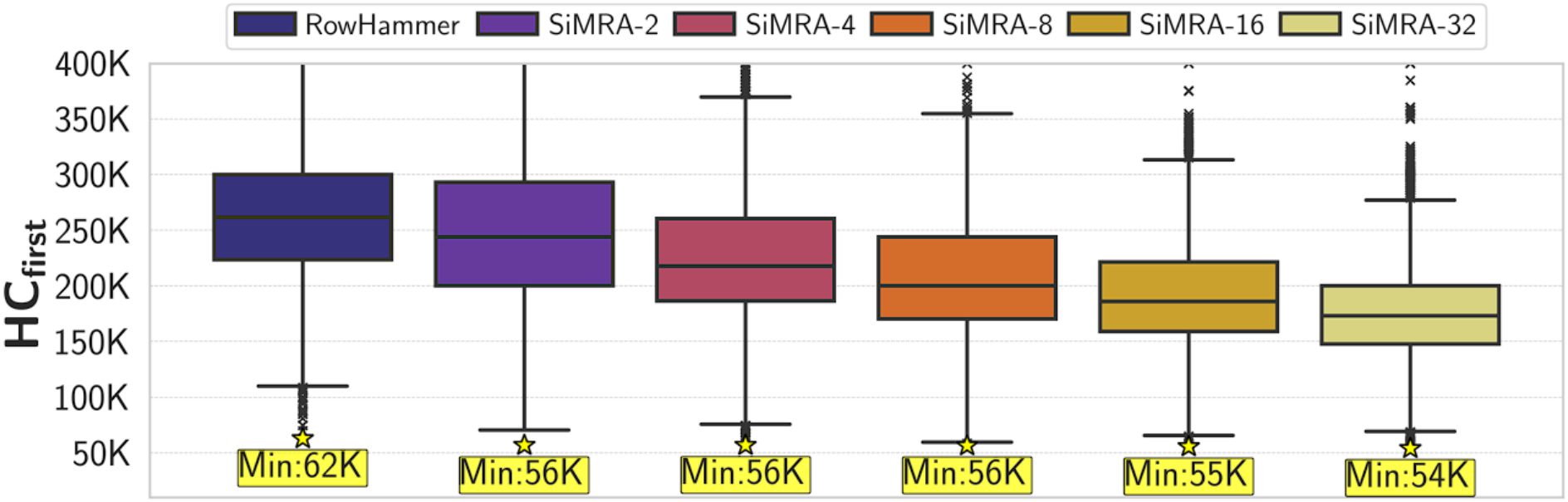
SiMRA is significantly affected by data pattern and directionality of SiMRA and RowHammer bitflips are opposite

SiMRA – Temperature



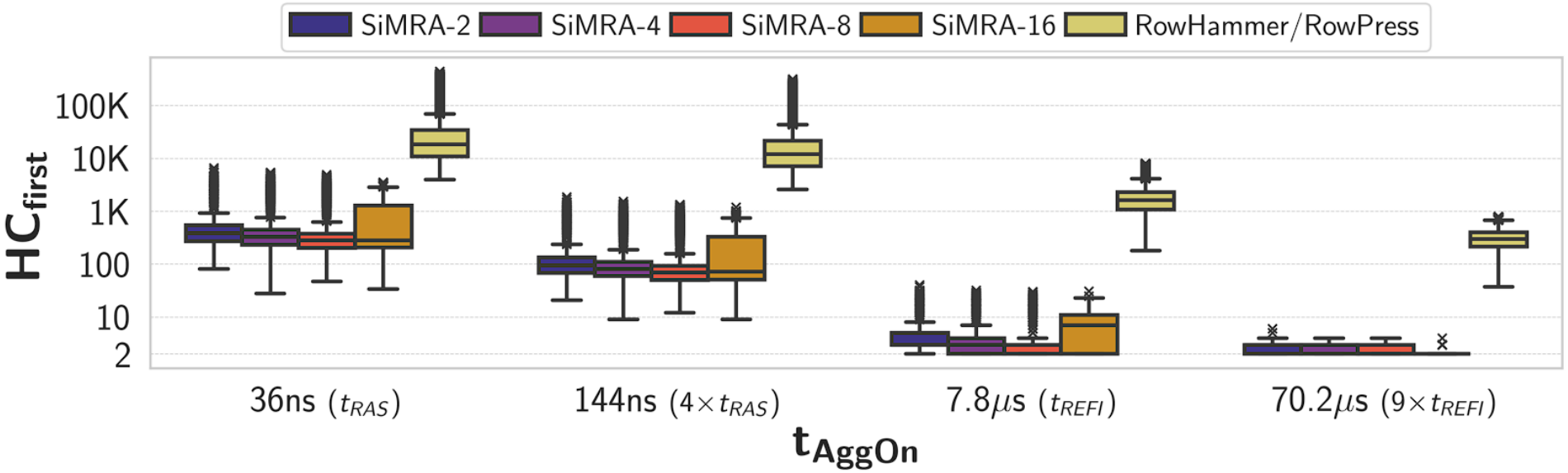
HCfirst decreases as temperature increases

Single-Sided SiMRA vs. RowHammer



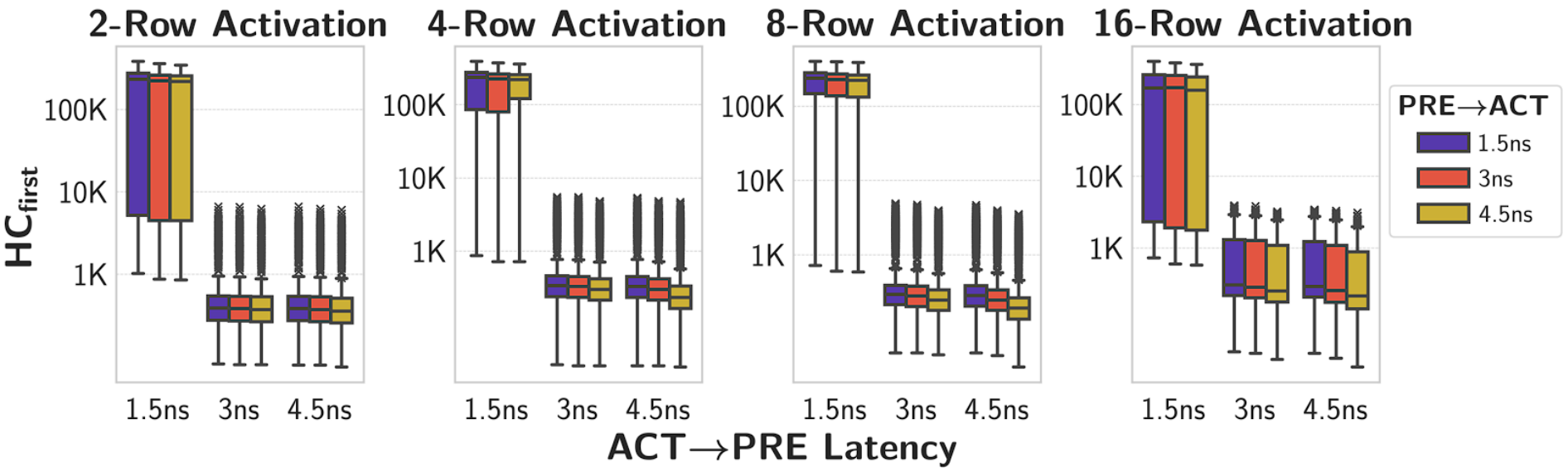
HCfirst consistently decreases as the number of simultaneously activated rows increases when performing single-sided SiMRA.

SiMRA vs. RowPress



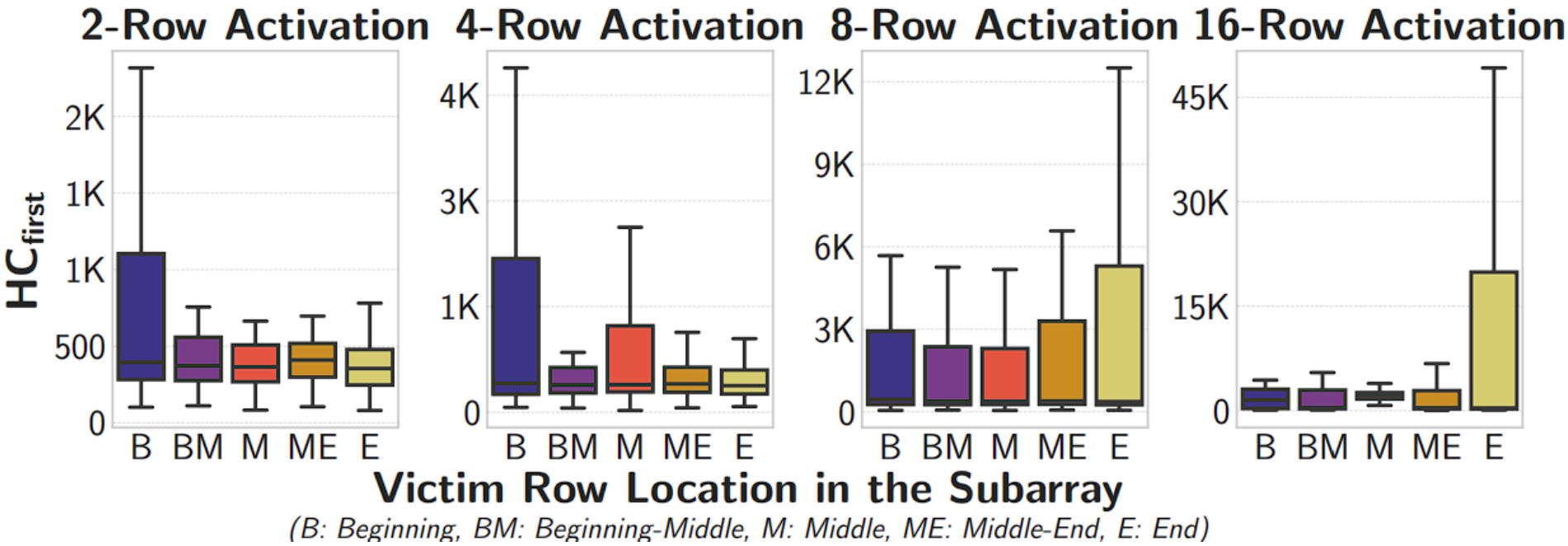
HCfirst decreases greatly as tAggOn increases

SiMRA -- Timing Delay



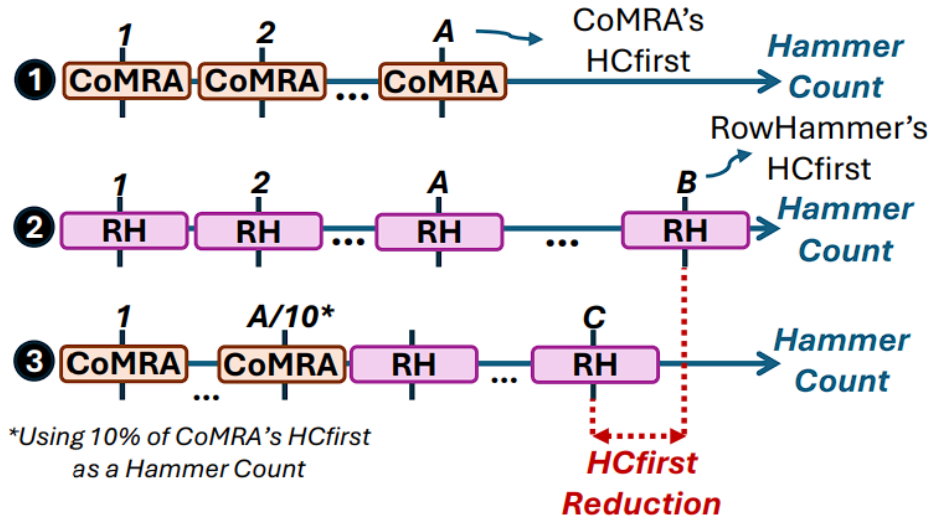
HCfirst decreases greatly as tAggOn increases

SiMRA – Spatial Variation

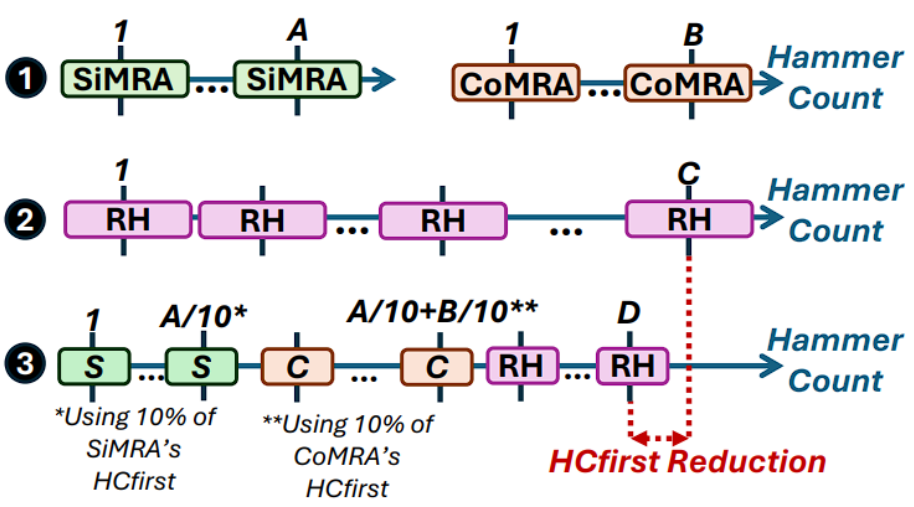


The effectiveness of SiMRA
depends on the victim row's location in a subarray.

Combined PuDHammer, RH & RP

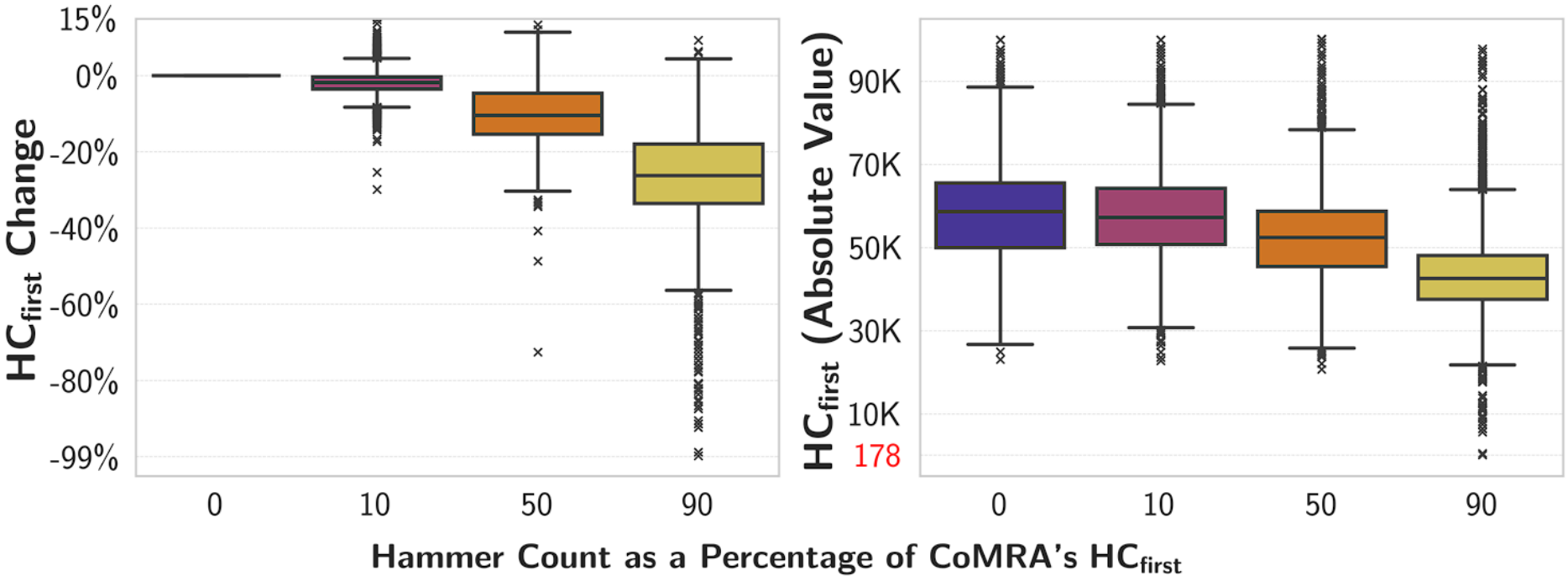


(a) Access Pattern for CoMRA (1), RowHammer (2), and Combined RowHammer and CoMRA (3)



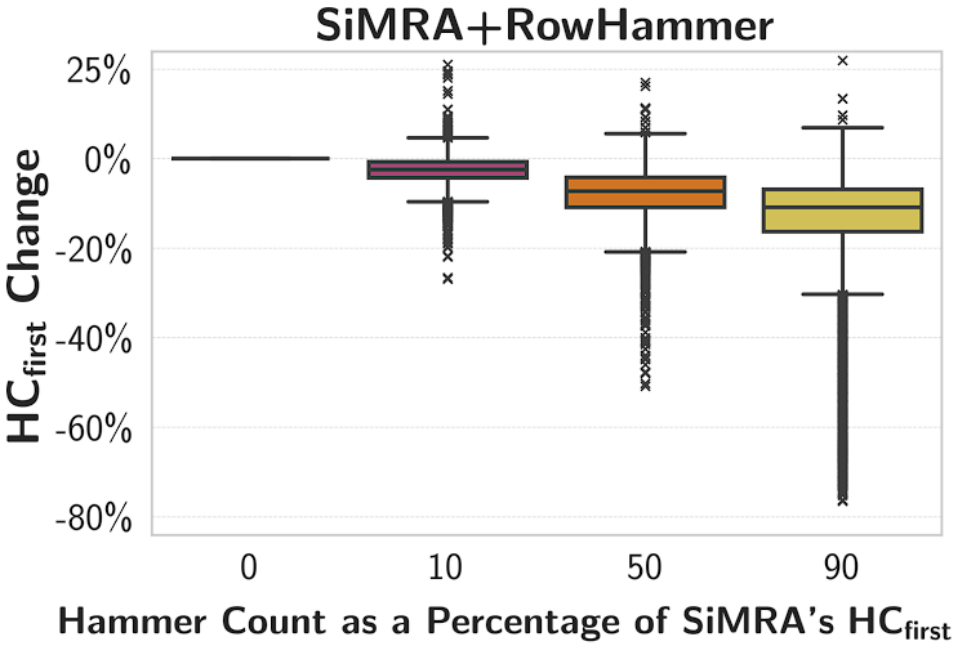
(b) Access Pattern for SiMRA and CoMRA (1), RowHammer (2), and Combined RowHammer, SiMRA and CoMRA (3)

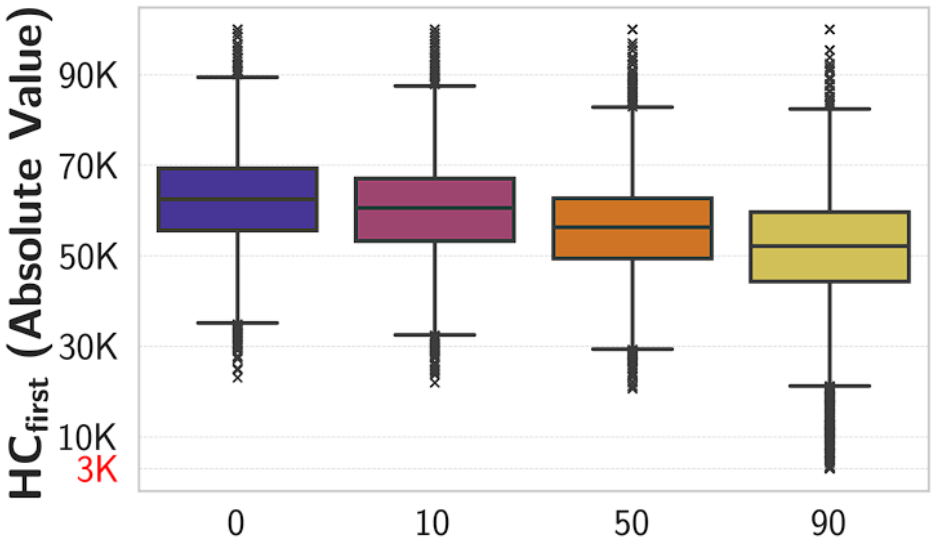
Combined CoMRA & RH



DRAM rows experience significantly lower HCfirst when performing combined RowHammer and CoMRA.

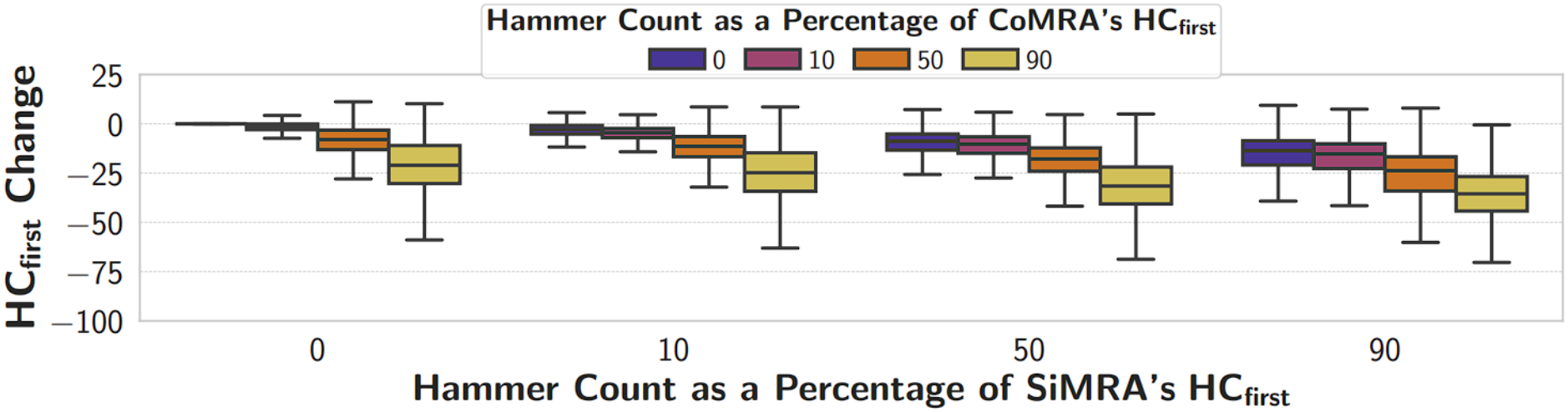
Combined SiMRA & RH





DRAM rows tend to experience lower HCfirst when performing combined RowHammer and SiMRA.

Combined CoMRA, SiMRA & RH



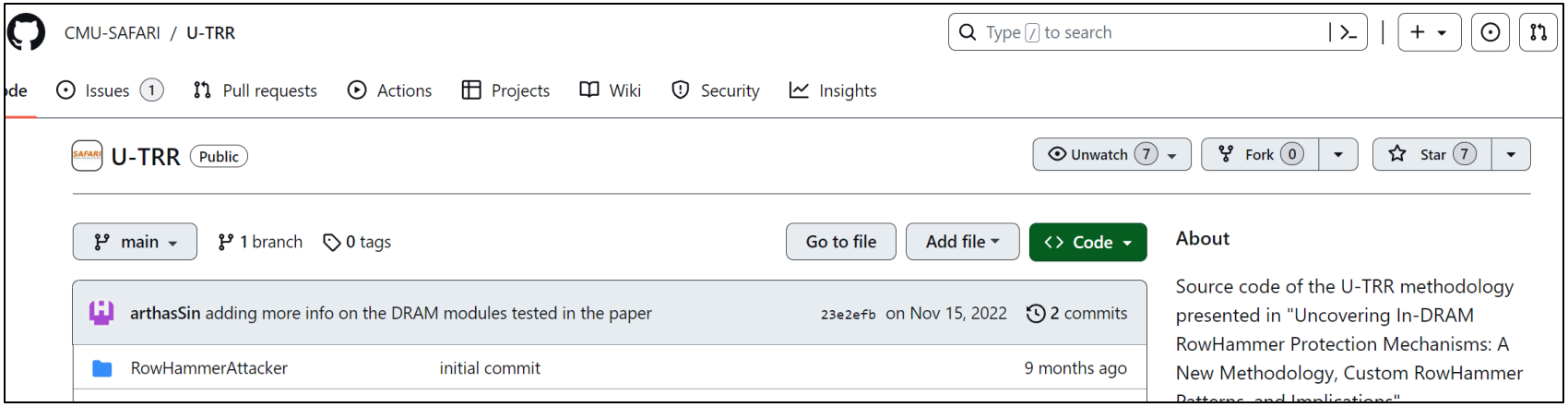
Combined RowHammer with CoMRA & SiMRA together is the most effective combined access pattern to decrease HCfirst.

Reverse Engineering TRR in Real DRAM

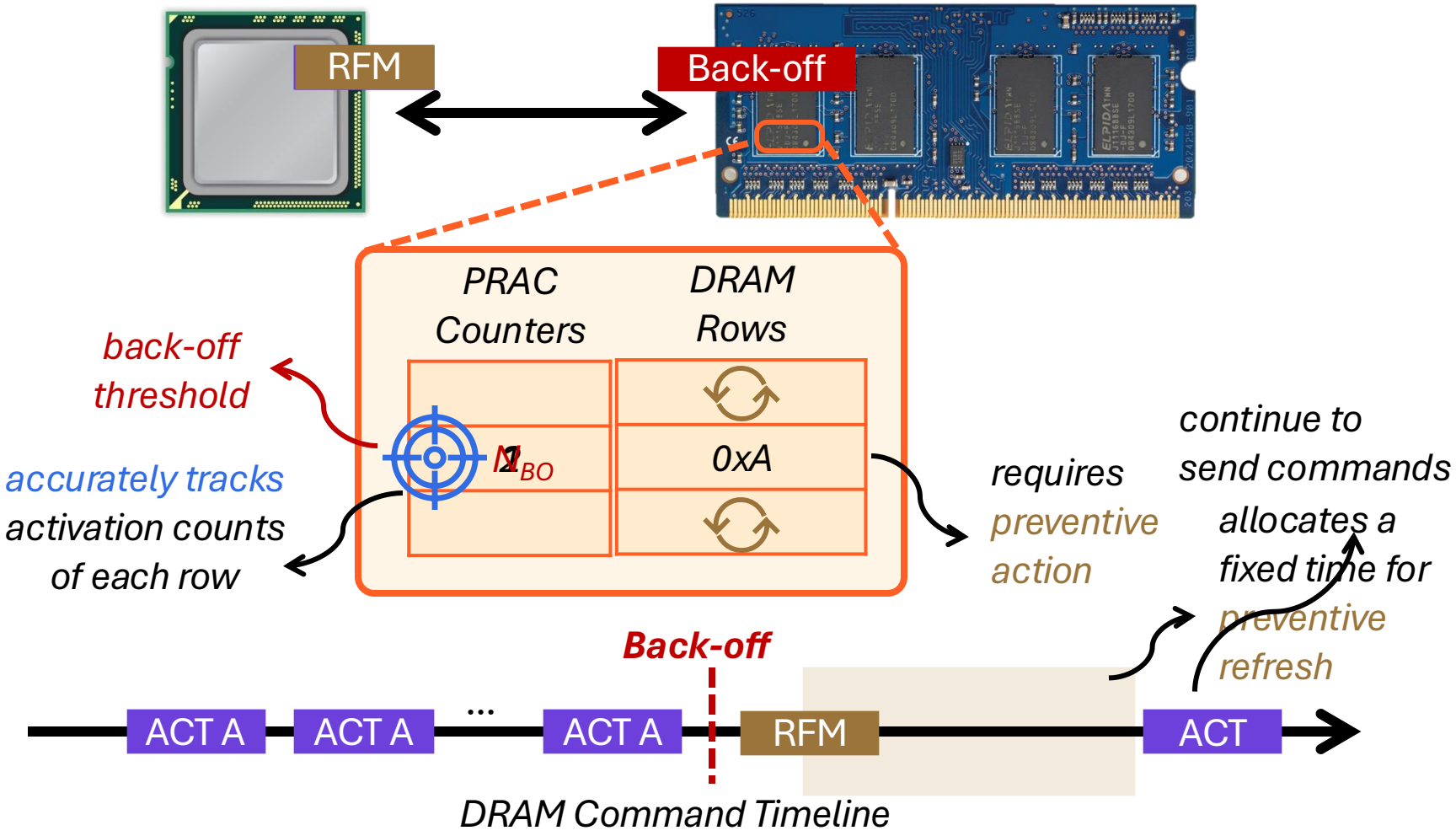
Key idea: Use data retention failures as a side channel to detect when a row is refreshed by on-die mitigation

Uncovering In-DRAM RowHammer Protection Mechanisms: A New Methodology, Custom RowHammer Patterns, and Implications

Hasan Hassan[†] Yahya Can Tuğrul[†‡] Jeremie S. Kim[†] Victor van der Veen[σ]
Kaveh Razavi[†] Onur Mutlu[†]
[†]ETH Zürich [‡]TOBB University of Economics & Technology [σ]Qualcomm Technologies Inc.



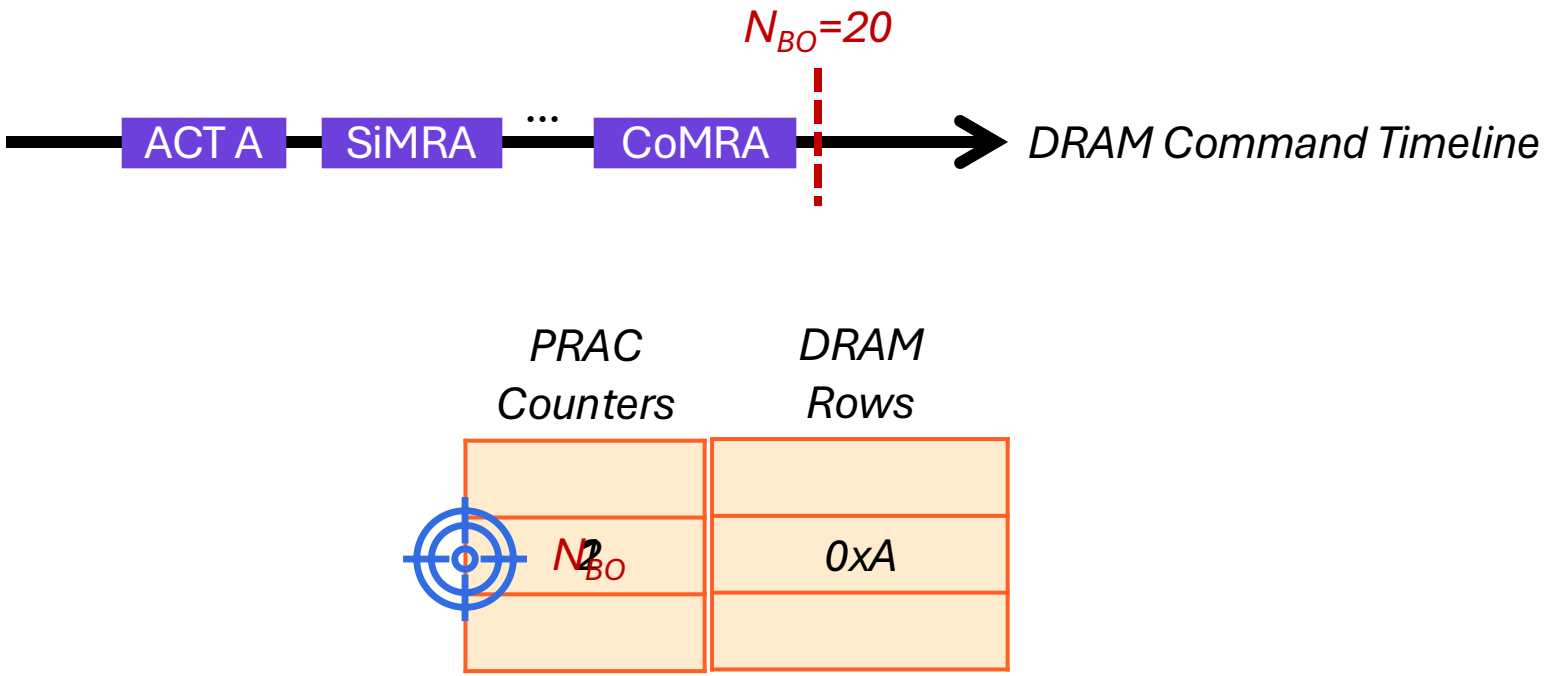
Per Row Activation Counting in DDR5 (April 2024)



A Naïve Approach (PRAC-Naïve)

Lowest HCfirst for tested DRAM chips

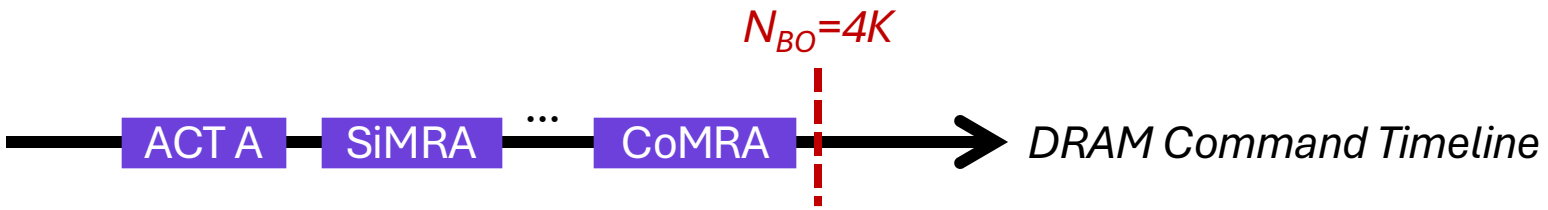
RH: ~4K	CoMRA: ~400	SiMRA: ~20

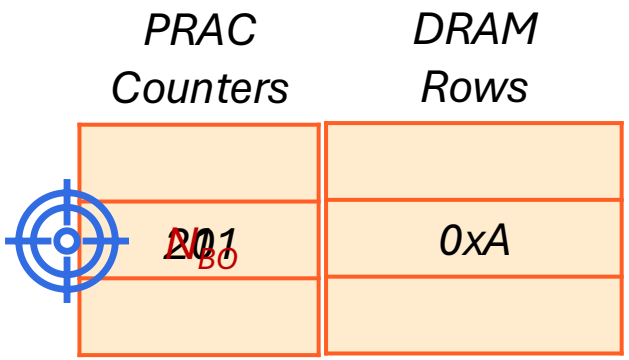


Weighted Counting (PRAC-WC)

Threshold is set to 4K (RH Threshold)
Each operation increases counters by

RH (ACT): 1	CoMRA: 10	SiMRA: 200





Updating Multiple Counters in PRAC

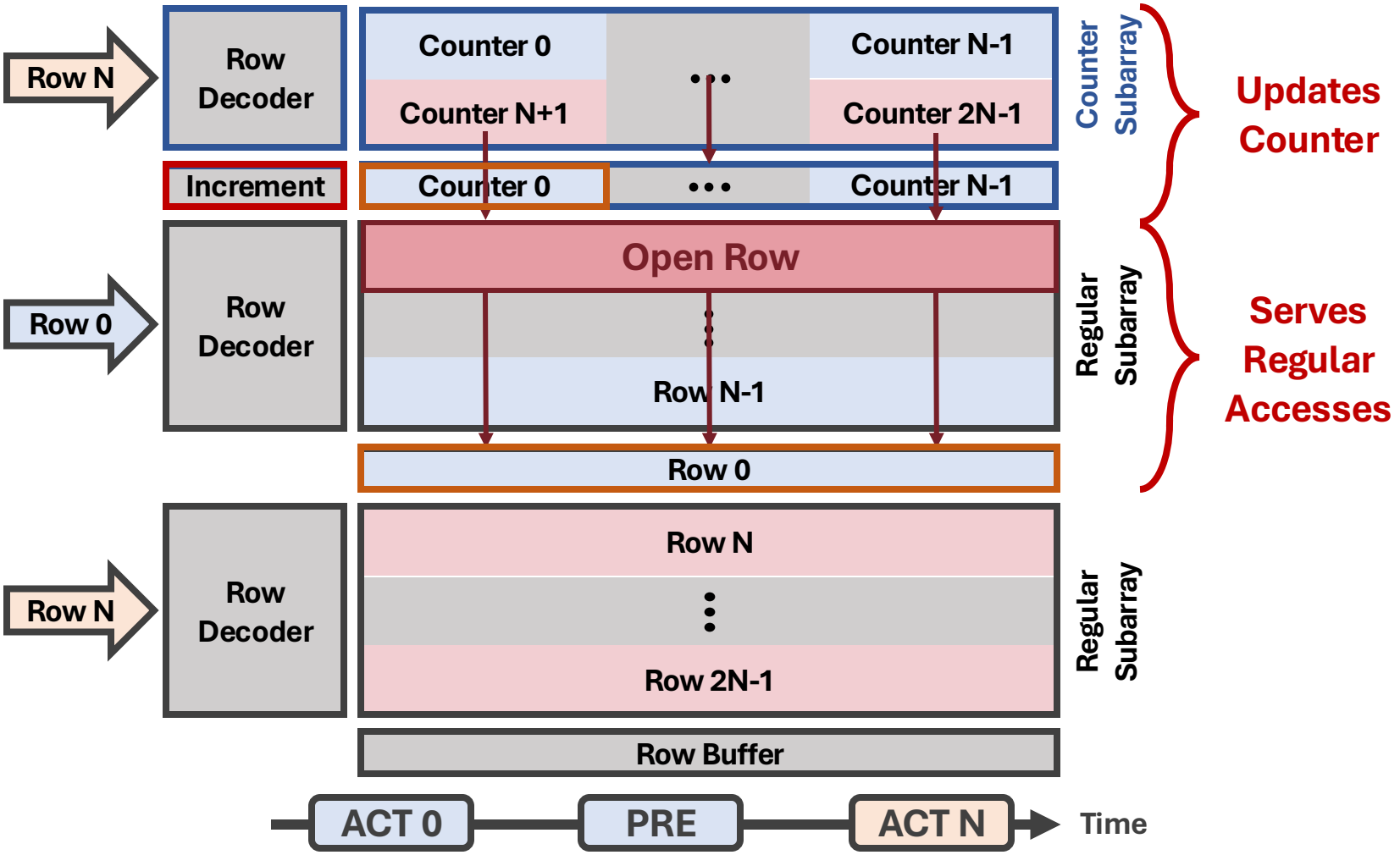
- SiMRA operation activates multiple rows simultaneously
  - Which requires updating multiple PRAC row activation counters simultaneously

- However, PRAC is designed to update only one counter upon an ACT command
  - We need to modify the counter organization and operation in PRAC to support updating multiple counters.

- Assuming we can activate up to N rows simultaneously
  - Simultaneous access to N different counter values
  - N incrementer circuits to update all the counters simultaneously

A Solution: Chronus + N Incrementers

Chronus concurrently updates counters while serving accesses



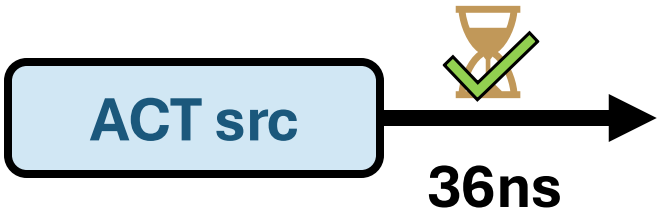
Limitations of Tested COTS DRAM Chips

- **Some COTS DRAM chips do not support SiMRA**
  - We do not observe SiMRA in all tested Samsung/Micron chips

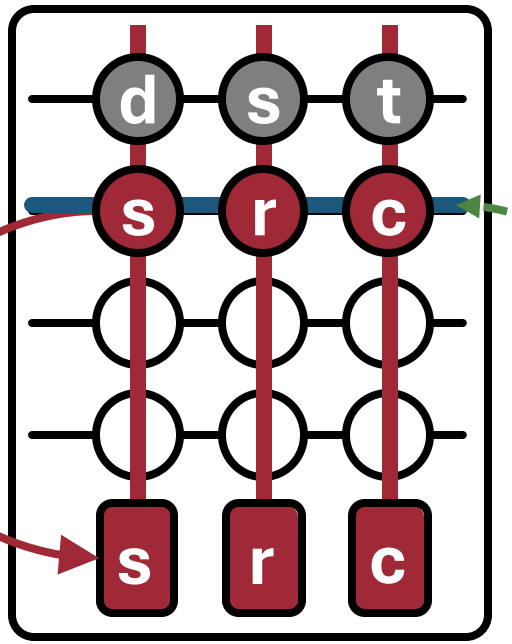
  - Hypothesis
    - Internal DRAM circuitry ignores the PRE command or the second ACT command when the timing parameters are greatly violated

If such a limitation were not imposed, we believe these DRAM chips are also fundamentally capable of performing the operations we examine in this work

In-DRAM Row-Copy (RowClone)

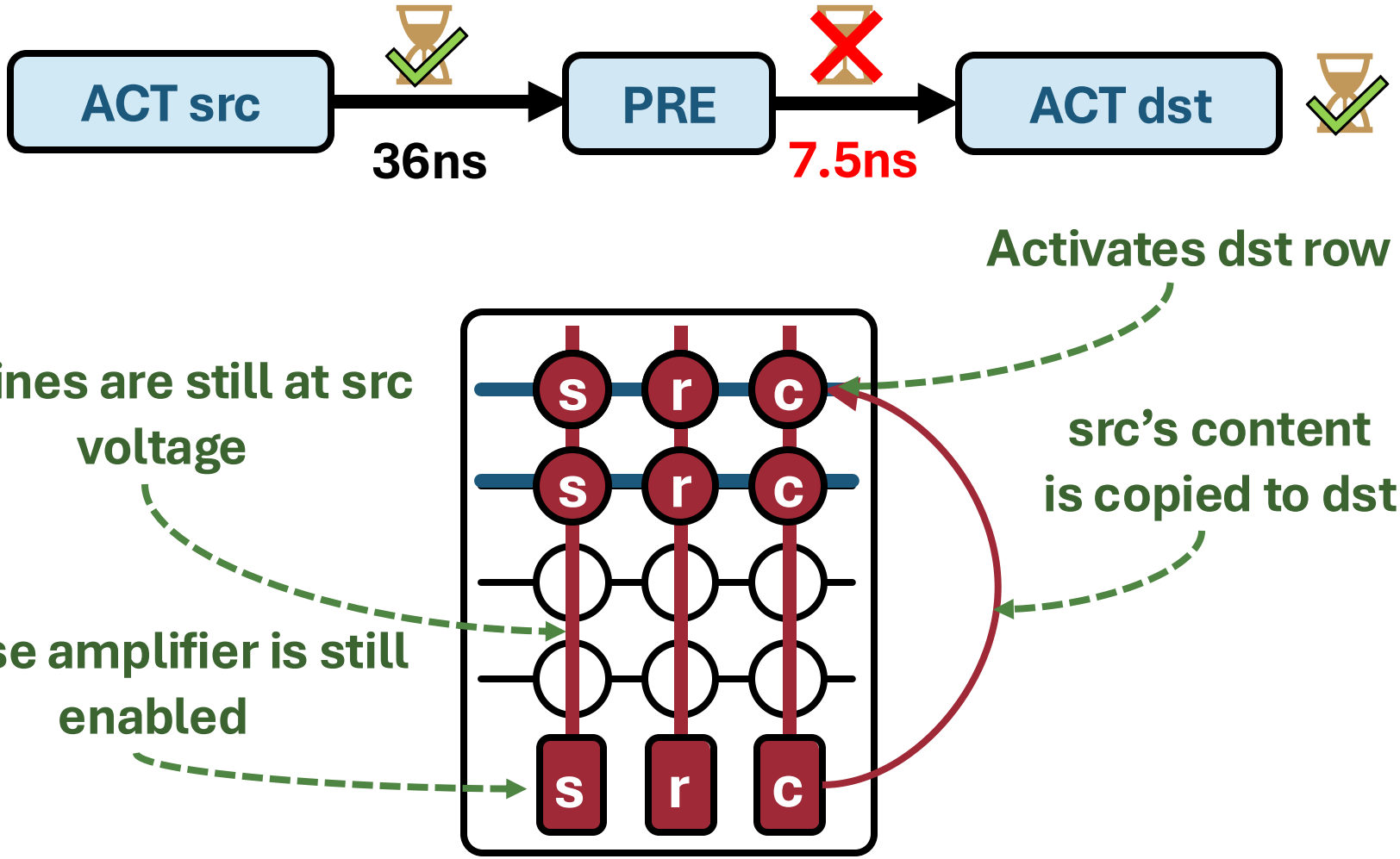


Fetch src's content into the sense amplifiers

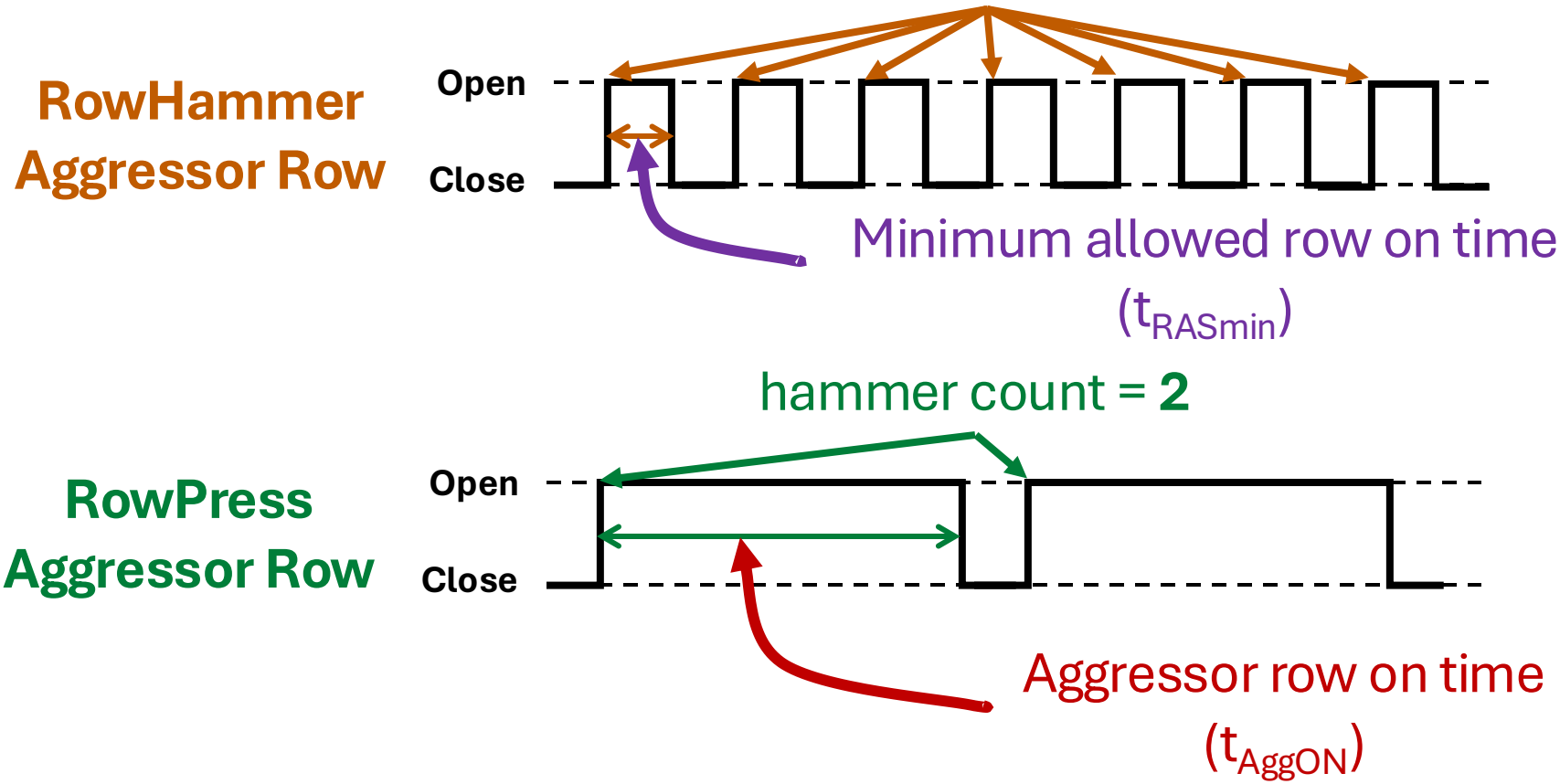


Assert the src's wordline

In-DRAM Row-Copy (RowClone)



Aggressor Row On Time & RowPress



Instead of using a high hammer count,

☞ increase the time that the aggressor row stays open

[Luo+, ISCA'23]